



**MISSOURI**  
DEPARTMENT OF  
NATURAL RESOURCES

Michael L. Parson  
Governor

Dru Buntin  
Director

October 28, 2024

Lou Rickers  
Chief Operations Officer  
Ag Processing, Inc.  
P.O. Box 2047  
Omaha, NE 68154

Re: Ag Processing, Inc., Facility ID: 021-0118  
Permit Number: OP102024-005

Dear Lou Rickers:

Enclosed with this letter is your Part 70 operating permit. Please review this document carefully. Operation of your installation in accordance with the rules and regulations cited in this document is necessary for continued compliance. It is very important that you read and understand the requirements contained in your permit.

This permit may include requirements with which you may not be familiar. If you would like the department to meet with you to discuss how to understand and satisfy the requirements contained in this permit, an appointment referred to as a Compliance Assistance Visit (CAV) can be set up with you. To request a CAV, please contact your local regional office or fill out an online request. The regional office contact information can be found at <https://dnr.mo.gov/about-us/division-environmental-quality/regional-office>. The online CAV request can be found at <https://dnr.mo.gov/compliance-assistance-enforcement/request-visit>.

You may appeal this permit to the Administrative Hearing Commission as provided in §§643.075.6 and 621.250 RSMo. To appeal, you must file a petition with the administrative hearing commission 30 days after the date this decision was mailed or the date it was delivered, whichever date was earlier. If any such petition is sent by registered mail or certified mail, it will be deemed filed on the date it is mailed; if it is sent by any method other than registered mail or certified mail, it will be deemed filed on the date it is received by the administrative hearing commission, whose contact information is: Administrative Hearing Commission, United States Post Office Building, 131 West High Street, Third Floor, PO Box 1557, Jefferson City, Missouri 65102, phone: 573-751-2422, fax: 573-751-5018, website: <https://ahc.mo.gov/>.



Lou Rickers  
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If you have any questions or need additional information regarding this permit, please contact the Air Pollution Control Program at (573) 751-4817, or you may write to the Department of Natural Resources, Air Pollution Control Program, PO Box 176, Jefferson City, MO 65102.

Sincerely,

AIR POLLUTION CONTROL PROGRAM

A handwritten signature in cursive script that reads "Alana L. Hess".

Alana Hess  
Unit Chief

AH:BKH

Enclosures

c: PAMS File: 2023-03-045  
Kansas City Regional Office



## PART 70

# PERMIT TO OPERATE

Under the authority of RSMo 643 and the Federal Clean Air Act the applicant is authorized to operate the air contaminant source(s) described below, in accordance with the laws, rules, and conditions set forth herein.

**Operating Permit Number:** OP102024-005  
**Expiration Date:** 10/28/2029  
**Installation ID:** 021-0118  
**Project Number:** 2023-03-045

### **Installation Name and Address**

Ag Processing, Inc.  
900 Lower Lake Road  
St. Joseph, MO 64502  
Buchanan County, S30, T57N, R35W

### **Installation Description:**

Ag Processing, Inc. is a 292 million pounds per year methyl ester (biodiesel) plant. The biodiesel process involves reacting semi-refined or refined vegetable oils with methanol in the presence of a catalyst to produce biodiesel (methyl ester) and crude glycerin. Excess methanol is recovered and recycled back into the process. Emissions from the process are controlled with a condenser and scrubber system.

This plant is adjacent to the Ag Processing, Inc. extraction/refining plant (021-0060). It is a major source of HAP emissions.

October 28, 2024

Effective Date

Director or Designee  
Department of Natural Resources

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## I. Installation Equipment Listing

### Emission Units with Limitations

The following list provides a description of the equipment at this installation that emits air pollutants and that are identified as having Emission Unit-Specific Emission Limitations in Section III of this permit. These emission units are also subject to the Plant Wide Emission Limitations in Section II of this permit.

| Emission Unit | Description                                       | Control Equipment                                                    | Applicable Requirements                         |
|---------------|---------------------------------------------------|----------------------------------------------------------------------|-------------------------------------------------|
| EP-100        | Biodiesel Process Vent<br>91 ft <sup>3</sup> /min | Closed Vent System to Precondenser, condenser, and scrubber (CD-100) | MACT FFFF,<br>Construction Permit<br>122014-009 |
| EP-101        | Light Liquid Valves                               | None                                                                 | MACT FFFF                                       |
| EP-102        | Light Liquid Pumps                                | None                                                                 |                                                 |
| EP-103        | Gas Pressure Relief Valves                        | Closed Vent System to Precondenser, condenser, and scrubber (CD-100) |                                                 |
|               | ME Wastewater                                     | None                                                                 |                                                 |
| EP-104        | Connectors                                        | None                                                                 |                                                 |
|               | Agitators                                         | None                                                                 |                                                 |
| EP-106        | Haul Roads                                        | None                                                                 | Construction Permit<br>122014-009               |
| EP-107        | Methanol Storage Tank<br>76,310 gallons           | Closed Vent System to Precondenser, condenser, and scrubber (CD-100) | MACT FFFF                                       |
|               | Catalyst Tank, 35,720 gallons                     |                                                                      |                                                 |

### Emission Units without Specific Limitations

The following list provides a description of the equipment that does not have unit specific limitations at the time of permit issuance. These emission units are subject to the Plant Wide Emission Limitations in Section II of this permit.

| Emission Unit | Description                                       |
|---------------|---------------------------------------------------|
| EP-105        | Biodiesel Truck & Rail Load Out                   |
| -             | Phosphoric Acid Tank, 9,300 gallons               |
| -             | Hydrochloric Acid Tank, 9,300 gallons             |
| -             | (4) Biodiesel Storage Tanks, 55,200 gallons each  |
| -             | (2) Biodiesel Storage Tanks, 525,000 gallons each |
| -             | (2) Glycerin Storage Tanks, 54,660 gallons each   |
| -             | Glycerin Day Tank, 26,172 gallons                 |
| -             | Fatty Acid Storage Tank, 36,320 gallons           |

## **II. Plant Wide Emission Limitations**

The installation shall comply with each of the following emission limitations. Consult the appropriate sections in the CFR, CSR, and Missouri's SIP for the full text of the applicable requirements. All citations, unless otherwise noted, are to the regulations in effect on the date of permit issuance. The plant wide conditions apply to all emission units at this installation. All emission units are listed in Section I under Emission Units with Limitations and Emission Units without Limitations.

None.

### III. Emission Unit Specific Emission Limitations

The installation shall comply with each of the following emission limitations. Consult the appropriate sections in the CFR, the CSR, and Missouri's SIP for the full text of the applicable requirements. All citations, unless otherwise noted, are to the regulations in effect on the date of permit issuance.

| <b>PERMIT CONDITION 001</b>                                                                                                                  |                                                  |                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|----------------------------------------------------------------------|
| <b>10 CSR 10-6.075 Maximum Achievable Control Technology Regulations</b>                                                                     |                                                  |                                                                      |
| <b>40 CFR Part 63, Subpart A – General Provisions</b>                                                                                        |                                                  |                                                                      |
| <b>40 CFR Part 63, Subpart FFFF – National Emission Standards for Hazardous Air Pollutants: Miscellaneous Organic Chemical Manufacturing</b> |                                                  |                                                                      |
| <b>Emission Unit</b>                                                                                                                         | <b>Description</b>                               | <b>Control Equipment</b>                                             |
| EP-100                                                                                                                                       | Biodiesel Process Vent (91 ft <sup>3</sup> /min) | Closed Vent System to Precondenser, condenser, and scrubber (CD-100) |
| EP-101                                                                                                                                       | Light Liquid Valves                              | None                                                                 |
| EP-102                                                                                                                                       | Light Liquid Pumps                               | None                                                                 |
| EP-103                                                                                                                                       | Gas Pressure Relief Valves                       | Closed Vent System to Precondenser, condenser, and scrubber (CD-100) |
|                                                                                                                                              | ME Wastewater                                    | None                                                                 |
| EP-104                                                                                                                                       | Connectors                                       | None                                                                 |
|                                                                                                                                              | Agitators                                        | None                                                                 |
| EP-107                                                                                                                                       | Methanol Storage Tank (76,310 gallons)           | Closed Vent System to Precondenser, condenser, and scrubber (CD-100) |
|                                                                                                                                              | Catalyst Tank (35,720 gallons)                   |                                                                      |

#### **MACT FFFF General Requirements:**

1. The permittee shall comply with the applicable requirements of 40 CFR Part 63, Subpart A – *General Provisions* as specified by Table 12 to MACT FFFF. [§63.2540]
2. The permittee must comply with the emission limits and work practice standards in Tables 2, 4, 6, and 7 to MACT FFFF at all times, and must meet the requirements specified in §63.2455 through §63.2490 (or the alternative means of compliance in §63.2495, §63.2500, or §63.2505), except as specified in §63.2450(b) through (v). The permittee must meet the notification, reporting, and recordkeeping requirements specified in §63.2515, §63.2520, and §63.2525. [§63.2450(a)(2)]
3. *Determine halogenated vent streams.* The permittee must determine if an emission stream is a halogenated vent stream, as defined in §63.2550, by calculating the mass emission rate of halogen atoms in accordance with §63.115(d)(2)(v). Alternatively, the permittee may elect to designate the emission stream as halogenated. [§63.2450(b)]
4. *Requirements for combined emission streams.* When organic HAP emissions from different emission types (e.g., continuous process vents, batch process vents, storage tanks, transfer operations, and waste management units) are combined, the permittee must comply with the requirements of either §63.2450(c)(1) or (2). [§63.2450(c)]
  - a) Comply with the applicable requirements of MACT FFFF for each kind of organic HAP emissions in the stream (e.g., the requirements of Table 2 to MACT FFFF for batch process vents and the requirements of Table 4 to MACT FFFF for emissions from storage tanks). [§63.2450(c)(1)]

- b) Determine the applicable requirements based on the hierarchy presented in §63.2450(c)(2)(i) through (vi). For a combined stream, the applicable requirements are specified in the highest-listed paragraph in the hierarchy that applies to any of the individual streams that make up the combined stream. For example, if a combined stream consists of emissions from Group 1 batch process vents and any other type of emission stream, then the permittee must comply with the requirements in §63.2450(c)(2)(i) for the combined stream; compliance with the requirements in §63.2450(c)(2)(i) constitutes compliance for the other emission streams in the combined stream. Exception recordkeeping requirements for Group 2 applicability or compliance are still required (e.g., the requirement in §63.2525(e)(3) and (4) to track the number of batches produced and calculate rolling annual emissions for processes with Group 2 batch process vents). [§63.2450(c)(2)]
  - 1) The requirements of Table 2 to MACT FFFF and §63.2460 for Group 1 batch process vents, including applicable monitoring, recordkeeping, and reporting. [§63.2450(c)(2)(i)]
  - 2) The requirements of Table 7 to MACT FFFF and §63.2485 for emissions from waste management units that are used to manage and treat Group 1 wastewater streams and residuals from Group 1 wastewater streams, including applicable monitoring, recordkeeping, and reporting. [§63.2450(c)(2)(iv)]
  - 3) The requirements of Table 4 to MACT FFFF and §63.2470 for control of emissions from storage tanks, including applicable monitoring, recordkeeping, and reporting. [§63.2450(c)(2)(v)]
- 5. *Requirements for control devices.* [§63.2450(e)]
  - a) Except when complying with §63.2485, if the permittee reduces organic HAP emissions by venting emissions through a closed-vent system to any combination of control devices (except a flare) or recovery devices, the permittee must meet the requirements of §63.982(c) and the requirements referenced therein. [§63.2450(e)(1)]
    - 1) *Closed vent system and nonflare control device.* If the permittee controls emissions through a closed vent system to a nonflare control device, the permittee shall meet the requirements in §63.983 for closed vent systems, the applicable recordkeeping and reporting requirements of §63.998 and §63.999, and the applicable requirements listed in §63.982(c)(1) through (3). [§63.982(c)]
      - (i) For storage tanks and low throughput transfer racks, the permittee shall meet the requirements in §63.985 for nonflare control devices and the monitoring, recordkeeping, and reporting requirements referenced therein. No other provisions of MACT SS apply to low throughput transfer rack emissions or storage tank emissions vented through a closed vent system to a nonflare control device unless specifically required in the monitoring plan submitted under §63.985(c). [§63.982(c)(1) and §63.2470(c)(2)]
      - (ii) For process vents, the permittee shall meet the requirements applicable to the control devices being used in §63.990 or §63.995; the applicable general monitoring requirements of §63.996 and the applicable performance test requirements and procedures of §63.997; and the monitoring, recordkeeping and reporting requirements referenced therein. The requirements of §63.984 through §63.986 do not apply to process vents. [§63.982(c)(2)]
  - b) The references provisions specified in §63.2450(e)(4)(i) through (xvi) do not apply when demonstrating compliance with MACT SS. [§63.2450(e)(4)]
    - 1) §63.996(c)(2)(ii). [§63.2450(e)(4)(vi)]

- 2) The last sentence of §63.997(e)(1)(i). [§63.2450(e)(4)(vii)]
- 3) §63.998(b)(2)(iii). [§63.2450(e)(4)(viii)]
- 4) The phrase “other than start-ups, shutdowns or malfunctions” in §63.998(b)(5)(i)(A). [§63.2450(e)(4)(ix)]
- 5) The phrase “other than a start-up, shutdown, or malfunction” from §63.998(b)(5)(i)(B)(3). [§63.2450(e)(4)(x)]
- 6) The phrase “other than start-ups, shutdowns or malfunctions” in §63.998(b)(5)(i)(C). [§63.2450(e)(4)(xi)]
- 7) The phrase “other than a start-up, shutdown, or malfunction” from §63.998(b)(5)(ii)(C). [§63.2450(e)(4)(xii)]
- 8) The phrase “except as provided in paragraphs (b)(6)(i)(A) and (B) of this section” in §63.998(b)(6)(i). [§63.2450(e)(4)(xiii)]
- 9) The second sentence of §63.998(b)(6)(ii). [§63.2450(e)(4)(xiv)]
- 10) §63.998(c)(1)(ii)(D), (E), (F), and (G). [§63.2450(e)(4)(xv)]
- 11) §63.998(d)(3). [§63.2450(e)(4)(xvi)]
- c) The use of a bypass line at any time on a closed vent system to divert emissions subject to the requirements in Table 2 to MACT FFFF to the atmosphere or to a control device not meeting the requirements specified in Table 2 to MACT FFFF is an emissions standards deviation. The permittee must also comply with the requirements specified in §63.2450(e)(6)(i) through (v), as applicable: [§63.2450(e)(6)]
  - 1) If subject to the bypass monitoring requirements of §63.983(a)(3), then the permittee must continue to comply with the requirements in §63.983(a)(3) and the recordkeeping and reporting requirements in §63.998(d)(1)(ii) and §63.999(c)(2), in addition to the requirements specified in §63.2450(e)(4), the recordkeeping requirements specified in § 63.2525(n), and the reporting requirements specified in §63.2520(e)(12). [§63.2450(e)(6)(iii)]
6. *Requirements for performance tests.* The requirements specified in §63.2450(g)(1) through (7) apply instead of or in addition to the requirements specified in MACT SS. [§63.2450(g)]
  - a) Conduct gas molecular weight analysis using Method 3, 3A, or 3B in NSPS Appendix A. [§63.2450(g)(1)]
  - b) Measure moisture content of the stack gas using Method 4 in NSPS Appendix A. [§63.2450(g)(2)]
  - c) As an alternative to using Method 18, Method 25/25A, or Method 26/26A of NSPS Appendix A, to comply with any of the emission limits specified in Tables 2, 4, 6, and 7 to MACT FFFF, the permittee may use Method 320 of NSPS Appendix A. When using Method 320, the permittee must follow the analyte spiking procedures of §13 of Method 320, unless the permittee demonstrates that the complete spiking procedure has been conducted at a similar source. [§63.2450(g)(4)]
  - d) §63.997(c)(1) does not apply. [§63.2450(g)(5)]
  - e) In lieu of the requirements specified in §63.7(e)(1) the permittee must conduct performance tests under such conditions as the Administrator specifies based on representative performance of the affected source for the period being tested. Representative conditions exclude periods of startup and shutdown. The permittee may not conduct performance tests during periods of malfunction. The permittee must record the process information that is necessary to document operating conditions during the test and include in such record an explanation to support that such conditions represent normal operation. Upon request, the permittee must make available to the Administrator such records as may be necessary to determine the conditions of performance tests.

[§63.2450(g)(6)]

- f) Comply with the requirements in § 63.2450(e)(4), as applicable. [§63.2450(g)(7)]
7. *Design evaluation.* To determine the percent reduction of a small control device that is used to comply with an emission limit specified in Table 2 to MACT FFFF, the permittee may elect to conduct a design evaluation as specified in §63.1257(a)(1) instead of a performance test as specified in MACT SS. The permittee must establish the value(s) and basis for the operating limits as part of the design evaluation. For continuous process vents, the design evaluation must be conducted at maximum representative operating conditions for the process, unless the Administrator specifies or approves alternate operating conditions. [§63.2450(h)]
8. *Startup, shutdown, and malfunction.* §63.152(f)(7)(ii) through (iv) and §63.998(b)(2)(iii) and (b)(6)(i)(A), which apply to the exclusion of monitoring data collected during periods of SSM from daily averages, do not apply for the purposes of MACT FFFF. [§63.2450(l)]
9. If an emission stream contains energetics or organic peroxides that, for safety reasons, cannot meet an applicable emission limit specified in Tables 2, 4, 6, and 7 to MACT FFFF, then the permittee must submit documentation in the precompliance report explaining why an undue safety hazard would be created if the air emission controls were installed, and the permittee must describe the procedures that the permittee will implement to minimize HAP emissions from these vent streams. [§63.2450(q)]
10. *Surge control vessels and bottoms receivers.* For each surge control vessel or bottoms receiver that meets the capacity and vapor pressure thresholds for a Group 1 storage tank, the permittee must meet emission limits and work practice standards specified in Table 4 to MACT FFFF. [§63.2450(r)]
11. For the purposes of determining Group status for continuous process vents, batch process vents, and storage tanks in §63.2455, §63.2460, and §63.2470, hydrazine is to be considered an organic HAP. [§63.2450(s)]
12. *New safety device requirements.* §63.2450(p) no longer applies. Instead, the permittee must comply with the requirements specified in §63.2480(e). [§63.2450(t)]
13. *General duty.* At all times, the permittee must operate and maintain any affected source, including associated air pollution control equipment and monitoring equipment, in a manner consistent with safety and good air pollution control practices for minimizing emissions. The general duty to minimize emissions does not require the permittee to make any further efforts to reduce emissions if levels required by the applicable standard have been achieved. Determination of whether a source is operating in compliance with operation and maintenance requirements will be based on information available to the Administrator which may include, but is not limited to, monitoring results, review of operation and maintenance procedures, review of operation and maintenance records, and inspection of the source. [§63.2450(u)]
14. *Maintenance vents.* The permittee may designate a process vent as a maintenance vent if the vent is only used because of startup, shutdown, maintenance, or inspection of equipment where equipment is emptied, depressurized, degassed, or placed into service. The permittee must comply with the applicable requirements in §63.2450(v)(1) through (3) for each maintenance vent. Any vent designated as a maintenance vent is only subject to the maintenance vent provisions in this paragraph and the associated recordkeeping and reporting requirements in §63.2525(p) and §63.2520(e)(14), respectively. The permittee does not need to designate a maintenance vent as a Group 1 or Group 2 process vent nor identify maintenance vents in a Notification of Compliance Status report. [§63.2450(v)]

- a) Prior to venting to the atmosphere, remove process liquids from the equipment as much as practical and depressurize the equipment to a non-flare control device meeting the requirements in §63.2450(e)(4) and the requirements specified in §63.982(c)(2) of MACT SS until one of the following conditions, as applicable, is met. [§63.2450(v)(1)]
  - 1) The vapor in the equipment served by the maintenance vent has a lower explosive limit (LEL) of less than 10% and has an outlet concentration less than or equal to 20 ppmv hydrogen halide and halogen HAP. [§63.2450(v)(1)(i)]
  - 2) If there is no ability to measure the LEL of the vapor in the equipment based on the design of the equipment, the pressure in the equipment served by the maintenance vent is reduced to 5 psig or less. Upon opening the maintenance vent, active purging of the equipment cannot be used until the LEL of the vapors in the maintenance vent (or inside the equipment if the maintenance is a hatch or similar type of opening) is less than 10%. [§63.2450(v)(1)(ii)]
  - 3) The equipment served by the maintenance vent contains less than 50 pounds of total VOC. [§63.2450(v)(1)(iii)]
  - 4) If, after applying best practices to isolate and purge equipment served by a maintenance vent, none of the applicable criterion in §63.2450(v)(1)(i) through (iii) can be met prior to installing or removing a blind flange or similar equipment blind, then the pressure in the equipment served by the maintenance vent must be reduced to 2 psig or less before installing or removing the equipment blind. During installation or removal of the equipment blind, active purging of the equipment may be used provided the equipment pressure at the location where purge gas is introduced remains at 2 psig or less. [§63.2450(v)(1)(iv)]
- b) Except for maintenance vents complying with the alternative in §63.2450(v)(1)(iii), the permittee must determine the LEL or, if applicable, equipment pressure using process instrumentation or portable measurement devices and follow procedures for calibration and maintenance according to manufacturer's specifications. [§63.2450(v)(2)]
- c) For maintenance vents complying with the alternative in §63.2450(v)(1)(iii), the permittee must determine mass of VOC in the equipment served by the maintenance vent based on the equipment size and contents after considering any contents drained or purged from the equipment. Equipment size may be determined from equipment design specifications. Equipment contents may be determined using process knowledge. [§63.2450(v)(3)]

**MACT FFFF Batch Process Vent Requirements:**

1. *General.* The permittee must meet each emission limit in Table 2 to MACT FFFF that applies, and the permittee must meet each applicable requirement specified in §63.2460(b) and (c). [§63.2460(a)]
2. *Group status.* If a process has batch process vents, as defined in §63.2550, the permittee must determine the group status of the batch process vents by determining and summing the uncontrolled organic HAP emissions from each of the batch process vents within the process using the procedures specified in §63.1257(d)(2)(i) and (ii), except as specified in §63.2460(b)(1) through (7). [§63.2460(b)]
  - a) To calculate emissions caused by the heating of a vessel without a process condenser to a temperature lower than the boiling point, the permittee must use the procedures in §63.1257(d)(2)(i)(C)(3). [§63.2460(b)(1)]
  - b) To calculate emissions from the depressurization of a vessel without a process condenser, the permittee must use the procedures in §63.1257(d)(2)(i)(D)(10). [§63.2460(b)(2)]

- c) To calculate emission from vacuum systems for the purposes of MACT FFFF, the receiving vessel is part of the vacuum system, and terms used in Equation 33 to MACT GGG, are defined as follows:  
 $P_{\text{system}}$  = absolute pressure of the receiving vessel  
 $P_i$  = partial pressure of the HAP determined at the exit temperature and exit pressure conditions of the condenser or at the conditions of the dedicated receiver  
 $P_j$  = partial pressure of the condensable (including HAP) determined at the exit temperature and exit pressure conditions of the condenser or at the conditions of the dedicated receiver  
 $MW_{\text{HAP}}$  = molecular weight of the HAP determined at the exit temperature and exit pressure conditions of the condenser or at the conditions of the dedicated receiver [§63.2460(b)(3)]
- d) To calculate uncontrolled emissions when a vessel is equipped with a process condenser, the permittee must use the procedures in §63.1257(d)(3)(i)(B), except as specified in §63.2460(b)(4)(i) through (vii). [§63.2460(b)(4)]
- 1) The permittee must determine the flowrate of gas (or volume of gas), partial pressures of condensables, temperature (T), and HAP molecular weight ( $MW_{\text{HAP}}$ ) at the exit temperature and exit pressure conditions of the condenser or at the conditions of the dedicated receiver. [§63.2460(b)(4)(i)]
  - 2) The permittee must assume that all components contained in the condenser exit vent stream are in equilibrium with the same components in the exit condensate stream (except for noncondensables). [§63.2460(b)(4)(ii)]
  - 3) The permittee must perform a material balance for each component. [§63.2460(b)(4)(iii)]
  - 4) For emissions from gas evolution, the term for time, t, must be used in Equation 12 to MACT GGG. [§63.2460(b)(4)(iv)]
  - 5) Emissions from empty vessel purging shall be calculated using Equation 36 to MACT GGG and the exit temperature and exit pressure conditions of the condenser or the conditions of the dedicated receiver. [§63.2460(b)(4)(v)]
  - 6) The permittee must conduct an engineering assessment as specified in §63.1257(d)(2)(ii) for each emission episode that is not due to vapor displacement, purging, heating, depressurization, vacuum operations, gas evolution, air drying, or empty vessel purging. The requirements of §63.2460(b)(3) through (4) shall apply. [§63.2460(b)(4)(vi)]
  - 7) The permittee may elect to conduct an engineering assessment if the permittee can demonstrate to the Administrator that the methods in §63.1257(d)(3)(i)(B) are not appropriate. [§63.2460(b)(4)(vii)]
- e) The permittee may elect to designate the batch process vents within a process as Group 1 and not calculate uncontrolled emissions if the permittee complies with one of the situations in §63.2460(b)(5)(i), (ii), or (iii). [§63.2460(b)(5)]
- 1) If the permittee complies with the alternative standard specified in §63.2505. [§63.2460(b)(5)(i)]
  - 2) If all Group 1 batch process vents within a process are controlled; the permittee conducts the performance test under hypothetical worst case conditions, as defined in §63.1257(b)(8)(i)(B); and the emission profile is based on capture and control system limitations as specified in §63.1257(b)(8)(ii)(C). [§63.2460(b)(5)(ii)]
- f) The permittee may change from Group 2 to Group 1 in accordance with either §63.2460(b)(6)(i) or (ii). The permittee must comply with the requirements of §63.2460

and submit the performance test report for the demonstration required in §63.1257(b)(8) in accordance with §63.2520(f). [§63.2460(b)(6)]

- 1) The permittee may switch at any time after operating as Group 2 for at least one year so that the permittee can show compliance with the 10,000 lbs./yr. threshold for Group 2 batch process vents for at least 365 days before the switch. The permittee may elect to start keeping records of emissions from Group 2 batch process vents before the compliance date. Report a switch based on this provision in the permittee's next compliance report in accordance with §63.2520(e)(10)(i). [§63.2460(b)(6)(i)]
- 2) If the conditions in §63.2460(b)(6)(i) are not applicable, the permittee must provide a 60-day advance notice in accordance with §63.2520(e)(10)(i) before switching. [§63.2460(b)(6)(ii)]
- g) As an alternative to determining the uncontrolled organic HAP emissions as specified in §63.1257(d)(2)(i) and (ii), the permittee may elect to demonstrate that non-reactive organic HAP are the only HAP used in the process and non-reactive HAP usage in the process is less than 10,000 lbs./yr. The permittee must provide data and supporting rationale in the notification of compliance status report explaining why the non-reactive organic HAP usage will be less than 10,000 lbs./yr. The permittee must keep records of the non-reactive organic HAP usage as specified in §63.2525(e)(2) and include information in compliance reports as specified in §63.2520(e)(5)(iv). [§63.2460(b)(7)]
3. Exceptions to the requirements in MACT SS and MACT WW are specified in §63.2460(c)(1) through (9). [§63.2460(c)]
  - a) *Process condensers*. Process condensers, as defined in §63.2550(i), are not considered to be control devices for batch process vents. The permittee must determine whether a condenser is a control device for a batch process vent or a process condenser from which the uncontrolled HAP emissions are evaluated as part of the initial compliance demonstration for each MCPU and report the results with supporting rationale in the notification of compliance status report. [§63.2460(c)(1)]
  - b) The permittee must conduct a subsequent performance test or compliance demonstration equivalent to an initial compliance demonstration within 180 days of a change in the worst-case conditions. [§63.2460(c)(2)(vi)]
  - c) *Averaging periods*. As an alternative to the requirement for daily averages in §63.998(b)(3), the permittee may determine averages for operating blocks. An operating block is a period of time that is equal to the time from the beginning to end of batch process operations within a process. [§63.2460(c)(4)]
  - d) *Outlet concentration correction for supplemental gases*. If the permittee uses a control device other than a combustion device to comply with a TOC, organic HAP, or hydrogen halide and halogen HAP outlet concentration emission limit for batch process vents, the permittee must correct the actual concentration for supplemental gases using Equation 1 to this paragraph; the permittee may use process knowledge and representative operating data to determine the fraction of the total flow due to supplemental gas.

$$C_a = C_m \left( \frac{Q_s + Q_a}{Q_a} \right) \text{ Equation 1}$$

Where:

$C_a$  = corrected outlet TOC, organic HAP, or hydrogen halide and halogen HAP concentration, dry basis, ppmv

$C_m$  = actual TOC, organic HAP, or hydrogen halide and halogen HAP concentration measured at control device outlet, dry basis, ppmv

$Q_a$  = total volumetric flowrate of all gas streams vented to the control device, except supplemental gases  
 $Q_s$  = total volumetric flowrate of supplemental gases. [§63.2460(c)(6)]

- e) If flow to a control device could be intermittent, the permittee must install, calibrate, and operate a flow indicator at the inlet or outlet of the control device to identify periods of no flow. Periods of no flow may not be used in daily or block averages, and it may not be used in fulfilling a minimum data availability requirement. [§63.2460(c)(7)]
- f) *Terminology.* When the term “storage vessel” is used in MACT WW, the term “process tank,” as defined in §63.2550(i), applies for the purposes of §63.2460. [§63.2460(c)(8)]

**Table 2 to MACT FFFF — Emission Limits and Work Practice Standards for Batch Process Vents**

| For each...                                 | Then the permittee must...                                                                                                                                                                                                                                                                                                                                                                        | And the permittee must...                                                                                                                             |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Process with Group 1 batch process vents | a. Reduce collective uncontrolled organic HAP emissions from the sum of all batch process vents within the process by ≥98% by weight by venting emissions from a sufficient number of the vents through one or more closed-vent systems to any combination of control devices (except a flare); or                                                                                                | Not applicable.                                                                                                                                       |
|                                             | b. Reduce collective uncontrolled organic HAP emissions from the sum of all batch process vents within the process by ≥95% by weight by venting emissions from a sufficient number of the vents through one or more closed-vent systems to any combination of recovery devices or a biofilter, except the permittee may elect to comply with the requirements of MACT WW for any process tank; or | Not applicable.                                                                                                                                       |
|                                             | c. Reduce uncontrolled organic HAP emissions from one or more batch process vents within the process by venting through one or more closed-vent systems to any combination of control devices (excluding a flare) that reduce organic HAP to an outlet concentration ≤20 ppmv as TOC or total organic HAP.                                                                                        | For all other batch process vents within the process, reduce collective organic HAP emissions as specified in Item 1.a and/or item 1.b of this table. |

**MACT FFFF Storage Tank Requirements:**

1. *General.* The permittee must meet each emission limit in Table 4 to MACT FFFF that applies to the storage tanks, except as specified in §63.2470(b), the permittee must also meet each applicable requirement specified in §63.2470(c) through (f) and §63.2492 and §63.2493(a) through (c). [63.2470(a)]
2. *Exceptions to MACT SS and MACT WW.* [§63.2470(c)]
  - a) Except as specified in §63.2470(c)(4)(ii), if the permittee conducts a performance test or design evaluation for a control device used to control emissions only from storage tanks, the permittee must establish operating limits, conduct monitoring, and keep records using the same procedures as required in MACT SS for control devices used to reduce emissions from process vents instead of the procedures specified in §63.985(c), §63.998(d)(2)(i), and §63.999(b)(2). The permittee must also comply with the requirements in §63.2450(e)(4), as applicable. [§63.2470(c)(1)]
  - b) Except as specified in §63.2470(c)(4), when the term “storage vessel” is used in MACT SS and MACT WW, the term “storage tank,” as defined in §63.2550 applies for the purposes of MACT FFFF. [§63.2470(c)(2)]

3. *Planned routine maintenance.* The emission limits in Table 4 to MACT FFFF for control devices used to control emissions from storage tanks do not apply during periods of planned routine maintenance. Periods of planned routine maintenance of each control device, during which the control device does not meet the emission limit specified in Table 4 to MACT FFFF, must not exceed 240 hr./yr. The permittee may submit an application to the Administrator requesting an extension of this time limit to a total of 360 hr./yr. The application must explain why the extension is needed, it must indicate that no material will be added to the storage tank between the time the 240-hr limit is exceeded and the control device is again operational, and it must be submitted at least 60 days before the 240-hr limit will be exceeded. [§63.2470(d)]
4. *Vapor balancing alternative.* As an alternative to the emission limits specified in Table 4 to MACT FFFF, the permittee may elect to implement vapor balancing in accordance with §63.1253(f), except as specified in §63.2470(e)(1) through (3). [§63.2470(e)]
  - a) When §63.1253(f)(6)(i) refers to a 90% reduction, 95% applies for the purposes of MACT FFFF. [§63.2470(e)(1)]
  - b) To comply with §63.1253(f)(6)(i), the permittee of an offsite cleaning or reloading facility must comply with §63.2445 through §63.2550 instead of complying with §63.1253(f)(7)(ii), except as specified in §63.2470(e)(2)(i) or (ii). [§63.2470(e)(2)]
    - 1) The reporting requirements in §63.2520 do not apply to the permittee of the offsite cleaning or reloading facility. [§63.2470(e)(2)(i)]
    - 2) As an alternative to complying with the monitoring, recordkeeping, and reporting provisions in §63.2445 through §63.2550, the permittee of an offsite cleaning or reloading facility may comply as specified in §63.2535(a)(2) with any other subpart of 40 CFR Part 63 which has monitoring, recordkeeping, and reporting provisions as specified in §63.2535(a)(2). [§63.2470(e)(2)(ii)]
  - c) The permittee may elect to set a pressure relief device to a value less than the 2.5 psig required in §63.1253(f)(5) if the permittee provides rationale in the notification of compliance status report explaining why the alternative value is sufficient to prevent breathing losses at all times. [§63.2470(e)(3)]
  - d) The permittee may elect to comply with the vapor balancing alternative in §63.1253(f) when the storage tank is filled from a barge. All requirements for tank trucks and railcars specified in §63.1253(f) also apply to barges, except as specified in §63.2470(e)(4)(i). [§63.2470(e)(4)]
    - 1) When §63.1253(f)(2) refers to pressure testing certifications, the requirements in §61.304(f) apply for barges. [§63.2470(e)(4)(i)]
5. *Storage tank degassing.* For each storage tank subject to Item 1 of Table 4 of MACT FFFF, the permittee must comply with §63.2470(f)(1) through (3) during storage tank shutdown operations (i.e., emptying and degassing of a storage tank) until the vapor space concentration in the storage tank is less than 10% of the LEL. The permittee must determine the LEL using process instrumentation or portable measurement devices and follow procedures for calibration and maintenance according to manufacturer's specifications. [§63.2470(f)]
  - a) Remove liquids from the storage tank as much as practicable. [§63.2470(f)(1)]
  - b) Comply with one of the following: [§63.2470(f)(2)]
    - 1) Reduce emissions of total organic HAP by venting emissions through a closed vent system to a flare. [§63.2470(f)(2)(i)]
    - 2) Reduce emissions of total organic HAP by 95 weight-percent by venting emissions through a closed vent system to any combination of non-flare control devices.

- [§63.2470(f)(2)(ii)]
- 3) Reduce emissions of total organic HAP by routing emissions to a fuel gas system or process and meet the requirements specified in §63.982(d) and the applicable requirements in §63.2450(e)(4). [§63.2470(f)(2)(iii)]
- c) Maintain records necessary to demonstrate compliance with the requirements in §63.2450(u) including, if appropriate, records of existing standard site procedures used to empty and degas (deinventory) equipment for safety purposes. [§63.2470(f)(3)]

**Table 4 to MACT FFFF — Emission Limits for Storage Tanks**

| For each...             | For which...                                                                                  | Then the permittee must...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Group 1 storage tank | a. The maximum true vapor pressure of total HAP at the storage temperature is $\geq 76.6$ kPa | i. Reduce total HAP emissions by $\geq 95\%$ by weight or to $\leq 20$ ppmv of TOC or organic HAP and $\leq 20$ ppmv of hydrogen halide and halogen HAP by venting emissions through a closed vent system to any combination of control devices (excluding a flare); or<br>ii. Reduce total organic HAP emission by venting emissions through a closed vent system to a flare; or<br>iii. Comply with the requirements of §63.2450(e)(4), as applicable; and reduce total HAP emission by venting emissions to a fuel gas system or process in accordance with §63.982(d) and the requirements referenced therein. <sup>1</sup>                                                                                      |
|                         | b. The maximum true vapor pressure of total HAP at the storage temperature is $< 76.6$ kPa    | i. Comply with the requirements of MACT WW, except as specified in §63.2470; or<br>ii. Reduce total HAP emissions by $\geq 95\%$ by weight or to $\leq 20$ ppmv of TOC or organic HAP and $\leq 20$ ppmv of hydrogen halide and halogen HAP by venting emissions through a closed vent system to any combination of control devices (excluding a flare); or<br>iii. Reduce total organic HAP emissions by venting emission through a closed vent system to a flare; or<br>iv. Comply with the requirements in §63.2450(e)(4), as applicable; and reduce total HAP emissions by venting emissions to a fuel gas system or process in accordance with §63.982(d) and the requirements references therein. <sup>1</sup> |

<sup>1</sup> Any flare using fuel gas from a fuel gas system, of which 50% or more of the fuel gas is derived from an MCPU that has processes and/or equipment in ethylene oxide service or that produces olefins or polyolefins, as determined on an annual average basis, must comply with §63.2450(e)(5). For purposes of compliance, an MCPU that “produces olefins or polyolefins” includes only those MCPUs that manufacturer ethylene, propylene, polyethylene, and/or polypropylene and a product. By-products and impurities as defined in §63.101, as well as wastes and trace contaminants, are not considered products.

**MACT SS Closed Vent System Requirements:**

1. *Closed vent system equipment and operating requirements.* Except for closed vent systems operated and maintained under negative pressure, the provisions of this paragraph apply to closed vent systems collecting regulated material from a regulated source. [§63.983(a)]
- a) *Collection of emissions.* Each closed vent system shall be designed and operated to collect the regulated material vapors from the emission point, and to route the collected vapors to a control device. [§63.983(a)(1)]

- b) *Period of operation.* Closed vent systems used to comply with the provisions of MACT SS shall be operated at all times when emissions are vented to, or collected by, them. [§63.983(a)(2)]
- c) *Bypass monitoring.* The permittee shall comply with the provisions of either §63.983(a)(3)(i) or (ii) for each closed vent system that contains bypass lines that could divert a vent stream to the atmosphere. [§63.983(a)(3) and §63.2450(e)(4)(i)]
  - 1) Properly install, maintain, and operate a flow indicator that can take periodic readings. Records shall be generated as specified in §63.998(d)(1)(ii)(A). The flow indicator shall be installed at the entrance to any bypass line. [§63.983(a)(3)(i)]
  - 2) Secure the bypass line valve in the non-diverting position with a car-seal or a lock-and-key type configuration. Records shall be generated as specified in §63.998(d)(1)(ii)(B). [§63.983(a)(3)(ii)]

**MACT SS Scrubber (Absorber) and Condenser Requirements:**

- 1. *Equipment and operating requirements.* [§63.990(a)]
  - a) If using absorbers, condensers, or carbon adsorbers to meet a weight-percent emission reduction or ppmv outlet concentration requirement specified in MACT FFFF, the permittee shall meet the requirements of §63.990. [§63.990(a)(1)]
  - b) Absorbers, condensers, and carbon adsorbers used to comply with the provisions of MACT FFFF and MACT SS shall be operated at all times when emissions are vented to them. [§63.990(a)(2)]
- 2. *Performance test requirements.* Except as specified in §63.997(b), the permittee shall conduct an initial performance test of any absorber, condenser, or carbon adsorber used as a control device to comply with the provisions of MACT FFFF and MACT SS according to the procedures in §63.997. Performance test records shall be kept as specified in §63.998(a)(2) and a performance test report shall be submitted as specified in §63.999(a)(2). [§63.990(b)]

**MACT FFFF Equipment Leak Requirements:**

- 1. The permittee must meet each requirement in Table 6 to MACT FFFF that applies to the permittee's equipment leaks, except as specified in §63.2480(b) through (f). [§63.2480(a)]
- 2. Except as specified in §63.2480(b)(6) and (7), if the permittee complies with either MACT H or MACT UU, the permittee may elect to comply with the provisions in §63.2480(b)(1) through (5) as an alternative to the referenced provisions in MACT H or MACT UU. [§63.2480(b)]
  - a) The requirements for pressure testing in §63.178(b) or §63.1036(b) may be applied to all processes, not just batch processes. [§63.2480(b)(1)]
  - b) For the purposes of MACT FFFF, pressure testing for leaks in accordance with §63.178(b) or §63.1036(b) is not required after reconfiguration of an equipment train if flexible hose connections are the only disturbed equipment. [§63.2480(b)(2)]
  - c) For an existing source, the permittee is not required to develop an initial list of identification numbers for connectors that would otherwise be required under §63.1022(b)(1) or §63.181(b)(1)(i). [§63.2480(b)(3)]
  - d) For connectors in gas/vapor and light liquid service at an existing source, the permittee may elect to comply with the requirements in §63.169 or §63.1029 for connectors in heavy liquid service, including all associated recordkeeping and reporting requirements, rather than the requirements of §63.174 or §63.1027. [§63.2480(b)(4)]
  - e) §63.2480(b)(5) no longer applies. [§63.2480(b)(6)]

- f) For each piece of equipment that is subject to Table 6 of MACT FFFF and is also subject to periodic monitoring with EPA Method 21 of NSPS Appendix A-7, and is added to an affected source after December 17, 2019, or replaces equipment at an affected source after December 17, 2019, the permittee must initially monitor for leaks within 30 days after initial startup of equipment. Equipment that is designated as unsafe- or difficult-to-monitor is not subject to this §63.2480(b)(7). [§63.2480(b)(7)]
- 3. Except as specified in §63.2480(c)(10) and (11), if the permittee complies with 40 CFR Part 65, Subpart F, the permittee may elect to comply with the provisions in §63.2480(c)(1) through (9) as an alternative to the referenced provisions in 40 CFR Part 65, Subpart F. [§63.2480(c)]
  - a) The requirements for pressure testing in §65.117(b) may be applied to all processes, not just batch processes. [§63.2480(c)(1)]
  - b) For the purposes of MACT FFFF, pressure testing for leaks in accordance with §65.117(b) is not required after reconfiguration of an equipment train if flexible hose connections are the only disturbed equipment. [§63.2480(c)(2)]
  - c) For an existing source, the permittee is not required to develop an initial list of identification numbers for connectors that would otherwise be required under §65.103(b)(1). [§63.2480(c)(3)]
  - d) The permittee may elect to comply with the monitoring and repair requirements specified in §65.108(e)(3) as an alternative to the requirements specified in §65.108(a) through (d) for any connectors at the affected source. [§63.2480(c)(4)]
  - e) When 40 CFR Part 65, Subpart F refers to the implantation date specified in §65.1(f), it means the compliance date specified in §63.2445. [§63.2480(c)(6)]
  - f) When §65.105(f) and §65.117(d)(3) refer to §65.4, it means §63.2525. [§63.2480(c)(7)]
  - g) When §65.120(a) refers to §65.5(d), it means §63.2515. [§63.2480(c)(8)]
  - h) When §65.120(b) refers to §65.5(e), it means §63.2520. [§63.2480(c)(9)]
  - i) §63.2480(c)(5) no longer applies. [§63.2480(c)(10)]
  - j) For each piece of equipment that is subject to Table 6 to MACT FFFF and is also subject to periodic monitoring with EPA Method 21 of NSPS Appendix A-7, and is added to an affected source after December 17, 2019, or replaces equipment at an affected source after December 17, 2019, the permittee must initially monitor for leaks within 30 days after August 12, 2020, or initial startup of the equipment, whichever is later. Equipment that is designated as unsafe- or difficult-to-monitor is not subject to this paragraph. [§63.2480(c)(11)]
- 4. The provisions of §63.2480 do not apply to bench-scale processes, regardless of whether the processes are located at the same plant site as a process subject to the provisions of MACT FFFF. [§63.2480(d)]
- 5. Except as specified in §63.2480(e)(4), the permittee must comply with the requirements specified in §63.2480(e)(1) and (2) for pressure relief devices, such as relief valves or rupture disks, in organic HAP gas or vapor service instead of the pressure relief device requirements of §63.1030 of MACT UU, §63.165 of MACT H, or §65.111 of MACT FFFF. Except as specified in §63.2480(e)(4) and (5), the permittee must also comply with the requirements specified in §63.2480(e)(3), (6), (7), and (8) for all pressure relief devices in organic HAP service. [§63.2480(e)]
  - a) *Pressure relief devices routed to a control device, process, fuel gas system, or drain system.* [§63.2480(e)(4)]
    - 1) If all releases and potential leaks from a pressure relief device are routed through a closed vent system to a control device, back into the process, to the fuel gas

- system, or to a drain system, then the permittee is not required to comply with §63.2480(e)(1), (2), or (3). [§63.2480(e)(4)(i)]
- 2) Both the closed vent system and control device referenced in §63.2480(e)(4)(i) must meet the applicable requirements specified in §63.982(c)(2) and §63.983 of MACT SS, and §63.2450(e)(4) through (6). [§63.2480(e)(4)(ii)]

**Table 6 to MACT FFFF — Requirements for Equipment Leaks**

| For all...                                                  | And that is part of... | The permittee must...                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Equipment that is in organic HAP service                 | a. Any MCPU            | i. Comply with the requirements of MACT UU and the requirements referenced therein, except as specified in §63.2480(b) and (d) through (f); or<br>ii. Comply with the requirements of MACT H and the requirements referenced therein, except as specified in §63.2480(b) and (d) through (f); or<br>iii. Comply with the requirements of 40 CFR Part 65, Subpart F, and the requirements referenced therein, except as specified in §63.2480(c) and (d) through (f) |
| 2. Equipment that is in organic HAP service at a new source | a. Any MCPU            | i. Comply with the requirements of MACT UU and the requirements referenced therein, except as specified in §63.2480(b)(6) and (7), (e), and (f); or<br>ii. Comply with the requirements of 40 CFR Part 65, Subpart F, except as specified in §63.2480(c)(1) and (11), (e), and (f)                                                                                                                                                                                  |

**MACT UU Applicability:**

1. The provisions of MACT UU apply to the control of air emissions from equipment leaks for which another subpart references the use of MACT UU for such air emission control. These air emission standards for equipment leaks are placed here for administrative convenience and only apply to facilities subject to a referencing subpart. The provisions of 40 CFR Part 63, subpart A (General Provisions) do not apply to MACT UU except as noted in MACT FFFF. [§63.1019(a)]
2. *Equipment subject to MACT UU.* The provisions of MACT UU and MACT FFFF apply to equipment that contains or contacts regulated material. MACT UU applies to pumps, compressors, agitators, pressure relief devices, sampling connection systems, open-ended valves or lines, valves, connectors, instrumentation systems, and closed vent systems and control devices used to meet the requirements of MACT UU. [§63.1019(b)]
3. *Equipment in vacuum service.* Equipment in vacuum service is excluded from the requirements of MACT UU. [§63.1019(c)]
4. *Equipment in service less than 300 hours per calendar year.* Equipment intended to be in regulated material service less than 300 hours per calendar year is excluded from the requirements of §63.1025 through §63.1034 and §63.1036 if it is identified as required in §63.1022(b)(5). [§63.1019(d)]
5. *Lines and equipment not containing process fluids.* Lines and equipment not containing process fluids are not subject to the provisions of MACT UU. Utilities, and other non-process lines, such as heating and cooling systems that do not combine their materials with those in the processes they serve, are not considered to be part of a process unit or affected facility. [§63.1019(e)]

**MACT UU Alternative Means of Emission Limitation:**

1. *Performance standard exemption.* The provisions of §63.1021(b) do not apply to the performance standards of §63.1030(b) for pressure relief devices or §63.1031(f) for compressors operating under the alternative compressor standard. [§63.1021(a)]
2. *Requests by the permittee.* The permittee may request a determination of alternative means of emission limitation to the requirements of §63.1025 through §63.1034 as provided in §63.1021(d). If the Administrator determines that a means of emission limitation is a permissible alternative, the permittee shall either comply with the alternative or comply with the requirements of §63.1025 through §63.1034. [§63.1021(b)]
3. *Requests by manufacturers of equipment.* [§63.1021(c)]
  - a) Manufacturers of equipment used to control equipment leaks of the regulated material may apply to the Administrator for permission for an alternative means of emission limitation that achieves a reduction in emissions of the regulated material achieved by the equipment, design, and operational requirements of MACT UU. [§63.1021(c)(1)]
  - b) The Administrator will grant permission according to the provisions of §63.1021(d). [§63.1021(c)(2)]
4. *Permission to use an alternative means of emission limitation.* Permission to use an alternative means of emission limitation shall be governed by the procedures in §63.1021(d)(1) through (d)(4). [§63.1021(d)]
  - a) Where the standard is an equipment, design, or operational requirement, the requirements of §63.1021(d)(1)(i) through (d)(1)(iii) apply. [§63.1021(d)(1)]
    - 1) The permittee shall be responsible for collecting and verifying emission performance test data for an alternative means of emission limitation. [§63.1021(d)(1)(i)]
    - 2) The Administrator will compare test data for the means of emission limitation to test data for the equipment, design, and operational requirements. [§63.1021(d)(1)(ii)]
    - 3) The Administrator may condition the permission on requirements that may be necessary to ensure operation and maintenance to achieve at least the same emission reduction as the equipment, design, and operational requirements of MACT UU. [§63.1021(d)(1)(iii)]
  - b) Where the standard is a work practice, the requirements of §63.1021(d)(2)(i) through (d)(2)(vi) apply. [§63.1021(d)(2)]
    - 1) The permittee shall be responsible for collecting and verifying test data for the alternative. [§63.1021(d)(2)(i)]
    - 2) For each kind of equipment for which permission is requested, the emission reduction achieved by the required work practices shall be demonstrated for a minimum period of 12 months. [§63.1021(d)(2)(ii)]
    - 3) For each kind of equipment for which permission is requested, the emission reduction achieved by the alternative means of emission limitation shall be demonstrated. [§63.1021(d)(2)(iii)]
    - 4) The permittee shall commit, in writing, for each kind of equipment to work practices that provide for emission reductions equal to or greater than the emission reductions achieved by the required work practices. [§63.1021(d)(2)(iv)]
    - 5) The Administrator will compare the demonstrated emission reduction for the alternative means of emission limitation to the demonstrated emission reduction for the required work practices and will consider the commitment in §63.1021(d)(2)(iv). [§63.1021(d)(2)(v)]

- 6) The Administrator may condition the permission on requirements that may be necessary to ensure operation and maintenance to achieve the same or greater emission reduction as the required work practices of MACT UU. [§63.1021(d)(2)(vi)]
- c) The permittee may offer a unique approach to demonstrate the alternative means of emission limitation. [§63.1021(d)(3)]
- d) If, in the judgement of the Administrator, an alternative means of emission limitation will be approved, the Administrator will publish a notice of the determination in the Federal Register using the procedures specified in the MACT FFFF. [§63.1021(d)(4)]

**MACT UU LDAR Equipment Identification:**

1. *General equipment identification.* Equipment subject to MACT UU shall be identified. Identification of the equipment does not require physical tagging of the equipment. For example, the equipment may be identified on a plant site plan, in log entries, by designation of process unit or affected facility boundaries by some form of weatherproof identification, or by other appropriate methods. [§63.1022(a)]
2. *Additional equipment identification.* In addition to the general identification required by §63.1022(a), equipment subject to any of the provisions in §63.1023 through §63.1034 shall be specifically identified as required in §63.1022(b)(2) through (b)(5), as applicable. This paragraph does not apply to a batch product process if the permittee elects to pressure test the batch product process equipment train pursuant to §63.1036. [§63.1022(b) and §63.2480(b)(3)]
  - a) *Equipped with a closed vent system and control device.* Identify the equipment that the permittee elects to route to equip with a closed vent system and control device, under the provisions of §63.1026(e)(3) (pumps in light liquid service), §63.1028(e)(3) (agitators), §63.1030(d) (pressure relief devices in gas and vapor service), §63.1031(e) (compressors), or §63.1037(a) (alternative means of emission limitation for enclosed-vented process units). [§63.1022(b)(2)]
  - b) *Pressure relief devices.* Identify the pressure relief devices equipped with rupture disks, under the provisions of §63.1030(e). [§63.1022(b)(3)]
  - c) *Instrumentation systems.* Identify instrumentation systems subject to the provisions of §63.1029. Individual components in an instrumentation system need not be identified. [§63.1022(b)(4)]
  - d) *Equipment in service less than 300 hours per calendar year.* The identity, either by list, location (area or group), or other method, of equipment in regulated material service less than 300 hours per calendar year within a process unit or affected facilities subject to the provisions of MACT UU shall be recorded. [§63.1022(b)(5)]
3. *Special equipment designations: Equipment that is unsafe or difficult-to-monitor —* [§63.1022(c)]
  - a) *Designation and criteria for unsafe-to-monitor.* Valves meeting the provisions of §63.1025(e)(1), pumps meeting the provisions of §63.1026(e)(6), connectors meeting the provisions of §63.1027(e)(1), and agitators meeting the provisions of §63.1028(e)(7) may be designated unsafe-to-monitor if the permittee determines that monitoring personnel would be exposed to an immediate danger as a consequence of complying with the monitoring requirements of MACT UU. Examples of unsafe-to-monitor equipment include, but is not limited to, equipment under extreme pressure or heat. [§63.1022(c)(1)]
  - b) *Designation and criteria for difficult-to-monitor.* Valves meeting the provisions of §63.1025(e)(2) may be designated difficult-to-monitor if the provisions of

§63.1022(c)(2)(i) apply. Agitators meeting the provisions of §63.1028(e)(5) may be designated difficult-to-monitor if the provisions of §63.1022(c)(2)(ii) apply.

[§63.1022(c)(2)]

1) *Valves.* [§63.1022(c)(2)(i)]

- (i) The permittee determines that the valve cannot be monitored without elevating the monitoring personnel more than 2 m (7 ft) above a support surface or it is not accessible in a safe manner when it is in regulated material service; and

[§63.1022(c)(2)(i)(A)]

- (ii) The process unit or affected facility within which the valve is located is an existing source, or the permittee designates less than 3% of the total number of valves in a new source as difficult-to-monitor. [§63.1022(c)(2)(i)(B)]

2) *Agitators.* The permittee determines that the agitator cannot be monitored without elevating the monitoring personnel more than 2 m (7 ft) above a support surface or it is not accessible in a safe manner when it is in regulated material service.

[§63.1022(c)(2)(ii)]

c) *Identification of unsafe or difficult-to-monitor equipment.* The permittee shall record the identity of equipment designated as unsafe-to-monitor according to the provisions of §63.1022(c)(1) and the planned schedule for monitoring this equipment. The permittee shall record the identity of equipment designated as difficult-to-monitor according to the provisions of §63.1022(c)(2), the planned schedule for monitoring this equipment, and an explanation why the equipment is unsafe or difficult-to-monitor. This record must be kept at the plant and be available for review by an inspector. [§63.1022(c)(3)]

d) *Written plan requirements.* [§63.1022(c)(4)]

- 1) For equipment designated as unsafe-to-monitor according to the provisions of §63.1022(c)(1), the permittee shall have a written plan that requires monitoring of the equipment as frequently as practical during safe-to-monitor times, but not more frequently than the periodic monitoring schedule otherwise applicable, and repair of the equipment according to the procedures in §63.1024 if a leak is detected.

[§63.1022(c)(4)(i)]

- 2) For equipment designated as difficult-to-monitor according to the provisions of §63.1022(c)(2), the permittee shall have a written plan that requires monitoring of the equipment at least once per calendar year and repair of the equipment according to the procedures in §63.1024 if a leak is detected. [§63.1022(c)(4)(ii)]

4. *Special equipment designations: Equipment that is unsafe-to-repair* — [§63.1022(d)]

- a) *Designation and criteria.* Connectors subject to the provisions of §63.1024(e) may be designated unsafe-to-repair if the permittee determines that repair personnel would be exposed to an immediate danger as a consequence of complying with the repair requirements of MACT UU, and if the connector will be repaired before the end of the next process unit or affected facility shutdown as specified in §63.1024(e)(2).

[§63.1022(d)(1)]

- b) *Identification of equipment.* The identity of connectors designated as unsafe-to-repair and an explanation why the connector is unsafe-to-repair shall be recorded.

[§63.1022(d)(2)]

5. *Special equipment designations: Compressors operating with an instrument reading of less than 500 ppm above background.* Identify the compressors that the permittee elects to designate as operating with an instrument reading of less than 500 ppm above background, under the provisions of §63.1031(f). [§63.1022(e)]

6. *Special equipment designations: Equipment in heavy liquid service.* For equipment in

- heavy liquid service, the permittee shall comply with the requirements of either §63.1022(f)(1) or (f)(2), as provided in §63.1022(f)(3). [§63.1022(f)]
- a) Retain information, data, and analyses used to determine that a piece of equipment is in heavy liquid service. [§63.1022(f)(1)]
  - b) When requested by the Administrator, demonstrate that the piece of equipment or process is in heavy liquid service. [§63.1022(f)(2)]
  - c) A determination or demonstration that a piece of equipment or process is in heavy liquid service shall include an analysis or demonstration that the process fluids do not meet the definition of "in light liquid service." Examples of information that could document this include, but are not limited to, records of chemicals purchased for the process, analyses of process stream composition, engineering calculations, or process knowledge. [§63.1022(f)(3)]

**MACT FFFF Wastewater Stream Requirements:**

1. The permittee must meet each requirement in Table 7 to MACT FFFF that applies to the wastewater streams and liquid streams in open systems within an MCPU, except as specified in §63.2480(b) and (q). [§63.2485(a)]
2. *Wastewater HAP.* Where §63.105 and §63.132 through §63.148 refer to compounds in Table 9 of MACT G, the compounds in Tables 8 and 9 to MACT FFFF apply for the purposes of MACT FFFF. [§63.2485(b)]
3. *Group 1 Wastewater.* §63.132(c)(1)(i) and (ii) do not apply. For the purposes of MACT FFFF, a process wastewater stream is Group 1 for the compounds in Tables 8 and 9 to MACT FFFF if any of the conditions specified in §63.2485(c)(1) through (3) are met. [§63.2485(c)]
  - a) The total annual average concentration of compounds in Table 8 to MACT FFFF is greater than or equal to 10,000 ppmw at any flowrate, and the total annual load of compounds in Table 8 to MACT FFFF is greater than or equal to 200 lbs./yr. [§63.2485(c)(1)]
  - b) The total annual average concentration of compounds in Table 8 to MACT FFFF is greater than or equal to 1,000 ppmw, and the annual average flowrate is greater than or equal to 1 L/min. [§63.2485(c)(2)]
  - c) The combined total annual average concentration of compounds in Tables 8 and 9 to MACT FFFF is greater than or equal to 30,000 ppmw, and the combined total annual load of compounds in Tables 8 and 9 to MACT FFFF is greater than or equal to 1 tpy. [§63.2485(c)(3)]
4. *Wastewater Tank Requirements.* [§63.2485(d)]
  - a) When §63.133 and §63.147 reference floating roof requirements in §63.119 and §63.120, the corresponding requirements in MACT WW may be applied for the purposes of MACT FFFF. [§63.2485(d)(1)]
  - b) When §63.133(a) refers to Table 10 of MACT G, the maximum true vapor pressure in the table shall be limited to the HAP listed in Tables 8 and 9 of MACT FFFF. [§63.2485(d)(2)]
  - c) For the purposes of MACT FFFF, the requirements of §63.133(a)(2) are satisfied by operating and maintaining a fixed roof if the permittee demonstrates that the total soluble and partially soluble HAP emissions from the wastewater tank are no more than 5% higher than the emissions would be if the contents of the wastewater tank were not heated, treated by an exothermic reaction, or sparged. [§63.2485(d)(3)]

- d) The emission limits specified in §63.133(b)(2) and §63.139 for control devices used to control emissions from wastewater tanks do not apply during periods of planned routine maintenance of the control device(s) of no more than 240 hr./yr. The permittee may request an extension of a total of 360 hr./yr. in accordance with the procedures specified in §63.2470(d). [§63.2485(d)(4)]
5. *Individual Drain Systems.* The provisions of §63.136(e)(3) apply except as specified in §63.2485(e)(1). [§63.2485(e)]
  - a) A sewer line connected to drains that comply with §63.136(e)(1) may be vented to the atmosphere, provided that the sewer line entrance to the first downstream junction box is water sealed and the sewer line vent pipe is designed as specified in §63.136(e)(2)(ii)(A). [§63.2485(e)(1)]
6. *Closed-Vent System Requirements.* Except as specified in §63.2450(e)(6), when §63.148(k) refers to closed vent systems that are subject to the requirements of §63.172, the requirements of either §63.172 or §63.1034 apply for the purposes of MACT FFFF. [§63.2485(f)]
7. *Halogenated vent stream requirements.* For each halogenated vent stream from a Group 1 wastewater stream or residual removed from a Group 1 wastewater stream that is vented through a closed-vent system to a combustion device to reduce organic HAP emissions, the permittee must meet the same emission limits as specified for batch process vents in Item 2 of Table 2 to MACT FFFF. [§63.2485(g)]
8. *Alternative Test Methods.* [§63.2485(h)]
  - a) As an alternative to the test methods specified in §63.144(b)(5)(i), the permittee may use Method 8260 or 8270 as specified in §63.1257(b)(10)(iii). [§63.2485(h)(1)]
  - b) As an alternative to using the methods specified in §63.144(b)(5)(i), the permittee may conduct wastewater analyses using Method 1666 or 1671 of 40 CFR Part 136 and comply with the sampling protocol requirements specified in §63.144(b)(5)(ii). The validation requirements specified in §63.144(b)(5)(iii) do not apply if the permittee uses Method 1666 or 1671 of 40 CFR Part 136. [§63.2485(h)(2)]
  - c) As an alternative to using NSPS Method 18, as specified in §63.139(c)(1)(ii) and §63.145(i)(2), the permittee may elect to use NSPS Method 25A as specified in §63.997. [§63.2485(h)(3)]
  - d) As an alternative to using EPA Method 624 of 40 CFR Part 136, appendix A, as specified in §63.144(b)(5)(i)(C), the permittee may use ASTM D5790–95 (Reapproved 2012) (incorporated by reference, see §63.14) for the analysis of total organic HAP in wastewater samples. If the permittee chooses to use ASTM D5790–95 (Reapproved 2012), then the permittee must also use the sampling procedures of EPA Method 25D NSPS Appendix A–7, or an equivalent method. [§63.2485(h)(4)]
9. *Offsite Management and Treatment Option.* [§63.2485(i)]
  - a) If the permittee ships wastewater to an offsite treatment facility that meets the requirements of §63.138(h), the permittee may elect to document in the notification of compliance status report that the wastewater will be treated as hazardous waste at a facility that meets the requirements of §63.138(h) as an alternative to having the offsite facility submit the certification specified in §63.132(g)(2). [§63.2482(i)(1)]
  - b) As an alternative to the management and treatment options specified in §63.132(g)(2), any affected wastewater stream (or residual removed from an affected wastewater stream) with a total annual average concentration of compounds in Table 8 to MACT FFFF less than 50 ppmw may be transferred offsite in accordance with §63.2485(i)(2) and (ii). [§63.2485(i)(2)]

- 1) The transferee (or the permittee) must demonstrate that less than 5% of the HAP in Table 9 to MACT FFFF is emitted from the waste management units up to the activated sludge unit. [§63.2485(i)(2)(i)]
- 2) The transferee must treat the wastewater stream or residual in a biological treatment unit in accordance with the requirement in §63.2485(i)(2)(iii) and the requirements of §63.138 and §63.145 and the requirements referenced therein. [§63.2485(i)(2)(ii)]
- 3) §63.145(a)(3) no longer applies. Instead, the transferee must comply with the conditions specified in §63.2450(g)(6). [§63.2485(i)(2)(iii)]
10. The permittee must determine the annual average concentration and annual average flowrate for wastewater streams for each MCPU. The procedures for flexible operation units specified in §63.144(b) and (c) do not apply for the purposes of MACT FFFF. [§63.2485(j)]
11. *Outlet Concentration Correction for Supplemental Gases.* The requirement to correct outlet concentrations from combustion devices to 3% oxygen in §63.139(c)(1)(ii) and §63.145(i)(6) applies only if supplemental gases are combined with a vent stream from a Group 1 wastewater stream. If emissions are controlled with a vapor recovery system as specified in §63.139(c)(2), the permittee must correct for supplemental gases as specified in §63.2460(c)(6). [§63.2485(k)]
12. *Requirements for Liquid Streams in Open Systems.* [§63.2485(l)]
  - a) References in §63.149 to §63.100(b) mean §63.2435(b) for the purposes of MACT FFFF. [§63.2485(l)(1)]
  - b) When §63.149(e) refers to §63.100(l)(1) or (2), §63.2445(a) applies for the purposes of MACT FFFF. [§63.2485(l)(2)]
  - c) When §63.149 uses the term “chemical manufacturing process unit,” the term “MCPU” applies for the purposes of MACT FFFF. [§63.2485(l)(3)]
  - d) When §63.149(e)(1) refers to characteristics of water that contain compounds in Table 9 to MACT G, the characteristics specified §63.2485(c)(1) through (3) apply for the purposes of MACT FFFF. [§63.2485(l)(4)]
  - e) When §63.149(e)(2) refers to characteristics of water that contain compounds in Table 9 to MACT G, the characteristics specified §63.2485(c)(2) apply for the purposes of MACT FFFF. [§63.2485(l)(5)]
13. When §63.132(f) refers to “a concentration of greater than 10,000 ppmw of Table 9 compounds,” the phrase “a concentration of greater than 30,000 ppmw of total partially soluble HAP (PSHAP) and soluble HAP (SHAP) or greater than 10,000 ppmw of PSHAP” shall apply for the purposes of MACT FFFF. [§63.2485(m)]
14. *Alternative Requirements for Wastewater that is Group 1 for Soluble HAP Only.* The option specified in this §63.2485(n) applies to wastewater that is Group 1 for soluble HAP in accordance with §63.2485(c)(3) and is discharged to biological treatment. Except as provided in §63.2485(n)(4), this option does not apply to wastewater that is Group 1 for partially soluble HAP in accordance with §63.2485(c)(1), (2), or (4). For wastewater that is Group 1 for SHAP, the permittee need not comply with §63.133 through §63.137 for any equalization unit, neutralization unit, and/or clarifier prior to the activated sludge unit, and the permittee need not comply with the venting requirements in §63.136(e)(2)(ii)(A) for lift stations with a volume larger than 10,000 gal, provided the permittee complies with the requirements specified in §63.2485(n)(1) through (3) and all otherwise applicable requirements specified in Table 7 to MACT FFFF. For this option, the treatment requirements in §63.138 and the performance testing requirements in §63.145 do not apply to the biological treatment unit, except as specified in §63.2485(n)(2)(i) through (iv).

[§63.2485(n)]

15. *Compliance Records after Date of Compliance.* §63.2485(o) no longer applies. Instead, for each CPMS used to monitor a nonflare control device for wastewater emissions, the permittee must keep records as specified in §63.998(c)(1) in addition to the records required in §63.147(d), except that the provisions of §63.998(c)(1)(ii)(D), (E), (F), and (G) do not apply. [§63.2485(p)]
16. *Startup, shutdown, and malfunction referenced provisions.* The referenced provisions specified in §63.2485(q)(1) through (5) do not apply when demonstrating compliance with §63.2485. [§63.2485(q)]
  - a) §63.105(d) and the phrase “as part of the start-up, shutdown, and malfunction plan required under §63.6(e)(3) of subpart A of this part” from §63.105(e). [§63.2485(q)(1)]
  - b) §63.132(b)(3)(i)(B). [§63.2485(q)(2)]
  - c) The phrase “or startup/shutdown/malfunction” in §63.132(f)(2). [§63.2485(q)(3)]
  - d) §63.148(f)(3). [§63.2485(q)(4)]
  - e) For flares complying with §63.2450(e)(5), the following provisions do not apply: [§63.2485(q)(5)]
    - 1) §63.139(c)(3); [§63.2485(q)(5)(i)]
    - 2) §63.139(d)(3); [§63.2485(q)(5)(ii)]
    - 3) §63.145(j); [§63.2485(q)(5)(iii)]
    - 4) §63.146(b)(7)(i); and [§63.2485(q)(5)(iv)]
    - 5) §63.147(d)(1). [§63.2485(q)(5)(v)]

**Table 7 to MACT FFFF — Requirements for Wastewater Streams and Liquid Streams in Open Systems within an MCPU**

| For each...                                       | The permittee must...                                                                                                             |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| 1. Process wastewater stream                      | Comply with the requirements in §63.132 through §63.148 and the requirements referenced therein, except as specified in §63.2485. |
| 2. Maintenance wastewater stream                  | Comply with the requirements in §63.105 and the requirements referenced therein, except as specified in §63.2485.                 |
| 3. Liquid streams in an open system within a MCPU | Comply with the requirements in §63.149 and the requirements referenced therein, except as specified in §63.2485.                 |

**Table 8 to MACT FFFF – Partially Soluble HAP**

| Chemical name...                              | CAS No.  |
|-----------------------------------------------|----------|
| 1. 1,1,1-Trichloroethane (methyl chloroform)  | 71-55-6  |
| 2. 1,1,2,2-Tetrachloroethane                  | 79-34-5  |
| 3. 1,1,2-Trichloroethane                      | 79-00-5  |
| 4. 1,1-Dichloroethylene (vinylidene chloride) | 75-35-4  |
| 5. 1,2-Dibromoethane                          | 106-93-4 |
| 6. 1,2-Dichloroethane (ethylene dichloride)   | 107-06-2 |
| 7. 1,2-Dichloropropane                        | 78-87-5  |
| 8. 1,3-Dichloropropene                        | 542-75-6 |
| 9. 2,4,5-Trichlorophenol                      | 95-95-4  |
| 10. 1,4-Dichlorobenzene                       | 106-46-7 |
| 11. 2-Nitropropane                            | 79-46-9  |
| 12. 4-Methyl-2-pentanone (MIBK)               | 108-10-1 |
| 13. Acetaldehyde                              | 75-07-0  |
| 14. Acrolein                                  | 107-02-8 |
| 15. Acrylonitrile                             | 107-13-1 |

| Chemical name...                              | CAS No.   |
|-----------------------------------------------|-----------|
| 16. Allyl chloride                            | 107-05-1  |
| 17. Benzene                                   | 71-43-2   |
| 18. Benzyl chloride                           | 100-44-7  |
| 19. Biphenyl                                  | 92-52-4   |
| 20. Bromoform (tribromomethane)               | 75-25-2   |
| 21. Bromomethane                              | 74-83-9   |
| 22. Butadiene                                 | 106-99-0  |
| 23. Carbon disulfide                          | 75-15-0   |
| 24. Chlorobenzene                             | 108-90-7  |
| 25. Chloroethane (ethyl chloride)             | 75-00-3   |
| 26. Chloroform                                | 67-66-3   |
| 27. Chloromethane                             | 74-87-3   |
| 28. Chloroprene                               | 126-99-8  |
| 29. Cumene                                    | 98-82-8   |
| 30. Dichloroethyl ether                       | 111-44-4  |
| 31. Dinitrophenol                             | 51-28-5   |
| 32. Epichlorohydrin                           | 106-89-8  |
| 33. Ethyl acrylate                            | 140-88-5  |
| 34. Ethylbenzene                              | 100-41-4  |
| 35. Ethylene oxide                            | 75-21-8   |
| 36. Ethylidene dichloride                     | 75-34-3   |
| 37. Hexachlorobenzene                         | 118-74-1  |
| 38. Hexachlorobutadiene                       | 87-68-3   |
| 39. Hexachloroethane                          | 67-72-1   |
| 40. Methyl methacrylate                       | 80-62-6   |
| 41. Methyl-t-butyl ether                      | 1634-04-4 |
| 42. Methylene chloride                        | 75-09-2   |
| 43. N-hexane                                  | 110-54-3  |
| 44. N,N-dimethylaniline                       | 121-69-7  |
| 45. Naphthalene                               | 91-20-3   |
| 46. Phosgene                                  | 75-44-5   |
| 47. Propionaldehyde                           | 123-38-6  |
| 48. Propylene oxide                           | 75-56-9   |
| 49. Styrene                                   | 100-42-5  |
| 50. Tetrachloroethylene (perchloroethylene)   | 127-18-4  |
| 51. Tetrachloromethane (carbon tetrachloride) | 56-23-5   |
| 52. Toluene                                   | 108-88-3  |
| 53. Trichlorobenzene (1,2,4-)                 | 120-82-1  |
| 54. Trichloroethylene                         | 79-01-6   |
| 55. Trimethylpentane                          | 540-84-1  |
| 56. Vinyl acetate                             | 108-05-4  |
| 57. Vinyl chloride                            | 75-01-4   |
| 58. Xylene (m)                                | 108-38-3  |
| 59. Xylene (o)                                | 95-47-6   |
| 60. Xylene (p)                                | 106-42-3  |

**Table 9 to MACT FFFF – Soluble HAP**

| Chemical name...                             | CAS No.  |
|----------------------------------------------|----------|
| 1. Acetonitrile                              | 75-05-8  |
| 2. Acetophenone                              | 98-86-2  |
| 3. Diethyl sulfate                           | 64-67-5  |
| 4. Dimethyl hydrazine (1,1)                  | 57-14-7  |
| 5. Dimethyl sulfate                          | 77-78-1  |
| 6. Dinitrotoluene (2,4)                      | 121-14-2 |
| 7. Dioxane (1,4)                             | 123-91-1 |
| 8. Ethylene glycol dimethyl ether            | 110-71-4 |
| 9. Ethylene glycol monobutyl ether acetate   | 112-07-2 |
| 10. Ethylene glycol monomethyl ether acetate | 110-49-6 |
| 11. Isophorone                               | 78-59-1  |
| 12. Methanol                                 | 67-56-1  |
| 13. Nitrobenzene                             | 98-95-3  |
| 14. Toluidine (o-)                           | 95-53-4  |
| 15. Triethylamine                            | 121-44-8 |

**MACT G Process Wastewater Provisions – General:**

1. *Existing sources.* This paragraph specifies the requirements applicable to process wastewater streams located at existing sources. The permittee shall comply with the requirements in §63.132(a)(1) through (a)(3). [§63.132(a)]
  - a) Determine wastewater streams to be controlled for Tables 8 and 9 to MACT FFFF compounds. Determine whether each wastewater stream requires control for Table 9 compounds by complying with the requirements in either §63.132(a)(1)(i) or (a)(1)(ii) and comply with the requirements in §63.132(a)(1)(iii). [§63.132(a)(1) and §63.2485(b)]
    - 1) Comply with §63.132(c), determining whether the wastewater stream is Group 1 or Group 2 for Tables 8 and 9 to MACT FFFF compounds; or [§63.132(a)(1)(i) and §63.2485(b)]
    - 2) Comply with §63.132(e), designating the wastewater stream as a Group 1 wastewater stream. [§63.132(a)(1)(ii)]
    - 3) Comply with §63.132(f). [§63.132(a)(1)(iii)]
  - b) *Requirements for Group 1 wastewater streams.* For wastewater streams that are Group 1 for Tables 8 and 9 to MACT FFFF compounds, comply with §63.132(a)(2)(i) through (a)(2)(iv). [§63.132(a)(2) and §63.2485(b)]
  - c) *Requirements for Group 2 wastewater streams.* For wastewater streams that are Group 2 for Tables 8 and 9 to MACT FFFF compounds, comply with the applicable recordkeeping and reporting requirements specified in §63.146(b)(1) and §63.147(b)(8). [§63.132(a)(3) and §63.2485(b)]
2. *How to determine Group 1 or Group 2 status for Tables 8 and 9 to MACT FFFF compounds.* This paragraph provides instructions for determining whether a wastewater stream is Group 1 or Group 2 for Tables 8 and 9 to MACT FFFF compounds. Total annual average concentration shall be determined according to the procedures specified in §63.144(b). Annual average flow rate shall be determined according to the procedures specified in §63.144(c). [§63.132(c) and §63.2485(b) and (c)]
  - a) A wastewater stream is a Group 1 wastewater stream if any of the conditions specified in §63.2485(c)(1) through (3) are met. [§63.132(c)(1) and §63.2485(c)]
  - b) A wastewater stream is a Group 2 wastewater stream if it is not a Group 1 wastewater stream by the criteria in §63.132(c)(1). [§63.132(c)(2)]

- c) The permittee shall re-determine group status for each Group 2 stream, as necessary, to determine whether the stream is Group 1 or Group 2 whenever process changes are made that could reasonably be expected to change the stream to a Group 1 stream. Examples of process changes include, but are not limited to, changes in production capacity, production rate, feedstock type, or whenever there is a replacement, removal, or addition of recovery or control equipment. For purposes of this paragraph, process changes do not include: Process upsets; unintentional, temporary process changes; and changes that are within the range on which the original determination was based. [§63.132(c)(3)]
3. *How to designate a Group 1 wastewater stream.* The permittee may elect to designate a wastewater stream a Group 1 wastewater stream to comply with §63.132(a)(1) or (b)(1). To designate a wastewater stream or a mixture of wastewater streams a Group 1 wastewater stream, the procedures specified in §63.132(e)(1) and (e)(2) and §63.144(a)(2) shall be followed. [§63.132(e)]
4. The permittee shall not discard liquid or solid organic materials with a concentration of greater than 30,000 ppmw of total partially soluble HAP (PSHAP) and soluble HAP (SHAP) or greater than 10,000 ppmw of PSHAP (as determined by analysis of the stream composition, engineering calculations, or process knowledge, according to the provisions of §63.144(b)) from a chemical manufacturing process unit to water or wastewater, unless the receiving stream is managed and treated as a Group 1 wastewater stream. This prohibition does not apply to materials from the activities listed in §63.132(f)(1) through (f)(4). [§63.132(f) and §63.2485(m)]
- a) Equipment leaks; [§63.132(f)(1)]
- b) Activities included in maintenance plans; [§63.132(c)(2) and §63.2485(q)(3)]
- c) Spills; or [§63.132(f)(3)]
- d) Samples of a size not greater than reasonably necessary for the method of analysis that is used. [§63.132(f)(4)]

**MACT G Process Wastewater Provisions — Test Methods and Procedures for Determining Applicability and Group 1/Group 2 Determinations (Determining Which Wastewater Streams Require Control):**

1. *Procedures to determine applicability.* The permittee shall comply with §63.144(a)(1) or (a)(2) for each wastewater stream to determine which wastewater streams require control for Tables 8 and 9 to MACT FFFF compounds. The permittee may use a combination of the approaches in §63.144(a)(1) and (a)(2) for different wastewater streams generated at the source. [§63.144(a) and §63.2485(b)]
- a) *Determine Group 1 or Group 2 status.* Determine whether a wastewater stream is a Group 1 or Group 2 wastewater stream in accordance with §63.144(b) and (c). [§63.144(a)(1)]
2. *Procedures to establish concentrations, when determining Group status under §63.144(a)(1).* The permittee shall determine the annual average concentration for Tables 8 and 9 to MACT FFFF compounds according to §63.144(b)(1) for existing sources. The annual average concentration shall be a flow weighted average representative of actual or anticipated operation of the chemical manufacturing process unit generating the wastewater over a designated 12-month period. The permittee is not required to determine the concentration of Tables 8 and 9 to MACT FFFF compounds that are not reasonably expected to be in the process. [§63.144(b) and §63.2485(b) and (j)]

- a) *Existing sources.* The permittee shall determine the flow weighted total annual average concentration for Tables 8 and 9 to MACT FFFF compounds. For the purposes of §63.144, the term concentration, whether concentration is used alone or with other terms, may be adjusted by multiplying by the compound-specific fraction measured ( $F_m$ ) factors listed in Table 34 of MACT G unless determined by the methods in §63.144(b)(5)(i)(A) and/or (B). When concentration is determined by Method 305 as specified in §63.144(b)(5)(i)(B), concentration may be adjusted by dividing by the compound specific  $F_m$  factors listed in Table 34 of MACT G. When concentration is determined by Method 25D as specified in §63.144(b)(5)(i)(A), concentration may not be adjusted by the compound specific  $F_m$  factors listed in Table 34 of MACT G. Compound-specific  $F_m$  factors may be used only when concentrations of individual compounds are determined or when only one compound is in the wastewater stream. Flow weighted total annual average concentration for Tables 8 and 9 to MACT FFFF compounds means the total mass of Tables 8 and 9 to MACT FFFF compounds occurring in the wastewater stream during the designated 12-month period divided by the total mass of the wastewater stream during the same designated 12-month period. The total annual average concentration shall be determined for each wastewater stream either at the point of determination, or downstream of the point of determination with adjustment for concentration changes made according to §63.144(b)(6). The procedures specified in §63.144(b)(3), (b)(4), and (b)(5) are considered acceptable procedures for determining the annual average concentration. They may be used in combination, and no one procedure shall take precedence over another. [§63.144(b)(1) and §63.2485(b)]
- b) *Knowledge of the wastewater.* Where knowledge is used to determine the annual average concentration, the permittee shall provide sufficient information to document the annual average concentration for wastewater streams determined to be Group 2 wastewater streams. Documentation to determine the annual average concentration is not required for Group 1 streams. Examples of acceptable documentation include material balances, records of chemical purchases, process stoichiometry, or previous test results. If test data are used, the permittee shall provide documentation describing the testing protocol and how any losses of volatile compounds during sampling, and the bias and accuracy of the analytical method, were accounted for in the determination. [§63.144(b)(3)]
- c) *Bench-scale or pilot-scale test data.* Where bench-scale or pilot-scale test data are used to determine the annual average concentration, the permittee shall provide sufficient information to document that the data are representative of the actual annual average concentration or are reliably indicative of another relevant characteristic of the wastewater stream that could be used to predict the annual average concentration. For concentration data, the permittee shall also provide documentation describing the testing protocol, and how any losses of volatile compounds during sampling, and the bias and accuracy of the analytical method, were accounted for in the determination of annual average concentration. [§63.144(b)(4)]
- d) *Test data from sampling at the point of determination or at a location downstream of the point of determination.* If the permittee elects to comply with §63.144(a)(1) by measuring the concentration for the relevant Tables 8 and 9 to MACT FFFF compounds, the permittee shall comply with the requirements of this paragraph. For each wastewater stream, measurements shall be made either at the point of determination, or downstream of the point of determination with adjustment for concentration changes

made according to §63.144(b)(6). A minimum of three samples from each wastewater stream shall be taken. Samples may be grab samples or composite samples.

[§63.144(b)(5) and §63.2485(b)]

- 1) *Methods*. The permittee shall use any of the methods specified in §63.144(b)(5)(i)(A) through (b)(5)(i)(H). [§63.144(b)(5)(i)]
- 2) *Sampling plan*. If the permittee is expressly referred to this paragraph by provisions of MACT G, the permittee shall prepare a sampling plan. Wastewater samples shall be collected using sampling procedures which minimize loss of organic compounds during sample collection and analysis and maintain sample integrity. The sample plan shall include procedures for determining recovery efficiency of the relevant HAP listed in Tables 8 and 9 of MACT FFFF. An example of an acceptable sampling plan would be one that incorporates similar sampling and sample handling requirements to those of Method 25D of NSPS Appendix A. The sampling plan shall be maintained at the facility. [§63.144(b)(5)(ii) and §63.2485(b)]
- 3) *Validation of methods*. The permittee shall validate EPA methods other than Methods 25D, 305, 624, 625, 1624, and 1625 using the procedures specified in §63.144(b)(5)(iii)(A) or (b)(5)(iii)(B). The permittee shall validate other methods as specified in §63.144(b)(5)(iii)(A). [§63.144(b)(5)(iii)]
- 4) *Calculations of average concentration*. The average concentration for each individually speciated Table 8 to MACT FFFF compound shall be calculated by adding the individual values determined for the specific compound in each sample and dividing by the number of samples. The total average concentration of Table 9 to MACT FFFF compounds shall be calculated by first summing the concentration of the individual compounds to obtain a total HAP concentration for the sample; add the sample totals and then divide by the number of samples in the run to obtain the sample average for the run. If the method used does not speciate the compounds, the sample results should be added, and this total divided by the number of samples in the run to obtain the sample average for the run. [§63.144(b)(5)(iv) and §63.2485(b)]

e) *Adjustment for concentrations determined downstream of the point of determination.*

The permittee shall make corrections to the annual average concentration or total annual average concentration when the concentration is determined downstream of the point of determination at a location where: two or more wastewater streams have been mixed; one or more wastewater streams have been treated; or losses to the atmosphere have occurred. The permittee shall make the adjustments either to the individual data points or to the final annual average concentration. [§63.144(b)(6)]

3. *Procedures to determine flow rate, when evaluating Group status under §63.144(a)(1).* To comply with §63.144(a)(1), the permittee shall determine the annual average flow rate of the wastewater stream either at the point of determination for each wastewater stream, or downstream of the point of determination with adjustment for flow rate changes made according to §63.144(c)(4). These procedures may be used in combination for different wastewater streams at the source. The annual average flow rate for the wastewater stream shall be representative of actual or anticipated operation of the chemical manufacturing process unit generating the wastewater over a designated 12-month period. The permittee shall consider the total annual wastewater volume generated by the chemical manufacturing process unit. The procedures specified in §63.144(c)(1), (c)(2), and (c)(3) are considered acceptable procedures for determining the flow rate. They may be used in combination, and no one procedure shall take precedence over another. [§63.144(c) and

§63.2485(j)]

- a) *Knowledge of the wastewater.* The permittee may use knowledge of the wastewater stream and/or the process to determine the annual average flow rate. The permittee shall use the maximum expected annual average production capacity of the process unit, knowledge of the process, and/or mass balance information to either: Estimate directly the annual average wastewater flow rate; or estimate the total annual wastewater volume and then divide total volume by 525,600 min/yr. Where knowledge is used to determine the annual average flow rate, the permittee shall provide sufficient information to document the flow rate for wastewater streams determined to be Group 2 wastewater streams. Documentation to determine the annual average flow rate is not required for Group 1 streams. [§63.144(c)(1)]
- b) *Historical records.* The permittee may use historical records to determine the annual average flow rate. Derive the highest annual average flow rate of wastewater from historical records representing the most recent five years of operation or, if the process unit has been in service for less than five years but at least one year, from historical records representing the total operating life of the process unit. Where historical records are used to determine the annual average flow rate, the permittee shall provide sufficient information to document the flow rate for wastewater streams determined to be Group 2 wastewater streams. Documentation to determine the annual average flow rate is not required for Group 1 streams. [§63.144(c)(2)]
- c) *Measurements of flow rate.* To comply with §63.144(a)(1) by measuring the flow rate, the permittee shall comply with the requirements of this paragraph. Measurements shall be made at the point of determination, or at a location downstream of the point of determination with adjustments for flow rate changes made according to §63.144(c)(4). Where measurement data are used to determine the annual average flow rate, the permittee shall provide sufficient information to document the flow rate for wastewater streams determined to be Group 2 wastewater streams. Documentation to determine the annual average flow rate is not required for Group 1 streams. [§63.144(c)(3)]
- d) *Adjustment for flow rates determined downstream of the point of determination.* The permittee shall make corrections to the annual average flow rate of a wastewater stream when it is determined downstream of the point of determination at a location where two or more wastewater streams have been mixed or one or more wastewater streams have been treated. The permittee shall make corrections for such changes in the annual average flow rate. [§63.144(c)(4)]

**MACT F Maintenance Wastewater Requirements:**

1. The permittee shall comply with the requirements of §63.105(b) through (e) for maintenance wastewaters containing those organic HAP's listed in Tables 8 and 9 of MACT FFFF. [§63.105(a) and §63.2485(b)]
2. The permittee shall prepare a description of maintenance procedures for management of wastewaters generated from the emptying and purging of equipment in the process during temporary shutdowns for inspections, maintenance, and repair (i.e., a maintenance-turnaround) and during periods which are not shutdowns (i.e., routine maintenance). The descriptions shall: [§63.105(b)]
  - a) Specify the process equipment or maintenance tasks that are anticipated to create wastewater during maintenance activities. [§63.105(b)(1)]
  - b) Specify the procedures that will be followed to properly manage the wastewater and control organic HAP emissions to the atmosphere; and [§63.105(b)(2)]

- c) Specify the procedures to be followed when clearing materials from process equipment. [§63.105(b)(3)]
3. The permittee shall modify and update the information required by §63.105(b) as needed following each maintenance procedure based on the actions taken and the wastewaters generated in the preceding maintenance procedure. [§63.105(c)]
4. The permittee shall maintain a record of the information required by §63.105(b) and (c). [§63.105(e) and §63.2485(q)(2)]

**MACT G Control Requirements for Certain Liquid Streams in Open Systems within a MCPU:**

1. The permittee shall comply with the provisions of Table 35 of MACT G, for each item of equipment meeting all the criteria specified in §63.149(b) through (d) and either §63.149(e)(1) or (e)(2). [§63.149(a)]
2. The item of equipment is of a type identified in Table 35 of MACT G; [§63.149(b)]
3. The item of equipment is part of a MCPU that meets the criteria of §63.2435(b); [§63.149(c) and §63.2485(l)(1) and (l)(3)]
4. The item of equipment is controlled less stringently than in Table 35 of MACT G and is not listed in §63.100(f), and the item of equipment is not otherwise exempt from controls by the provisions of MACT A, MACT F, MACT G, or MACT H; and [§63.149(d)]
5. The item of equipment: [§63.149(e)]
  - a) Is a drain, drain hub, manhole, lift station, trench, pipe, or oil/water separator that conveys water containing compounds in Tables 8 and 9 to MACT FFFF if any of the conditions specified in §63.2485(c)(1) through (3) are met. At MCPUs subject to the new source requirements of §63.2445(a), the criteria of this paragraph are also met if the item of equipment conveys water with an annual average concentration greater than or equal to 10 ppm by weight of any Table 8 compound at an annual average flow rate greater than or equal to 0.02 L/min, or [§63.149(e)(1) and §63.2485(b), (l)(2), (l)(3), and (l)(4)]
  - b) Is a tank that receives one or more streams that contain water with a total annual average concentration of Table 8 to MACT FFFF compounds greater than or equal to 1,000 ppmw, and the annual average flowrate is greater than or equal to 1 L/min. At MCPUs subject to the new source requirements of §63.2445(a), the criteria of this paragraph are also met if the tank receives one or more streams that contain water with an annual average concentration greater than or equal to 10 ppm by weight of any Table 8 compound at an annual average flow rate greater than or equal to 0.02 L/min. The permittee shall determine the characteristics of the stream as specified in §63.149(e)(2)(i) and (ii). [§63.149(e)(2) and §63.2485(b), (l)(2), (l)(3), and (l)(5)]
    - 1) The characteristics of the stream being received shall be determined at the inlet to the tank. [§63.149(e)(2)(i)]
    - 2) The characteristics shall be determined according to the procedures in §63.144(b) and (c). [§63.149(e)(2)(ii)]

**Table 35 to MACT G – Control Requirements for Items of Equipment That Meet the Criteria of §63.149**

| Item of equipment    | Control requirement <sup>a</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Drain or drain hub   | (a) Tightly fitting solid cover (TFSC); or<br>(b) TFSC with a vent to either a process, or to a fuel gas system, or to a control device meeting the requirements of §63.139(c); or<br>(c) Water seal with submerged discharge or barrier to protect discharge from wind.                                                                                                                                                                                                                                                                 |
| Manhole <sup>b</sup> | (a) TFSC: or<br>(b) TFSC with a vent to either a process, or to a fuel gas system, or to a control device meeting the requirements of §63.139(c); or<br>(c) If the item is vented to the atmosphere, use a TFSC with a properly operating water seal at the entrance or exit to the item to restrict ventilation in the collection system. The vent pipe shall be at least 90 cm in length and not exceeding 10.2 cm in nominal inside diameter.                                                                                         |
| Lift station         | (a) TFSC: or<br>(b) TFSC with a vent to either a process, or to a fuel gas system, or to a control device meeting the requirements of §63.139(c); or<br>(c) If the lift station is vented to the atmosphere, use a TFSC with a properly operating water seal at the entrance or exit to the item to restrict ventilation in the collection system. The vent pipe shall be at least 90 cm in length and not exceeding 10.2 cm in nominal inside diameter. The lift station shall be level controlled to minimize changes in liquid level. |
| Trench               | (a) TFSC: or<br>(b) TFSC with a vent to either a process, or to a fuel gas system, or to a control device meeting the requirements of §63.139(c); or<br>(c) If the item is vented to the atmosphere, use a TFSC with a properly operating water seal at the entrance or exit to the item to restrict ventilation in the collection system. The vent pipe shall be at least 90 cm in length and not exceeding 10.2 cm in nominal inside diameter.                                                                                         |
| Pipe                 | Each pipe shall have no visible gaps in joints, seals, or other emission interfaces.                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Oil/Water separator  | (a) Equip with a fixed roof and route vapors to a process or to a fuel gas system, or equip with a closed vent system that routes vapors to a control device meeting the requirements of §63.139(c); or<br>(b) Equip with a floating roof that meets the equipment specifications of §60.693(a)(1)(i), (a)(1)(ii), (a)(2), (a)(3), and (a)(4).                                                                                                                                                                                           |
| Tank <sup>c</sup>    | Maintain a fixed roof. <sup>d</sup> If the tank is sparged <sup>e</sup> or used for heating or treating by means of an exothermic reaction, a fixed roof and a system shall be maintained that routes the organic HAP vapors to other process equipment or a fuel gas system, or a closed vent system that routes vapors to a control device that meets the requirements of §63.119(e)(1) or (e)(2).                                                                                                                                     |

<sup>a</sup> Where a tightly fitting solid cover is required, it shall be maintained with no visible gaps or openings, except during periods of sampling, inspection, or maintenance.

<sup>b</sup> Manhole includes sumps and other points of access to a conveyance system.

<sup>c</sup> Applies to tanks with capacities of 38 m<sup>3</sup> or greater.

<sup>d</sup> A fixed roof may have openings necessary for proper venting of the tank, such as pressure/vacuum vent, j-pipe vent.

<sup>e</sup> The liquid in the tank is agitated by injecting compressed air or gas.

**MACT FFFF Closed Vent System, Scrubber (Absorber), and Condenser Monitoring:**

1. *Continuous Parameter Monitoring.* The provisions in §63.2450(k)(1) through (8) apply in addition to the requirements for continuous parameter monitoring system (CPMS) in MACT SS. [§63.2450(k)]
  - a) The permittee must comply with §63.2450(k)(1)(i) and (ii). [§63.2450(k)(1)]
    - 1) §63.2450(k)(1)(i) no longer applies. Instead, the permittee must record the results of each calibration check and all maintenance performed on the CPMS as specified in §63.998(c)(1)(ii)(A), except the permittee must record all maintenance, not just preventative maintenance. [§63.2450(k)(1)(ii)]
  - b) When MACT SS uses the term “a range” or “operating range” of a monitored parameter, it means an “operating limit” for a monitored parameter for the purposes of MACT FFFF. [§63.2450(k)(2)]
  - c) For absorbers that control organic compounds and use water as the scrubbing fluid, the permittee must conduct monitoring and recordkeeping as specified in §63.2450(k)(5)(i) through (iii) instead of the monitoring and recordkeeping requirements specified in §63.990(c)(1), §63.993(c)(1), and §63.998(a)(2)(ii)(C). [§63.2450(k)(5)]
    - 1) The permittee must use a flow meter capable of providing a continuous record of the absorber influent liquid flow. [§63.2450(k)(5)(i)]
    - 2) The permittee must determine gas stream flow using one of the procedures specified in §63.994(c)(1)(ii)(A) through (D). [§63.2450(k)(5)(ii)]
    - 3) The permittee must record the absorber liquid-to-gas ratio average over the time period of any performance test. [§63.2450(k)(5)(iii)]
  - d) The manufacturer’s specifications or the permittee’s procedures must include a schedule for calibrations, preventative maintenance procedures, a schedule for preventative maintenance, and corrective actions to be taken if a calibration fails. If a CPMS calibration fails, the CPMS is considered to be inoperative until the permittee takes corrective action, and the system passes calibration. The permittee must record the nature and cause of instances when the CPMS is inoperative, and the corrective action taken. [§63.2450(k)(7)]
  - e) The permittee must comply with the requirements of §63.2450(e)(4), as applicable. [§63.2450(k)(8)]

**MACT SS Closed Vent System Inspection and Monitoring Requirements:**

1. *Closed vent system inspection and monitoring requirements.* The provisions of MACT SS apply to closed vent systems collecting regulated material from a regulated source. Inspection records shall be generated as specified in §63.998(d)(1)(iii) and (iv). [§63.983(b)]
  - a) Except for any closed vent systems that are designated as unsafe or difficult to inspect as provided in §63.983(b)(2) and (3), each closed vent system shall be inspected as specified in §63.983(b)(1)(i) or (ii). [§63.983(b)(1)]
    - 1) If the closed vent system is constructed of hard-piping, the permittee shall comply with the requirements specified in §63.983(b)(1)(i)(A) and (B). [§63.983(b)(1)(i)]
      - (i) Conduct annual inspections for visible, audible, or olfactory indications of leaks. [§63.983(b)(1)(i)(B)]
    - 2) If the closed vent system is constructed of ductwork, the permittee shall conduct an annual inspection according to the procedures in §63.983(c). [§63.983(b)(1)(ii)]
  - b) Any parts of the closed vent system that are designated, as described in §63.998(d)(1)(i), as unsafe to inspect are exempt from the inspection requirements of

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- §63.983(b)(1) if the conditions of §63.983(b)(2)(i) and (ii) are met. [§63.983(b)(2)]
- 1) The permittee determines that the equipment is unsafe-to-inspect because inspecting personnel would be exposed to an imminent or potential danger because of complying with §63.983(b)(1); and [§63.983(b)(2)(i)]
  - 2) The permittee has a written plan that requires inspection of the equipment as frequently as practical during safe-to-inspect times. Inspection is not required more than once annually. [§63.983(b)(2)(ii)]
- c) Any parts of the closed vent system that are designated, as described in §63.998(d)(1)(i), as difficult-to-inspect are exempt from the inspection requirements of §63.983(b)(1) if the provisions of §63.983(b)(3)(i) and (ii) apply. [§63.983(b)(3)]
- 1) The permittee determines that the equipment cannot be inspected without elevating the inspecting personnel more than 2 m (7 ft) above a support surface; and [§63.983(b)(3)(i)]
  - 2) The permittee has a written plan that requires inspection of the equipment at least once every five years. [§63.983(b)(3)(ii)]
- d) For each bypass line, the permittee shall comply with §63.983(b)(4)(i) or (ii). [§63.983(b)(4)]
- 1) If a flow indicator is used, take a reading at least once every 15 minutes. [§63.983(b)(4)(i)]
  - 2) If the bypass line valve is secured in the non-diverting position, visually inspect the seal or closure mechanism at least once every month to verify that the valve is maintained in the non-diverting position, and the vent stream is not diverted through the bypass line. [§63.983(b)(4)(ii)]
2. *Closed vent system inspection procedures.* The provisions of this paragraph apply to closed vent systems collecting regulated material from a regulated source. [§63.983(c)]
- a) Each closed vent system subject to this paragraph shall be inspected according to the procedures specified in §63.983(c)(1)(i) through (vii). [§63.983(c)(1)]
- 1) Inspections shall be conducted in accordance with Method 21 of NSPS Appendix A, except as specified in §63.983. [§63.983(c)(1)(i)]
  - 2) Except as provided in §63.983(c)(1)(iii), the detection instrument shall meet the performance criteria of Method 21 of NSPS Appendix A, except the instrument response factor criteria in §3.1.2(a) of Method 21 must be for the representative composition of the process fluid and not of each individual VOC in the stream. For process streams that contain nitrogen, air, water, or other inerts that are not organic HAP or VOC, the representative stream response factor must be determined on an inert-free basis. The response factor may be determined at any concentration for which the monitoring for leaks will be conducted. [§63.983(c)(1)(ii)]
  - 3) If no instrument is available at the plant site that will meet the performance criteria of Method 21 specified in §63.983(c)(1)(ii), the instrument readings may be adjusted by multiplying by the representative response factor of the process fluid, calculated on an inert-free basis as described in §63.983(c)(1)(ii). [§63.983(c)(1)(iii)]
  - 4) The detection instrument shall be calibrated before use on each day of its use by the procedures specified in Method 21 of NSPS Appendix A. [§63.983(c)(1)(iv)]
  - 5) Calibration gases shall be as specified in §63.983(c)(1)(v)(A) through (C). [§63.983(c)(1)(v)]
    - (i) Zero air (less than 10 ppm hydrocarbon in air); and [§63.983(c)(1)(v)(A)]
    - (ii) Mixtures of methane in air at a concentration less than 10,000 ppm. A calibration gas other than methane in air may be used if the instrument does not respond to

- methane or if the instrument does not meet the performance criteria specified in §63.983(c)(1)(ii). In such cases, the calibration gas may be a mixture of one or more of the compounds to be measured in air. [§63.983(c)(1)(v)(B)]
- (iii) If the detection instrument's design allows for multiple calibration scales, then the lower scale shall be calibrated with a calibration gas that is no higher than 2,500 ppm. [§63.983(c)(1)(v)(C)]
- 6) The permittee may elect to adjust or not adjust instrument readings for background. If the permittee elects not to adjust readings for background, all such instrument readings shall be compared directly to 500 ppm to determine whether there is a leak. If the permittee elects to adjust instrument readings for background, the permittee shall measure background concentration using the procedures in §63.983. The permittee shall subtract the background reading from the maximum concentration indicated by the instrument. [§63.983(c)(1)(vi)]
- 7) If the permittee elects to adjust for background, the arithmetic difference between the maximum concentration indicated by the instrument and the background level shall be compared with 500 ppm for determining whether there is a leak. [§63.983(c)(1)(vii)]
- b) The instrument probe shall be traversed around all potential leak interfaces as described in Method 21 of NSPS Appendix A. [§63.983(c)(2)]
- c) Except as provided in §63.983(c)(4), inspections shall be performed when the equipment is in regulated material service, or in use with any other detectable gas or vapor. [§63.983(c)(3)]
3. *Closed vent system leak repair provisions.* The provisions of this paragraph apply to closed vent systems collecting regulated material from a regulated source. [§63.983(d)]
- a) If there are visible, audible, or olfactory indications of leaks at the time of the annual visual inspections required by §63.983(b)(1)(i)(B), the permittee shall follow the procedure specified in either §63.983(d)(1)(i) or (ii). [§63.983(d)(1)]
- 1) The permittee shall eliminate the leak. [§63.983(d)(1)(i)]
- 2) The permittee shall monitor the equipment according to the procedures in §63.983(c). [§63.983(d)(1)(ii)]
- b) Leaks, as indicated by an instrument reading greater than 500 ppmv above background or by visual inspections, shall be repaired as soon as practical, except as provided in §63.983(d)(3). Records shall be generated as specified in §63.998(d)(1)(iii) when a leak is detected. [§63.983(d)(2)]
- 1) A first attempt at repair shall be made no later than five days after the leak is detected. [§63.983(d)(2)(i)]
- 2) Except as provided in §63.983(d)(3), repairs shall be completed no later than 15 days after the leak is detected or at the beginning of the next introduction of vapors to the system, whichever is later. [§63.983(d)(2)(ii)]
- c) Delay of repair of a closed vent system for which leaks have been detected is allowed if repair within 15 days after a leak is detected is technically infeasible or unsafe without a closed vent system shutdown, as defined in §63.981, or if the permittee determines that emissions resulting from immediate repair would be greater than the emissions likely to result from delay of repair. Repair of such equipment shall be completed as soon as practical, but not later than the end of the next closed vent system shutdown. [§63.983(d)(3)]

**MACT SS Control Device Monitoring Requirements:**

1. *Monitoring requirements.* Where an absorber, condenser, or carbon adsorber is used as a control device, either an organic monitoring device capable of providing a continuous record, or the monitoring devices specified in §63.990(c)(1) through (3), as applicable, shall be used. Monitoring results shall be recorded as specified in §63.998(b) and (c), as applicable. General requirements for monitoring and continuous parameter monitoring systems are contained in MACT FFFF and §63.996. [§63.990(c)]
  - a) Where an absorber is used, a scrubbing liquid temperature monitoring device and a specific gravity monitoring device, each capable of providing a continuous record, shall be used. If the difference between the specific gravity of the saturated scrubbing fluid and specific gravity of the fresh scrubbing fluid is less than 0.02 specific gravity units, an organic monitoring device capable of providing a continuous record shall be used. [§63.990(c)(1)]
  - b) Where a condenser is used, a condenser exit (product side) temperature monitoring device capable of providing a continuous record shall be used. [§63.990(c)(2)]

**MACT SS General Monitoring Requirements:**

1. *General monitoring requirements applicability.* [§63.996(a)]
  - a) §63.996 applies to permittees required to monitor under MACT S. [§63.996(a)(1)]
  - b) Flow indicators are not subject to the requirements of §63.996. [§63.996(a)(3)]
2. *Conduct of monitoring.* [§63.996(b)]
  - a) Monitoring shall be conducted as set forth in §63.996 and in the relevant sections of MACT SS unless the provision in either §63.996(b)(1)(i) or (ii) applies. [§63.996(b)(1)]
    - 1) The Administrator specifies or approves the use of minor changes in methodology for the specified monitoring requirements and procedures; or [§63.996(b)(1)(i)]
    - 2) The Administrator approves the use of alternatives to any monitoring requirements or procedures as provided in MACT FFFF or §63.996(d). [§63.996(b)(1)(ii)]
  - b) When one CPMS is used as a backup to another CPMS, the permittee shall report the results from the CPMS used to meet the monitoring requirements of MACT SS. If both such CPMS's are used during a particular reporting period to meet the monitoring requirements of MACT SS, then the permittee shall report the results from each CPMS for the time during the six-month period that the instrument was relied upon to demonstrate compliance. [§63.996(b)(2)]
3. *Operation and maintenance of continuous parameter monitoring systems.* [§63.996(c)]
  - a) All monitoring equipment shall be installed, calibrated, maintained, and operated according to manufacturer's specifications or other written procedures that provide adequate assurance that the equipment would reasonably be expected to monitor accurately. [§63.996(c)(1)]
  - b) The permittee shall maintain and operate each CPMS as specified in §63.996, or in MACT FFFF, and in a manner consistent with good air pollution control practices. [§63.996(c)(2)]
    - 1) The permittee shall ensure the immediate repair or replacement of CPMS parts to correct "routine" or otherwise predictable CPMS malfunctions. The necessary parts for routine repairs of the affected equipment shall be readily available. [§63.996(c)(2)(i)]
    - 2) The Administrator's determination of whether acceptable operation and maintenance procedures are being used for the CPMS will be based on information that may include, but is not limited to, review of operation and maintenance

procedures, operation and maintenance records as specified in §63.998(c)(1)(i), manufacturer's recommendations and specifications, and inspection of the CPMS. [§63.996(c)(2)(iii) and §63.2450(e)(4)(vi)]

- c) All CPMS's shall be installed and operational, and the data verified as specified in MACT SS either prior to or in conjunction with conducting performance tests. Verification of operational status shall, at a minimum, include completion of the manufacturer's written specifications or recommendations for installation, operation, and calibration of the system or other written procedures that provide adequate assurance that the equipment would reasonably be expected to monitor accurately. [§63.996(c)(3)]
  - d) All CPMS's shall be installed such that representative measurements of parameters from the regulated source are obtained. [§63.996(c)(4)]
  - e) In accordance with MACT FFFF, except for system breakdowns, repairs, maintenance periods, instrument adjustments, or checks to maintain precision and accuracy, calibration checks, and zero and span adjustments, all continuous parameter monitoring systems shall be in continuous operation when emissions are being routed to the monitored device. [§63.996(c)(5)]
  - f) The permittee shall establish a range for monitored parameters that indicates proper operation of the control or recovery device. To establish the range, the information required in §63.999(b)(3) shall be submitted in the Notification of Compliance Status or the operating permit application or amendment. The range may be based upon a prior performance test meeting the specifications of §63.997(b)(1) or a prior TRE index value determination, as applicable, or upon existing ranges or limits established under MACT FFFF. [§63.996(c)(6)]
4. *Alternatives to monitoring requirements* — [§63.996(d)]
- a) *Alternatives to the continuous operating parameter monitoring and recordkeeping provisions.* The permittee may request approval to use alternatives to the continuous operating parameter monitoring and recordkeeping provisions listed in §63.990(c), §63.998(a)(2) through (4), §63.998(c)(2) and (3), as specified in §63.999(d)(1). [§63.996(d)(1)]
  - b) *Monitoring a different parameter than those listed.* The permittee may request approval to monitor a different parameter than those established in §63.996(c)(6), as specified in §63.999(d)(2). [§63.996(d)(2)]

**MACT UU Closed Vent System Requirements:**

1. *Compliance standard.* [§63.1034(b)]
- a) For closed vent systems and control devices used to comply with the provisions of MACT UU, the permittee shall comply with the provisions of MACT SS and §63.1034(b)(2)(i) through (b)(2)(iii), except as provided in §63.1002(b). [§63.1034(b)(2)]
  - 1) Nonflare control devices shall be designed and operated to reduce emissions of regulated material vented to them with an efficiency of 95% or greater, or to an exit concentration of 20 ppmv, whichever is less stringent. The 20 ppmv standard is not applicable to the provisions of §63.1016. [§63.1034(b)(2)(i)]

**MACT UU Equipment Leak Monitoring:**

1. *Instrument and sensory monitoring for leaks.* [§63.1023]
- a) *Monitoring for leaks.* The permittee shall monitor regulated equipment as specified in §63.1023(a)(1) for instrument monitoring and §63.1023(a)(2) for sensory monitoring. [§63.1023(a)]

- 1) *Instrument monitoring for leaks.* [§63.1023(a)(1)]
  - (i) Valves in gas and vapor service and in light liquid service shall be monitored pursuant to §63.1025(b). [§63.1023(a)(1)(i)]
  - (ii) Pumps in light liquid service shall be monitored pursuant to §63.1026(b). [§63.1023(a)(1)(ii)]
  - (iii) Connectors in gas and vapor service and in light liquid service shall be monitored pursuant to §63.1027(b). [§63.1023(a)(1)(iii)]
  - (iv) Agitators in gas and vapor service and in light liquid service shall be monitored pursuant to §63.1028(c). [§63.1023(a)(1)(iv)]
  - (v) Pressure relief devices in gas and vapor service shall be monitored pursuant to §63.1030(c). [§63.1023(a)(1)(v)]
  - (vi) Compressors designated to operate with an instrument reading less than 500 ppm above background, as described in §63.1022(e), shall be monitored pursuant to §63.1031(f). [§63.1023(a)(1)(vi)]
- 2) *Sensory monitoring for leaks.* [§63.1023(a)(2)]
  - (i) Pumps in light liquid service shall be observed pursuant to §63.1026(b)(4) and (e)(1)(v). [§63.1023(a)(2)(i)]
  - (ii) Agitators in gas and vapor service and in light liquid service shall be observed pursuant to §63.1028(c)(3) or (e)(1)(iv). [§63.1023(a)(2)(iii)]
- b) *Instrument monitoring methods.* Instrument monitoring, as required under MACT UU, shall comply with the requirements specified in §63.1023(b)(1) through (b)(6). [§63.1023(b)]
  - 1) *Monitoring method.* Monitoring shall comply with Method 21 of NSPS Appendix A, except as otherwise provided in §63.1023. [§63.1023(b)(1)]
  - 2) *Detection instrument performance criteria.* [§63.1023(b)(2)]
    - (i) Except as provided for in §63.1023(b)(2)(ii), the detection instrument shall meet the performance criteria of Method 21 of NSPS Appendix A, except the instrument response factor criteria in §3.1.2, paragraph (a) of Method 21 shall be for the representative composition of the process fluid not each individual VOC in the stream. For process streams that contain nitrogen, air, water or other inerts that are not HAP or VOC, the representative stream response factor shall be determined on an inert-free basis. The response factor may be determined at any concentration for which monitoring for leaks will be conducted. [§63.1023(b)(2)(i)]
    - (ii) If there is no instrument commercially available that will meet the performance criteria specified in §63.1023(b)(2)(i), the instrument readings may be adjusted by multiplying by the representative response factor of the process fluid, calculated on an inert-free basis as described in §63.1023(b)(2)(i). [§63.1023(b)(2)(ii)]
  - 3) *Detection instrument calibration procedure.* The detection instrument shall be calibrated before use on each day of its use by the procedures specified in Method 21 of NSPS Appendix A. [§63.1023(b)(3)]
  - 4) *Detection instrument calibration gas.* Calibration gases shall be zero air (less than 10 ppm of hydrocarbon in air); and the gases specified in §63.1023(b)(4)(i) except as provided in §63.1023(b)(4)(ii). [§63.1023(b)(4)]
    - (i) Mixtures of methane in air at a concentration no more than 2,000 ppm greater than the leak definition concentration of the equipment monitored. If the monitoring instrument's design allows for multiple calibration scales, then the lower scale shall be calibrated with a calibration gas that is no higher than 2,000

- ppm above the concentration specified as a leak, and the highest scale shall be calibrated with a calibration gas that is approximately equal to 10,000 ppm. If only one scale on an instrument will be used during monitoring, the permittee need not calibrate the scales that will not be used during that day's monitoring. [§63.1023(b)(4)(i)]
- (ii) A calibration gas other than methane in air may be used if the instrument does not respond to methane or if the instrument does not meet the performance criteria specified in §63.1023(b)(2)(i). In such cases, the calibration gas may be a mixture of one or more of the compounds to be measured in air. [§63.1023(b)(4)(ii)]
- 5) *Monitoring performance.* Monitoring shall be performed when the equipment is in regulated material service or is in use with any other detectable material. [§63.1023(b)(5)]
- c) *Instrument monitoring using background adjustments.* The permittee may elect to adjust or not to adjust the instrument readings for background. If the permittee elects not to adjust instrument readings for background, the permittee shall monitor the equipment according to the procedures specified in §63.1023(b)(1) through (b)(5). In such cases, all instrument readings shall be compared directly to the applicable leak definition for the monitored equipment to determine whether there is a leak or to determine compliance with §63.1030(b) (pressure relief devices) or §63.1031(f) (alternative compressor standard). If the permittee elects to adjust instrument readings for background, the permittee shall monitor the equipment according to the procedures specified in §63.1023(c)(1) through (c)(4). [§63.1023(c)]
- 1) The requirements of §63.1023(b)(1) through (b)(5) shall apply. [§63.1023(c)(1)]
- 2) The background level shall be determined, using the procedures in Method 21 of NSPS Appendix A. [§63.1023(c)(2)]
- 3) The instrument probe shall be traversed around all potential leak interfaces as close to the interface as possible as described in Method 21 of NSPS Appendix A. [§63.1023(c)(3)]
- 4) The arithmetic difference between the maximum concentration indicated by the instrument and the background level shall be compared to the applicable leak definition for the monitored equipment to determine whether there is a leak or to determine compliance with §63.1030(b) (pressure relief devices) or §63.1031(f) (alternative compressor standard). [§63.1023(c)(4)]
- d) *Sensory monitoring methods.* Sensory monitoring consists of visual, audible, olfactory, or any other detection method used to determine a potential leak to the atmosphere. [§63.1023(d)]
- e) *Leaking equipment identification and records.* [§63.1023(e)]
- 1) When each leak is detected pursuant to the monitoring specified in §63.1023(a), a weatherproof and readily visible identification, shall be attached to the leaking equipment. [§63.1023(e)(1)]
- 2) When each leak is detected, the information specified in §63.1024(f) shall be recorded and kept pursuant to MACT FFFF, except for the information for connectors complying with the eight-year monitoring period allowed under §63.1027(b)(3)(iii) shall be kept five years beyond the date of its last use. [§63.1023(e)(2)]

**MACT UU Leak Repair Requirements:**

1. ***Leak Repair Schedule.*** The permittee shall repair each leak detected as soon as practical, but not later than 15 calendar days after it is detected, except as provided by §63.1024(d) and (e). A first attempt at repair as defined in MACT FFFF shall be made no later than five calendar days after the leak is detected. First attempt at repair for pumps includes, but is not limited to, tightening the packing gland nuts and/or ensuring that the seal flush is operating at design pressure and temperature. First attempt at repair for valves includes, but is not limited to, tightening the bonnet bolts, and/or replacing the bonnet bolts, and/or tightening the packing gland nuts, and/or injecting lubricant into the lubricated packing. [§63.1024(a)]
2. ***Leak identification removal*** — [§63.1024(c)]
  - a) ***Valves and connectors in gas/vapor and light liquid service.*** The leak identification on a valve in gas/vapor or light liquid service may be removed after it has been monitored as specified in §63.1025(d)(2), and no leak has been detected during that monitoring. The leak identification on a connector in gas/vapor or light liquid service may be removed after it has been monitored as specified in §63.1027(b)(3)(iv) and no leak has been detected during that monitoring. [§63.1024(c)(1)]
  - b) ***Other equipment.*** The identification that has been placed, pursuant to §63.1023(e)(1), on equipment determined to have a leak, except for a valve or for a connector in gas/vapor or light liquid service that is subject to the provisions of §63.1027(b)(3)(iv), may be removed after it is repaired. [§63.1024(c)(2)]
3. ***Delay of repair.*** Delay of repair is allowed for any of the conditions specified in §63.1024(d)(1) through (d)(5). The permittee shall maintain a record of the facts that explain any delay of repairs and, where appropriate, why the repair was technically infeasible without a process unit shutdown. [§63.1024(d)]
  - a) Delay of repair of equipment for which leaks have been detected is allowed if repair within 15 days after a leak is detected is technically infeasible without a process unit or affected facility shutdown. Repair of this equipment shall occur as soon as practical, but no later than the end of the next process unit or affected facility shutdown, except as provided in §63.1024(d)(5). [§63.1024(d)(1)]
  - b) Delay of repair of equipment for which leaks have been detected is allowed for equipment that is isolated from the process and that does not remain in regulated material service. [§63.1024(d)(2)]
  - c) Delay of repair for valves, connectors, and agitators is also allowed if the provisions of §63.1024(d)(3)(i) and (d)(3)(ii) are met. [§63.1024(d)(3)]
    - 1) The permittee determines that emissions of purged material resulting from immediate repair would be greater than the fugitive emissions likely to result from delay of repair, and [§63.1024(d)(3)(i)]
    - 2) When repair procedures are affected, the purged material is collected and destroyed, collected, and routed to a fuel gas system or process, or recovered in a control device complying with either §63.1034 or §63.1021(b). [§63.1024(d)(3)(ii)]
  - d) Delay of repair for pumps is also allowed if the provisions of §63.1024(d)(4)(i) and (d)(4)(ii) are met. [§63.1024(d)(4)]
    - 1) Repair requires replacing the existing seal design with a new system that the permittee has determined under the provisions of §63.1035(d) will provide better performance or one of the specifications of §63.1024(d)(4)(i)(A) through (d)(4)(i)(C) are met. [§63.1024(d)(4)(i)]
      - (i) A dual mechanical seal system that meets the requirements of §63.1026(e)(1)

- will be installed; [§63.1024(d)(4)(i)(A)]
- (ii) A pump that meets the requirements of §63.1026(e)(2) will be installed; or [§63.1024(d)(4)(i)(B)]
- (iii) A system that routes emissions to a process or a fuel gas system or a closed vent system and control device that meets the requirements of § 63.1026(e)(3) will be installed; and [§63.1024(d)(4)(i)(C)]
- 2) Repair is completed as soon as practical, but not later than six months after the leak was detected. [§63.1024(d)(4)(ii)]
- e) Delay of repair beyond a process unit or affected facility shutdown will be allowed for a valve if valve assembly replacement is necessary during the process unit or affected facility shutdown, and valve assembly supplies have been depleted, and valve assembly supplies had been sufficiently stocked before the supplies were depleted. Delay of repair beyond the second process unit or affected facility shutdown will not be allowed unless the third process unit or affected facility shutdown occurs sooner than six months after the first process unit or affected facility shutdown. [§63.1024(d)(5)]
- 4. *Unsafe-to-repair—connectors.* Any connector that is designated, as described in §63.1022(d), as an unsafe-to-repair connector is exempt from the requirements of §63.1027(d), and §63.1024(a). [§63.1024(e)]
- 5. *Leak repair records.* For each leak detected, the information specified in §63.1024(f)(1) through (f)(5) shall be recorded and maintained pursuant to MACT FFFF. [§63.1024(f)]
  - a) The date of first attempt to repair the leak. [§63.1024(f)(1)]
  - b) The date of successful repair of the leak. [§63.1024(f)(2)]
  - c) Maximum instrument reading measured by Method 21 of NSPS Appendix A at the time the leak is successfully repaired or determined to be nonrepairable. [§63.1024(f)(3)]
  - d) "Repair delayed" and the reason for the delay if a leak is not repaired within 15 calendar days after discovery of the leak as specified in §63.1024(f)(4)(i) and (f)(4)(ii). [§63.1024(f)(4)]
    - 1) The permittee may develop a written procedure that identifies the conditions that justify a delay of repair. The written procedures may be included as part of the startup, shutdown, and malfunction plan, as required by MACT FFFF for the source, or may be part of a separate document that is maintained at the plant site. In such cases, reasons for delay of repair may be documented by citing the relevant sections of the written procedure. [§63.1024(f)(4)(i)]
    - 2) If delay of repair was caused by depletion of stocked parts, there must be documentation that the spare parts were sufficiently stocked on-site before depletion and the reason for depletion. [§63.1024(f)(4)(ii)]
  - e) Dates of process unit or affected facility shutdowns that occur while the equipment is unrepaired. [§63.1024(f)(5)]

**MACT UU Valves in Gas and Vapor Service or in Light Liquid Service Standards:**

- 1. *Leak detection.* Unless otherwise specified in §63.1021(b) or §63.1025(e), or MACT FFFF, the permittee shall monitor all valves at intervals specified in §63.1025(b)(3) and/or (b)(4) and shall comply with all other provisions of §63.1025. [§63.1025(b)]
  - a) *Monitoring method.* The valves shall be monitored to detect leaks by the method specified in §63.1023(b) and, as applicable, §63.1023(c). [§63.1025(b)(1)]
  - b) *Instrument reading that defines a leak.* The instrument reading that defines a leak is 500 ppm or greater. [§63.1025(b)(2)]
  - c) *Monitoring frequency.* The permittee shall monitor valves for leaks at the intervals

specified in §63.1025(b)(3)(i) through (b)(3)(v) and shall keep the record specified in §63.1025(b)(3)(vi). [§63.1025(b)(3)]

- 1) If at least the greater of two valves or 2% of the valves in a process unit leak, as calculated according to §63.1025(c), the permittee shall monitor each valve once per month. [§63.1025(b)(3)(i)]
  - 2) At process units with less than the greater of two leaking valves or 2% leaking valves, the permittee shall monitor each valve once per quarter. [§63.1025(b)(3)(ii)]
  - 3) At process units with less than 1% leaking valves, the permittee may elect to monitor each valve once every two quarters. [§63.1025(b)(3)(iii)]
  - 4) At process units with less than 0.5% leaking valves, the permittee may elect to monitor each valve once every four quarters. [§63.1025(b)(3)(iv)]
  - 5) At process units with less than 0.25% leaking valves, the permittee may elect to monitor each valve once every two years. [§63.1025(b)(3)(v)]
- d) The permittee shall keep a record of the monitoring schedule for each process unit. [§63.1025(b)(3)(vi)]
- e) *Valve subgrouping.* For a process unit or a group of process units to which MACT UU applies, the permittee may choose to subdivide the valves in the applicable process unit or group of process units and apply the provisions of §63.1025(b)(3) to each subgroup. If the permittee elects to subdivide the valves in the applicable process unit or group of process units, then the provisions of §63.1025(b)(4)(i) through (b)(4)(viii) apply. [§63.1025(b)(4)]
- 1) The overall performance of total valves in the applicable process unit or group of process units to be subdivided shall be less than 2% leaking valves, as detected according to §63.1025(b)(1) and (b)(2) and as calculated according to §63.1025(c)(1)(ii) and (c)(2). [§63.1025(b)(4)(i)]
  - 2) The initial assignment or subsequent reassignment of valves to subgroups shall be governed by the provisions of §63.1025(b)(4)(ii)(A) through (b)(4)(ii)(C). [§63.1025(b)(4)(ii)]
    - (i) The permittee shall determine which valves are assigned to each subgroup. Valves with less than one year of monitoring data or valves not monitored within the last 12 months must be placed initially into the most frequently monitored subgroup until at least one year of monitoring data have been obtained. [§63.1025(b)(4)(ii)(A)]
    - (ii) Any valve or group of valves can be reassigned from a less frequently monitored subgroup to a more frequently monitored subgroup provided that the valves to be reassigned were monitored during the most recent monitoring period for the less frequently monitored subgroup. The monitoring results must be included with that less frequently monitored subgroup's associated percent leaking valves calculation for that monitoring event. [§63.1025(b)(4)(ii)(B)]
    - (iii) Any valve or group of valves can be reassigned from a more frequently monitored subgroup to a less frequently monitored subgroup provided that the valves to be reassigned have not leaked for the period of the less frequently monitored subgroup (e.g., for the last 12 months, if the valve or group of valves is to be reassigned to a subgroup being monitored annually). Nonrepairable valves may not be reassigned to a less frequently monitored subgroup. [§63.1025(b)(4)(ii)(C)]
  - 3) The permittee shall determine every six months if the overall performance of total valves in the applicable process unit or group of process units is less than 2%

leaking valves and so indicate the performance in the next Periodic Report. If the overall performance of total valves in the applicable process unit or group of process units is 2% leaking valves or greater, the permittee shall no longer subgroup and shall revert to the program required in §63.1025(b)(1) through (b)(3) for that applicable process unit or group of process units. The permittee can again elect to comply with the valve subgrouping procedures of §63.1025(b)(4) if future overall performance of total valves in the process unit or group of process units is again less than 2%. The overall performance of total valves in the applicable process unit or group of process units shall be calculated as a weighted average of the percent leaking valves of each subgroup according to Equation 1:

$$\%V_{LO} = \frac{\sum_{i=1}^n (\%V_{Li} \times V_i)}{\sum_{i=1}^n V_i} \text{ Equation 1}$$

Where:

$\%V_{LO}$  = Overall performance of total valves in the applicable process unit or group of process units

$\%V_{Li}$  = Percent leaking valves in subgroup i, most recent value calculated according to the procedures in §63.1025(c)(1)(ii) and (c)(2).

$V_i$  = Number of valves in subgroup i.

n = Number of subgroups. [§63.1025(b)(4)(iii)]

- 4) The permittee shall maintain records specified in §63.1025(b)(4)(iv)(A) through (b)(4)(iv)(D). [§63.1025(b)(4)(iv)]
    - (i) Which valves are assigned to each subgroup, [§63.1025(b)(4)(iv)(A)]
    - (ii) Monitoring results and calculations made for each subgroup for each monitoring period, [§63.1025(b)(4)(iv)(B)]
    - (iii) Which valves are reassigned, the last monitoring result prior to reassignment, and when they were reassigned, and [§63.1025(b)(4)(iv)(C)]
    - (iv) The results of the semiannual overall performance calculation required in §63.1025(b)(4)(iii). [§63.1025(b)(4)(iv)(D)]
  - 5) The permittee shall notify the Administrator no later than 30 days prior to the beginning of the next monitoring period of the decision to subgroup valves. The notification shall identify the participating process units and the number of valves assigned to each subgroup, if applicable, and may be included in the next Periodic Report. [§63.1025(b)(4)(v)]
  - 6) The permittee shall submit in the periodic reports the information specified in §63.1025(b)(4)(vi)(A) and (b)(4)(vi)(B). [§63.1025(b)(4)(vi)]
    - (i) Total number of valves in each subgroup, and [§63.1025(b)(4)(vi)(A)]
    - (ii) Results of the semiannual overall performance calculation required by §63.1025(b)(4)(iii). [§63.1025(b)(4)(vi)(B)]
  - 7) To determine the monitoring frequency for each subgroup, the calculation procedures of §63.1025(c)(2) shall be used. [§63.1025(b)(4)(vii)]
  - 8) Except for the overall performance calculations required by §63.1025(b)(4)(i) and (iii), each subgroup shall be treated as if it were a process unit for the purposes of applying the provisions of §63.1025. [§63.1025(b)(4)(viii)]
2. *Percent leaking valves calculation* — [§63.1025(c)]
- a) *Calculation basis and procedures.* [§63.1025(c)(1)]

- 1) The permittee shall decide no later than the compliance date of MACT FFFF or upon revision of an operating permit whether to calculate percent leaking valves on a process unit or group of process units basis. Once the permittee has decided, all subsequent percentage calculations shall be made on the same basis, and this shall be the basis used for comparison with the subgrouping criteria specified in §63.1025(b)(4)(i). [§63.1025(c)(1)(i)]
- 2) The percent leaking valves for each monitoring period for each process unit or valve subgroup, as provided in §63.1025(b)(4) of this section, shall be calculated using the following equation:

$$\%V_L = (V_L / V_T) \times 100 \quad \text{Equation 2}$$

Where:

$\%V_L$  = Percent leaking valves.

$V_L$  = Number of valves found leaking, excluding nonrepairable valves, as provided in §63.1025(c)(3), and including those valves found leaking pursuant to §63.1025(d)(2)(iii)(A) and (d)(2)(iii)(B).

$V_T$  = The sum of the total number of valves monitored. [§63.1025(c)(1)(ii)]

- b) *Calculation for monitoring frequency.* When determining monitoring frequency for each process unit or valve subgroup subject to monthly, quarterly, or semiannual monitoring frequencies, the percent leaking valves shall be the arithmetic average of the percent leaking valves from the last two monitoring periods. When determining monitoring frequency for each process unit or valve subgroup subject to annual or biennial (once every two years) monitoring frequencies, the percent leaking valves shall be the arithmetic average of the percent leaking valves from the last three monitoring periods. [§63.1025(c)(2)]
- c) *Nonrepairable valves.* [§63.1025(c)(3)]
  - 1) Nonrepairable valves shall be included in the calculation of percent leaking valves the first time the valve is identified as leaking and nonrepairable and as required to comply with §63.1025(c)(3)(ii). Otherwise, a number of nonrepairable valves (identified and included in the percent leaking valves calculation in a previous period) up to a maximum of 1% of the total number of valves in regulated material service at a process unit or affected facility may be excluded from calculation of percent leaking valves for subsequent monitoring periods. [§63.1025(c)(3)(i)]
  - 2) If the number of nonrepairable valves exceeds 1% of the total number of valves in regulated material service at a process unit or affected facility, the number of nonrepairable valves exceeding 1% of the total number of valves in regulated material service shall be included in the calculation of percent leaking valves. [§63.1025(c)(3)(ii)]
3. *Leak repair.* [§63.1025(d)]
  - a) If a leak is determined pursuant to §63.1025(b), (e)(1), or (e)(2), then the leak shall be repaired using the procedures in §63.1024, as applicable. [§63.1025(d)(1)]
  - b) After a leak has been repaired, the valve shall be monitored at least once within the first three months after its repair. The monitoring required by this paragraph is in addition to the monitoring required to satisfy the definition of repaired and first attempt at repair. [§63.1025(d)(2)]
    - 1) The monitoring shall be conducted as specified in §63.1023(b) and (c), as appropriate, to determine whether the valve has resumed leaking. [§63.1025(d)(2)(i)]

- 2) Periodic monitoring required by §63.1025(b) may be used to satisfy the requirements of this paragraph, if the timing of the monitoring period coincides with the time specified in this paragraph. Alternatively, other monitoring may be performed to satisfy the requirements of this paragraph, regardless of whether the timing of the monitoring period for periodic monitoring coincides with the time specified in this paragraph. [§63.1025(d)(2)(ii)]
- 3) If a leak is detected by monitoring that is conducted pursuant to §63.1025(d)(2), the permittee shall follow the provisions of §63.1025(d)(2)(iii)(A) and (d)(2)(iii)(B), to determine whether that valve must be counted as a leaking valve for purposes of §63.1025(c)(1)(ii). [§63.1025(d)(2)(iii)]
  - (i) If the permittee elected to use periodic monitoring required by §63.1025(b) to satisfy the requirements of §63.1025(d)(2), then the valve shall be counted as a leaking valve. [§63.1025(d)(2)(iii)(A)]
  - (ii) If the permittee elected to use other monitoring, prior to the periodic monitoring required by §63.1025(b), to satisfy the requirements of §63.1025(d)(2), then the valve shall be counted as a leaking valve unless it is repaired and shown by periodic monitoring not to be leaking. [§63.1025(d)(2)(ii)(B)]
4. *Special provisions for valves* — [§63.1025(e)]
  - a) *Unsafe-to-monitor valves*. Any valve that is designated, as described in §63.1022(c)(1), as an unsafe-to-monitor valve is exempt from the requirements of §63.1025(b) and (d)(2) and the permittee shall monitor the valve according to the written plan specified in §63.1022(c)(4). [§63.1025(e)(1)]
  - b) *Difficult-to-monitor valves*. Any valve that is designated, as described in §63.1022(c)(2), as a difficult-to-monitor valve is exempt from the requirements of §63.1025(b) and the permittee shall monitor the valve according to the written plan specified in §63.1022(c)(4). [§63.1025(e)(2)]
  - c) *Fewer than 250 valves*. Any equipment located at a plant site with fewer than 250 valves in regulated material service is exempt from the requirements for monthly monitoring specified in §63.1025(b)(3)(i). Instead, the permittee shall monitor each valve in regulated material service for leaks once each quarter, as provided in §63.1025(e)(1) and (e)(2). [§63.1025(e)(3)]

**MACT UU Pumps in Light Liquid Service Standards:**

1. *Leak detection*. Unless otherwise specified in §63.1021(b), §63.1036, §63.1037, or §63.1026(e), the permittee shall monitor each pump to detect leaks and shall comply with all other provisions of §63.1026. [§63.1026(b)]
  - a) *Monitoring method and frequency*. The pumps shall be monitored monthly to detect leaks by the method specified in §63.1023(b), and, as applicable, §63.1023(c). [§63.1026(b)(1)]
  - b) *Instrument reading that defines a leak*. The instrument reading that defines a leak is specified in §63.1026(b)(2)(i) through (b)(2)(iii). [§63.1026(b)(2)]
    - 1) 5,000 ppm or greater for pumps handling polymerizing monomers; [§63.1026(b)(2)(i)]
    - 2) 2,000 ppm or greater for pumps in food/medical service; and [§63.1026(b)(2)(ii)]
    - 3) 1,000 ppm or greater for all other pumps. [§63.1026(b)(2)(iii)]
  - c) *Leak repair exception*. For pumps to which a 1,000-ppm leak definition applies, repair is not required unless an instrument reading of 2,000 ppm or greater is detected. [§63.1026(b)(3)]

- d) *Visual inspection.* Each pump shall be checked by visual inspection each calendar week for indications of liquid dripping from the pump seal. The permittee shall document that the inspection was conducted and the date of inspection. If there are indications of liquids dripping from the pump seal at the time of the weekly inspection, the permittee shall follow the procedure specified in either §63.1026(b)(4)(i) or (b)(4)(ii).

[§63.1026(b)(4)]

- 1) The permittee shall monitor the pump as specified in §63.1023(b), and, as applicable, §63.1023(c). If the instrument reading indicates a leak as specified in §63.1026(b)(2), a leak is detected and it shall be repaired using the procedures in §63.1024, except as specified in §63.1026(b)(3); or [§63.1026(b)(4)(i)]

- 2) The permittee shall eliminate the visual indications of liquids dripping.

[§63.1026(b)(4)(ii)]

2. *Percent leaking pumps calculation.* [§63.1026(c)]

- a) The permittee shall decide no later than the compliance date of MACT FFFF or upon revision of an operating permit whether to calculate percent leaking pumps on a process unit basis or group of process units basis. Once the permittee has decided, all subsequent percentage calculations shall be made on the same basis. [§63.1026(c)(1)]

- b) If, when calculated on a six-month rolling average, at least the greater of either 10% of the pumps in a process unit or three pumps in a process unit leak, the permittee shall implement a quality improvement program for pumps that complies with the requirements of §63.1035. [§63.1026(c)(2)]

- c) The number of pumps at a process unit or affected facility shall be the sum of all the pumps in regulated material service, except that pumps found leaking in a continuous process unit or affected facility within one month after start-up of the pump shall not count in the percent leaking pumps calculation for that one monitoring period only.

[§63.1026(c)(3)]

- d) Percent leaking pumps shall be determined by the following equation:

$$\%P_L = [(P_L - P_S) / (P_T - P_S)] \times 100 \quad \text{Equation 3}$$

Where:

$\%P_L$  = Percent leaking pumps

$P_L$  = Number of pumps found leaking as determined through monthly monitoring as required in §63.1026(b)(1). Do not include results from inspection of unsafe-to-monitor pumps pursuant to §63.1026(e)(6).

$P_S$  = Number of pumps leaking within one month of start-up during the current monitoring period.

$P_T$  = Total pumps in regulated material service, including those meeting the criteria in §63.1026(e)(1), (e)(2), (e)(3), and (e)(6). [§63.1026(c)(4)]

3. *Leak repair.* If a leak is detected pursuant to §63.1026(b), then the leak shall be repaired using the procedures in §63.1024, as applicable, unless otherwise specified in §63.1026(b)(5) for leaks identified by visual indications of liquids dripping. [§63.1026(d)]

4. *Special provisions for pumps* — [§63.1026(e)]

- a) *Dual mechanical seal pumps.* Each pump equipped with a dual mechanical seal system that includes a barrier fluid system is exempt from the requirements of §63.1026(b), provided the requirements specified in §63.1026(e)(1)(i) through (e)(1)(viii) are met.

[§63.1026(e)(1)]

- 1) The permittee determines, based on design considerations and operating experience, criteria applicable to the presence and frequency of drips and to the

sensor that indicates failure of the seal system, the barrier fluid system, or both. The permittee shall keep records at the plant of the design criteria and an explanation of the design criteria, and any changes to these criteria and the reasons for the changes. This record must be available for review by an inspector.

[§63.1026(e)(1)(i)]

- 2) Each dual mechanical seal system shall meet the requirements specified in §63.1026(e)(1)(ii)(A), (e)(1)(ii)(B), or (e)(1)(ii)(C). [§63.1026(e)(1)(ii)]
    - (i) Each dual mechanical seal system is operated with the barrier fluid at a pressure that is at all times (except periods of startup, shutdown, or malfunction) greater than the pump stuffing box pressure; or [§63.1026(e)(1)(ii)(A)]
    - (ii) Equipped with a barrier fluid degassing reservoir that is routed to a process or fuel gas system or connected by a closed-vent system to a control device that complies with the requirements of either §63.1034 or §63.1021(b); or [§63.1026(e)(1)(ii)(B)]
    - (iii) Equipped with a closed-loop system that purges the barrier fluid into a process stream. [§63.1026(e)(1)(ii)(C)]
  - 3) The barrier fluid is not in light liquid service. [§63.1026(e)(1)(iii)]
  - 4) Each barrier fluid system is equipped with a sensor that will detect failure of the seal system, the barrier fluid system, or both. [§63.1026(e)(1)(iv)]
  - 5) Each pump is checked by visual inspection each calendar week for indications of liquids dripping from the pump seal. The permittee shall document that the inspection was conducted and the date of the inspection. If there are indications of liquids dripping from the pump seal at the time of the weekly inspection, the permittee shall follow the procedure specified in §63.1026(e)(1)(v)(A) or (e)(1)(v)(B) prior to the next required inspection. [§63.1026(e)(1)(v)]
    - (i) The permittee shall monitor the pump as specified in §63.1023(b) and, as applicable, §63.1023(c), to determine if there is a leak of regulated material in the barrier fluid. If an instrument reading of 1,000 ppm or greater is measured, a leak is detected, and it shall be repaired using the procedures in §63.1024; or [§63.1026(e)(1)(v)(A)]
    - (ii) The permittee shall eliminate the visual indications of liquids dripping. [§63.1026(e)(1)(v)(B)]
  - 6) If indications of liquids dripping from the pump seal exceed the criteria established in §63.1026(e)(1)(i), or if based on the criteria established in §63.1026(e)(1)(i) the sensor indicates failure of the seal system, the barrier fluid system, or both, a leak is detected. [§63.1026(e)(1)(vi)]
  - 7) Each sensor as described in §63.1026(e)(1)(iv) is observed daily or is equipped with an alarm unless the pump is located within the boundary of an unmanned plant site. [§63.1026(e)(1)(vii)]
  - 8) When a leak is detected pursuant to §63.1026(e)(1)(vi), it shall be repaired as specified in §63.1024. [§63.1026(e)(1)(viii)]
- b) *No external shaft.* Any pump that is designed with no externally actuated shaft penetrating the pump housing is exempt from the requirements of §63.1026(b). [§63.1026(e)(2)]
- c) *Routed to a process or fuel gas system or equipped with a closed vent system.* Any pump that is routed to a process or fuel gas system or equipped with a closed vent system capable of capturing and transporting leakage from the pump to a control device

meeting the requirements of §63.1034 or §63.1021(b) is exempt from the requirements of §63.1026(b). [§63.1026(e)(3)]

- d) *Unmanned plant site.* Any pump that is located within the boundary of an unmanned plant site is exempt from the weekly visual inspection requirement of §63.1026(b)(4) and (e)(1)(v), and the daily requirements of §63.1026(e)(1)(vii), provided that each pump is visually inspected as often as practical and at least monthly. [§63.1026(e)(4)]
- e) *90% exemption.* If more than 90% of the pumps at a process unit or affected facility meet the criteria in either §63.1026(e)(1) or (e)(2), the process unit or affected facility is exempt from the percent leaking calculation in §63.1026(c). [§63.1026(e)(5)]
- f) *Unsafe-to-monitor pumps.* Any pump that is designated, as described in §63.1022(c)(1), as an unsafe-to-monitor pump is exempt from the requirements of §63.1026(b), the monitoring and inspection requirements of §63.1026(e)(1)(v) through (viii), and the permittee shall monitor and inspect the pump according to the written plan specified in §63.1022(c)(4). [§63.1026(e)(6)]

**MACT UU Connectors in Gas and Vapor Service and in Light Liquid Service Standards:**

1. *Leak detection.* Except as allowed in §63.1021(b), §63.1036, §63.1037, or §63.1027(e), the permittee shall monitor all connectors in gas and vapor and light liquid service as specified in §63.1027(a) and (b)(3). [§63.1027(b)]
  - a) *Monitoring method.* The connectors shall be monitored to detect leaks by the method specified in §63.1023(b) and, as applicable, §63.1023(c). [§63.1027(b)(1)]
  - b) *Instrument reading that defines a leak.* If an instrument reading greater than or equal to 500 ppm is measured, a leak is detected. [§63.1027(b)(2)]
  - c) *Monitoring periods.* The permittee shall perform monitoring, after the initial monitoring required in §63.1027(a), as specified in §63.1027(b)(3)(i) through (b)(3)(iii) and shall comply with the requirements of §63.1027(b)(3)(iv) and (b)(3)(v). The required period in which monitoring must be conducted shall be determined from §63.1027(b)(3)(i) through (b)(3)(iii) using the monitoring results from the preceding monitoring period. The percent leaking connectors shall be calculated as specified in §63.1027(c). [§63.1027(b)(3)]
    - 1) If the percent leaking connectors in the process unit was greater than or equal to 0.5%, then monitor within 12 months (one year). [§63.1027(b)(3)(i)]
    - 2) If the percent leaking connectors in the process unit was greater than or equal to 0.25% but less than 0.5%, then monitor within four years. The permittee may comply with the requirements of this paragraph by monitoring at least 40% of the connectors within two years of the start of the monitoring period, provided all connectors have been monitored by the end of the four-year monitoring period. [§63.1027(b)(3)(ii)]
    - 3) If the percent leaking connectors in the process unit was less than 0.25%, then monitor as provided in §63.1027(b)(3)(iii)(A) and either §63.1027(b)(3)(iii)(B) or (b)(3)(iii)(C), as appropriate. [§63.1027(b)(3)(iii)]
      - (i) A permittee shall monitor at least 50% of the connectors within four years of the start of the monitoring period. [§63.1027(b)(3)(iii)(A)]
      - (ii) If the percent leaking connectors calculated from the monitoring results in §63.1027(b)(3)(iii)(A) is greater than or equal to 0.35% of the monitored connectors, the permittee shall monitor as soon as practical, but within the next six months, all connectors that have not yet been monitored during the monitoring period. At the conclusion of monitoring, a new monitoring period shall be started pursuant to §63.1027(b)(3), based on the percent leaking connectors

- of the total monitored connectors. [§63.1027(b)(3)(iii)(B)]
- (iii) If the percent leaking connectors calculated from the monitoring results in §63.1027(b)(3)(iii)(A) is less than 0.35% of the monitored connectors, the permittee shall monitor all connectors that have not yet been monitored within eight years of the start of the monitoring period. [§63.1027(b)(3)(iii)(C)]
- 4) If, during the monitoring conducted pursuant to §63.1027(b)(3)(i) through (b)(3)(iii), a connector is found to be leaking, it shall be re-monitored once within 90 days after repairs to confirm that it is not leaking. [§63.1027(b)(3)(iv)]
- 5) The permittee shall keep a record of the start date and end date of each monitoring period under §63.1027 for each process unit. [§63.1027(b)(3)(v)]
2. **Percent leaking connectors calculation.** For use in determining the monitoring frequency, as specified in §63.1027(a) and (b)(3), the percent leaking connectors as used in §63.1027(a) and (b)(3) shall be calculated by using Equation 4.

$$\%C_L = C_L / C_t \times 100 \quad \text{Equation 4}$$

Where:

$\%C_L$  = Percent leaking connectors as determined through periodic monitoring required in §63.1027(a) and (b)(3)(i) through (b)(3)(iii).

$C_L$  = Number of connectors measured at 500 ppm or greater, by the method specified in §63.1023(b).

$C_t$  = Total number of monitored connectors in the process unit or affected facility. [§63.1027(c)]

3. **Leak repair.** If a leak is detected pursuant to §63.1027(a) and (b), then the leak shall be repaired using the procedures in §63.1024, as applicable. [§63.1027(d)]
4. **Special provisions for connectors** — [§63.1027(e)]
- a) **Unsafe-to-monitor connectors.** Any connector that is designated, as described in §63.1022(c)(1), as an unsafe-to-monitor connector is exempt from the requirements of §63.1027(a) and (b) and the permittee shall monitor according to the written plan specified in §63.1022(c)(4). [§63.1027(e)(1)]
- b) **Inaccessible, ceramic, or ceramic-lined connectors.** [§63.1027(e)(2)]
- 1) Any connector that is inaccessible or that is ceramic or ceramic-lined (e.g., porcelain, glass, or glass-lined), is exempt from the monitoring requirements of §63.1027(a) and (b), from the leak repair requirements of §63.1027(d), and from the recordkeeping and reporting requirements of §63.1038 and §63.1039. An inaccessible connector is one that meets any of the provisions specified in §63.1027(e)(2)(i)(A) through (e)(2)(i)(F), as applicable. [§63.1027(e)(2)(i)]
- (i) Buried; [§63.1027(e)(2)(i)(A)]
- (ii) Insulated in a manner that prevents access to the connector by a monitor probe; [§63.1027(e)(2)(i)(B)]
- (iii) Obstructed by equipment or piping that prevents access to the connector by a monitor probe; [§63.1027(e)(2)(i)(C)]
- (iv) Unable to be reached from a wheeled scissor-lift or hydraulic-type scaffold that would allow access to connectors up to 7.6 m (25 ft) above the ground. [§63.1027(e)(2)(i)(D)]
- (v) Inaccessible because it would require elevating the monitoring personnel more than 2 m (7 ft) above a permanent support surface or would require the erection of scaffold; [§63.1027(e)(2)(i)(E)]
- (vi) Not able to be accessed at any time in a safe manner to perform monitoring. Unsafe access includes, but is not limited to, the use of a wheeled scissor-lift on

- unstable or uneven terrain, the use of a motorized man-lift basket in areas where an ignition potential exists, or access would require near proximity to hazards such as electrical lines or would risk damage to equipment. [§63.1027(e)(2)(i)(F)]
- 2) If any inaccessible, ceramic, or ceramic-lined connector is observed by visual, audible, olfactory, or other means to be leaking, the visual, audible, olfactory, or other indications of a leak to the atmosphere shall be eliminated as soon as practical. [§63.1027(e)(2)(ii)]

**MACT UU Agitators in Gas and Vapor Service and in Light Liquid Service Standards:**

1. *Leak detection.* [§63.1028(c)]
- a) *Monitoring method.* Each agitator seal shall be monitored monthly to detect leaks by the methods specified in §63.1023(b) and, as applicable, §63.1023(c), except as provided in §63.1021(b), §63.1036, §63.1037, or §63.1028(e). [§63.1028(c)(1)]
- b) *Instrument reading that defines a leak.* If an instrument reading equivalent of 10,000 ppm or greater is measured, a leak is detected. [§63.1028(c)(2)]
- c) *Visual inspection.* [§63.1028(c)(3)]
- 1) Each agitator seal shall be checked by visual inspection each calendar week for indications of liquids dripping from the agitator seal. The permittee shall document that the inspection was conducted and the date of the inspection. [§63.1028(c)(3)(i)]
- 2) If there are indications of liquid dripping from the agitator seal, the permittee shall follow the procedures specified in §63.1028(c)(3)(ii)(A) or (c)(3)(ii)(B) prior to the next required inspection. [§63.1028(c)(3)(ii)]
- (i) The permittee shall monitor the agitator seal as specified in §63.1023(b) and, as applicable, §63.1023(c), to determine if there is a leak of regulated material. If an instrument reading of 10,000 ppm or greater is measured, a leak is detected, and it shall be repaired according to §63.1028(d); or [§63.1028(c)(3)(ii)(A)]
- (ii) The permittee shall eliminate the indications of liquids dripping from the agitator seal. [§63.1028(c)(3)(ii)(B)]
2. *Leak repair.* If a leak is detected, then the leak shall be repaired using the procedures in §63.1024. [§63.1028(d)]
3. *Special provisions for agitators* — [§63.1028(e)]
- a) *Dual mechanical seal.* Each agitator equipped with a dual mechanical seal system that includes a barrier fluid system is exempt from the requirements of §63.1028(c), provided the requirements specified in §63.1028(e)(1)(i) through (e)(1)(vi) are met. [§63.1028(e)(1)]
- 1) Each dual mechanical seal system shall meet the applicable requirements specified in §63.1028(e)(1)(i)(A), (e)(1)(i)(B), or (e)(1)(i)(C). [§63.1028(e)(1)(i)]
- (i) Operated with the barrier fluid at a pressure that is at all times (except during periods of startup, shutdown, or malfunction) greater than the agitator stuffing box pressure; or [§63.1028(e)(1)(i)(A)]
- (ii) Equipped with a barrier fluid degassing reservoir that is routed to a process or fuel gas system or connected by a closed-vent system to a control device that meets the requirements of either §63.1034 or §63.1021(b); or [§63.1028(e)(1)(i)(B)]
- (iii) Equipped with a closed-loop system that purges the barrier fluid into a process stream. [§63.1028(e)(1)(i)(C)]
- 2) The barrier fluid is not in light liquid service. [§63.1028(e)(1)(ii)]
- 3) Each barrier fluid system is equipped with a sensor that will detect failure of the seal

- system, the barrier fluid system, or both. [§63.1028(e)(1)(iii)]
- 4) Each agitator seal is checked by visual inspection each calendar week for indications of liquids dripping from the agitator seal. If there are indications of liquids dripping from the agitator seal at the time of the weekly inspection, the permittee shall follow the procedure specified in §63.1028(e)(1)(iv)(A) or (e)(1)(iv)(B) prior to the next required inspection. [§63.1028(e)(1)(iv)]
    - (i) The permittee shall monitor the agitator seal as specified in §63.1023(b) and, as applicable, §63.1023(c), to determine the presence of regulated material in the barrier fluid. If an instrument reading equivalent to or greater than 10,000 ppm is measured, a leak is detected and it shall be repaired using the procedures in §63.1024, or [§63.1028(e)(1)(iv)(A)]
    - (ii) The permittee shall eliminate the visual indications of liquids dripping. [§63.1028(e)(1)(iv)(B)]
  - 5) Each sensor as described in §63.1028(e)(1)(iii) is observed daily or is equipped with an alarm unless the agitator seal is located within the boundary of an unmanned plant site. [§63.1028(e)(1)(v)]
  - 6) For each dual mechanical seal system, the permittee shall meet the requirements specified in §63.1028(e)(1)(vi)(A) and (e)(1)(vi)(B). [§63.1028(e)(1)(vi)]
    - (i) The permittee shall determine, based on design considerations and operating experience, criteria that indicates failure of the seal system, the barrier fluid system, or both and applicable to the presence and frequency of drips. If indications of liquids dripping from the agitator seal exceed the criteria, or if, based on the criteria the sensor indicates failure of the seal system, the barrier fluid system, or both, a leak is detected and shall be repaired pursuant to §63.1024, as applicable. [§63.1028(e)(1)(vi)(A)]
    - (ii) The permittee shall keep records of the design criteria and an explanation of the design criteria, and any changes to these criteria and the reasons for the changes. [§63.1028(e)(1)(vi)(B)]
  - b) *No external shaft.* Any agitator that is designed with no externally actuated shaft penetrating the agitator housing is exempt from §63.1028(c). [§63.1028(e)(2)]
  - c) *Routed to a process or fuel gas system or equipped with a closed vent system.* Any agitator that is routed to a process or fuel gas system that captures and transports leakage from the agitator to a control device meeting the requirements of either §63.1034 or §63.1021(b) is exempt from the requirements of §63.1028(c). [§63.1028(e)(3)]
  - d) *Unmanned plant site.* Any agitator that is located within the boundary of an unmanned plant site is exempt from the weekly visual inspection requirement of §63.1028(c)(3) and (e)(1)(iv), and the daily requirements of §63.1028(e)(1)(v), provided that each agitator is visually inspected as often as practical and at least monthly. [§63.1028(e)(4)]
  - e) *Difficult-to-monitor agitator seals.* Any agitator seal that is designated, as described in §63.1022(c)(2), as a difficult-to-monitor agitator seal is exempt from the requirements of §63.1028(c) and the permittee shall monitor the agitator seal according to the written plan specified in §63.1022(c)(4). [§63.1028(e)(5)]
  - f) *Equipment obstructions.* Any agitator seal that is obstructed by equipment or piping that prevents access to the agitator by a monitor probe is exempt from the monitoring requirements of §63.1028(c). [§63.1028(e)(6)]
  - g) *Unsafe-to-monitor agitator seals.* Any agitator seal that is designated, as described in §63.1022(c)(1), as an unsafe-to-monitor agitator seal is exempt from the requirements

of §63.1028(c) and the permittee shall monitor the agitator seal according to the written plan specified in §63.1022(c)(4). [§63.1028(e)(7)]

**MACT UU Pressure Relief Devices in Gas and Vapor Service Standards:**

1. *Compliance standard.* Except during pressure releases as provided for in §63.1030(c), or as otherwise specified in §63.1036, §63.1037, or §63.1030(d) and (e), each pressure relief device in gas and vapor service shall be operated with an instrument reading of less than 500 ppm as measured by the method specified in §63.1023(b) and, as applicable, §63.1023(c). [§63.1030(b)]
2. *Pressure relief requirements.* [§63.1030(c)]
  - a) After each pressure release, the pressure relief device shall be returned to a condition indicated by an instrument reading of less than 500 ppm, as soon as practical, but no later than five calendar days after each pressure release, except as provided in §63.1024(d). [§63.1030(c)(1)]
  - b) The pressure relief device shall be monitored no later than five calendar days after the pressure release to confirm the condition indicated by an instrument reading of less than 500 ppm above background, as measured by the method specified in §63.1023(b) and, as applicable, §63.1023(c). [§63.1030(c)(2)]
  - c) The permittee shall record the dates and results of the monitoring required by §63.1030(c)(2) following a pressure release including the background level measured and the maximum instrument reading measured during the monitoring. [§63.1030(c)(3)]
3. *Pressure relief devices routed to a process or fuel gas system or equipped with a closed vent system and control device.* Any pressure relief device that is routed to a process or fuel gas system or equipped with a closed vent system capable of capturing and transporting leakage from the pressure relief device to a control device meeting the requirements of §63.1034 is exempt from the requirements of §63.1030(b) and (c). [§63.1030(d)]
4. *Rupture disk exemption.* Any pressure relief device that is equipped with a rupture disk upstream of the pressure relief device is exempt from the requirements of §63.1030(b) and (c) provided the permittee installs a replacement rupture disk upstream of the pressure relief device as soon as practical after each release but no later than five calendar days after each pressure release, except as provided in §63.1024(d). [§63.1030(e)]

**MACT UU Sampling Connection Systems Standards:**

1. *Equipment requirement.* Each sampling connection system shall be equipped with a closed-purge, closed-loop, or closed vent system, except as provided in §63.1021(b), §63.1036, §63.1037, or §63.1032(d). Gases displaced during filling of the sample container are not required to be collected or captured. [§63.1032(b)]
2. *Equipment design and operation.* Each closed-purge, closed-loop, or closed vent system as required in §63.1032(b) shall meet the applicable requirements specified in §63.1032(c)(1) through (c)(5). [§63.1032(c)]
  - a) The system shall return the purged process fluid directly to a process line or to a fuel gas system that meets the requirements of either §63.1034 or §63.1021(b); or [§63.1032(c)(1)]
  - b) Be designed and operated to capture and transport all the purged process fluid to a control device that meets the requirements of either §63.1034 or §63.1021(b); or [§63.1032(c)(3)]
  - c) Collect, store, and transport the purged process fluid to a system or facility identified in

§63.1032(c)(4)(i), (c)(4)(ii), or (c)(4)(iii). [§63.1032(c)(4)]

- i) A waste management unit as defined in §63.111, if the waste management unit is subject to and operating in compliance with the provisions of MACT G, applicable to group 1 wastewater streams. If the purged process fluid does not contain any regulated material listed in Table 9 of MACT G, the waste management unit need not be subject to, and operated in compliance with the requirements of MACT G, applicable to group 1 wastewater streams provided the facility has a National Pollution Discharge Elimination System (NPDES) permit or sends the wastewater to an NPDES-permitted facility. [§63.1032(c)(4)(i)]
  - ii) A treatment, storage, or disposal facility subject to regulation under 40 CFR Parts 262, 264, 265, or 266; or [§63.1032(c)(4)(ii)]
  - iii) A facility permitted, licensed, or registered by a State to manage municipal or industrial solid waste, if the process fluids are not hazardous waste as defined in 40 CFR Part 261. [§63.1032(c)(4)(iii)]
  - d) Containers that are part of a closed purge system must be covered or closed when not being filled or emptied. [§63.1032(c)(5)]
3. *In-situ sampling systems.* In-situ sampling systems and sampling systems without purges are exempt from the requirements of §63.1032(b) and (c). [§63.1032(d)]

**MACT UU Open-ended Valves or Lines Standards:**

1. *Equipment and operational requirements.* [§63.1033(b)]
  - a) Each open-ended valve or line shall be equipped with a cap, blind flange, plug, or a second valve, except as provided in §63.1021(b), §63.1036, §63.1037, and §63.1033(c) and (d). The cap, blind flange, plug, or second valve shall seal the open end at all times except during operations requiring process fluid flow through the open-ended valve or line, or during maintenance. The operational provisions of §63.1033(b)(2) and (b)(3) also apply. [§63.1033(b)(1)]
  - b) Each open-ended valve or line equipped with a second valve shall be operated in a manner such that the valve on the process fluid end is closed before the second valve is closed. [§63.1033(b)(2)]
  - c) When a double block and bleed system is being used, the bleed valve or line may remain open during operations that require venting the line between the block valves but shall comply with §63.1033(b)(1) at all other times. [§63.1033(b)(3)]
2. *Emergency shutdown exemption.* Open-ended valves or lines in an emergency shutdown system that are designed to open automatically in the event of a process upset are exempt from the requirements of §63.1033(b). [§63.1033(c)]
3. *Polymerizing materials exemption.* Open-ended valves or lines containing materials that would autocatalytically polymerize or, would present an explosion, serious overpressure, or other safety hazard if capped or equipped with a double block and bleed system as specified in §63.1033(b) are exempt from the requirements of §63.1033(b). [§63.1033(d)]

**MACT UU Quality Improvement Program for Pumps:**

1. *Criteria.* If, on a six-month rolling average, at least the greater of either 10% of the pumps in a process unit or affected facility (or plant site) or three pumps in a process unit or affected facility (or plant site) leak, the permittee shall comply with the requirements specified in §63.1035(a)(1) and (a)(2). [§63.1035(a)]

- a) Pumps that are in food and medical service or in polymerizing monomer service shall comply with all requirements except for those specified in §63.1035(d)(8).  
[§63.1035(a)(1)]
- b) Pumps that are not in food and medical or polymerizing monomer service shall comply with all the requirements of §63.1035. [§63.1035(a)(2)]
2. *Exiting the QIP.* The permittee shall comply with the requirements of §63.1035 until the number of leaking pumps is less than the greater of either 10% of the pumps or three pumps, calculated as a six-month rolling average, in the process unit or affected facility (or plant site). Once the performance level is achieved, the permittee shall comply with the requirements in § 63.1026. [§63.1035(b)]
3. *Resumption of QIP.* If, in a subsequent monitoring period, the process unit or affected facility (or plant site) has greater than either 10% of the pumps leaking or three pumps leaking (calculated as a six-month rolling average), the permittee shall resume the quality improvement program starting at performance trials. [§63.1035(c)]
4. *QIP requirements.* The quality improvement program shall meet the requirements specified in §63.1035(d)(1) through (d)(8). [§63.1035(d)]
  - a) The permittee shall comply with the requirements in §63.1026. [§63.1035(d)(1)]
  - b) *Data collection.* The permittee shall collect the data specified in §63.1035(d)(2)(i) through (d)(2)(v) and maintain records for each pump in each process unit or affected facility (or plant site) subject to the quality improvement program. The data may be collected, and the records may be maintained on a process unit, affected facility, or plant site basis. [§63.1035(d)(2)]
    - i) Pump type (e.g., piston, horizontal or vertical centrifugal, gear, bellows); pump manufacturer; seal type and manufacturer; pump design (e.g., external shaft, flanged body); materials of construction; if applicable, barrier fluid or packing material; and year installed. [§63.1035(d)(2)(i)]
    - ii) Service characteristics of the stream such as discharge pressure, temperature, flow rate, corrosivity, and annual operating hours. [§63.1035(d)(2)(ii)]
    - iii) The maximum instrument readings observed in each monitoring observation before repair, response factor for the stream if appropriate, instrument model number, and date of the observation. [§63.1035(d)(2)(iii)]
    - iv) If a leak is detected, the repair methods used and the instrument readings after repair. [§63.1035(d)(2)(iv)]
    - v) If the data will be analyzed as part of a larger analysis program involving data from other plants or other types of process units or affected facilities, a description of any maintenance or quality assurance programs used in the process unit or affected facility that are intended to improve emission performance. [§63.1035(d)(2)(v)]
  - c) The permittee shall continue to collect data on the pumps as long as the process unit or affected facility (or plant site) remains in the quality improvement program.  
[§63.1035(d)(3)]
  - d) *Pump or pump seal inspection.* The permittee shall inspect all pumps or pump seals that exhibited frequent seal failures and were removed from the process unit or affected facility due to leaks. The inspection shall determine the probable cause of the pump seal failure or of the pump leak and shall include recommendations, as appropriate, for design changes or changes in specifications to reduce leak potential. [§63.1035(d)(4)]
  - e) *Data analysis.* The permittee shall analyze the data collected to comply with the requirements of §63.1035(d)(2) to determine the services, operating or maintenance practices, and pump or pump seal designs or technologies that have poorer than

average emission performance and those that have better than average emission performance. The analysis shall determine if specific trouble areas can be identified based on service, operating conditions or maintenance practices, equipment design, or other process-specific factors. [§63.1035(d)(5)(i)]

- f) The analysis shall also be used to determine if there are superior performing pump or pump seal technologies that are applicable to the service(s), operating conditions, or pump or pump seal designs associated with poorer than average emission performance. A superior performing pump or pump seal technology is one with a leak frequency of less than 10% for specific applications in the process unit, affected facility, or plant site. A candidate superior performing pump or pump seal technology is one demonstrated or reported in the available literature or through a group study as having low emission performance and as being capable of achieving less than 10% leaking pumps in the process unit or affected facility (or plant site). [§63.1035(d)(5)(ii)]
- g) The analysis shall include consideration of the information specified in §63.1035(d)(5)(iii)(A) through (d)(5)(iii)(C). [§63.1035(d)(5)(iii)]
  - i) The data obtained from the inspections of pumps and pump seals removed from the process unit or affected facility due to leaks; [§63.1035(d)(5)(iii)(A)]
  - ii) Information from the available literature and from the experience of other plant sites that will identify pump designs or technologies and operating conditions associated with low emission performance for specific services; and [§63.1035(d)(5)(iii)(B)]
  - iii) Information on limitations on the service conditions for the pump seal technology operating conditions as well as information on maintenance procedures to ensure continued low emission performance. [§63.1035(d)(5)(iii)(C)]
- h) The data analysis may be conducted through an inter- or intra-company program (or through some combination of the two approaches) and may be for a single process unit, a plant site, a company, or a group of process units. [§63.1035(d)(5)(iv)]
- i) The first analysis of the data shall be completed no later than 18 months after the start of the quality improvement program. The first analysis shall be performed using data collected for a minimum of six months. An analysis of the data shall be done each year the process unit or affected facility is in the quality improvement program. [§63.1035(d)(5)(v)]
- j) *Trial evaluation program.* A trial evaluation program shall be conducted at each plant site for which the data analysis does not identify use of superior performing pump seal technology or pumps that can be applied to the areas identified as having poorer than average performance, except as provided in §63.1035(d)(6)(v). The trial program shall be used to evaluate the feasibility of using in the process unit or affected facility (or plant site) the pump designs or seal technologies and operating and maintenance practices that have been identified by others as having low emission performance. [§63.1035(d)(6)]
  - i) The trial evaluation program shall include on-line trials of pump seal technologies or pump designs and operating and maintenance practices that have been identified in the available literature or in analysis by others as having the ability to perform with leak rates below 10% in similar services, as having low probability of failure, or as having no external actuating mechanism in contact with the process fluid. If any of the candidate superior performing pump seal technologies or pumps is not included in the performance trials, the reasons for rejecting specific technologies from consideration shall be documented as required in §63.1035(e)(3)(ii). [§63.1035(d)(6)(i)]

- ii) The number of pump seal technologies or pumps in the trial evaluation program shall be the lesser of 1% or two pumps for programs involving single process units or affected facilities and the lesser of 1% or five pumps for programs involving a plant site or groups of process units or affected facilities. The minimum number of pumps or pump seal technologies in a trial program shall be one. [§63.1035(d)(6)(ii)]
- iii) The trial evaluation program shall specify and include documentation of the information specified in §63.1035(d)(6)(iii)(A) through (d)(6)(iii)(D). [§63.1035(d)(6)(iii)]
  - (1) The candidate superior performing pump seal designs or technologies to be evaluated, the stages for evaluating the identified candidate pump designs or pump seal technologies, including the time period necessary to test the applicability; [§63.1035(d)(6)(iii)(A)]
  - (2) The frequency of monitoring or inspection of the equipment; [§63.1035(d)(6)(iii)(B)]
  - (3) The range of operating conditions over which the component will be evaluated; and [§63.1035(d)(6)(iii)(C)]
  - (4) Conclusions regarding the emission performance and the appropriate operating conditions and services for the trial pump seal technologies or pumps. [§63.1035(d)(6)(iii)(D)]
- iv) The performance trials shall initially be conducted, at least, for a six-month period beginning not later than 18 months after the start of the quality improvement program. No later than 24 months after the start of the quality improvement program, the permittee shall have identified pump seal technologies or pump designs that, combined with appropriate process, operating, and maintenance practices, operate with low emission performance for specific applications in the process unit or affected facility. The permittee shall continue to conduct performance trials as long as no superior performing design or technology has been identified, except as provided in §63.1035(d)(6)(vi). The initial list of superior emission performance pump designs or pump seal technologies shall be amended in the future, as appropriate, as additional information and experience are obtained. [§63.1035(d)(6)(iv)]
- v) Any plant site with fewer than 400 valves and owned by a corporation with fewer than 100 employees shall be exempt from trial evaluations of pump seals or pump designs. Plant sites exempt from the trial evaluations of pumps shall begin the pump seal or pump replacement program at the start of the fourth year of the quality improvement program. [§63.1035(d)(6)(v)]
- vi) If the permittee has conducted performance trials on all alternative superior emission performance technologies suitable for the required applications in the process unit or affected facility, the permittee may stop conducting performance trials provided that a superior performing design or technology has been demonstrated or there are no technically feasible alternative superior technologies remaining. The permittee shall prepare an engineering evaluation documenting the physical, chemical, or engineering basis for the judgment that the superior emission performance technology is technically infeasible or demonstrating that it would not reduce emissions. [§63.1035(d)(6)(vi)]
- k) *Quality assurance program.* The permittee shall prepare and implement a pump quality assurance program that details purchasing specifications and maintenance procedures for all pumps and pump seals in the process unit or affected facility. The quality assurance program may establish any number of categories, or classes, of pumps as

needed to distinguish among operating conditions and services associated with poorer than average emission performance as well as those associated with better than average emission performance. The quality assurance program shall be developed considering the findings of the data analysis required under §63.1035(d)(5); and, if applicable, the findings of the trial evaluation required in §63.1035(d)(6); and the operating conditions in the process unit or affected facility. The quality assurance program shall be updated each year as long as the process unit or affected facility has the greater of either 10% or more leaking pumps or has three leaking pumps.

[§63.1035(d)(7)]

- i) The quality assurance program shall meet the requirements specified in §63.1035(d)(7)(i)(A) through (d)(7)(i)(D). [§63.1035(d)(7)(i)]
  - (1) Establish minimum design standards for each category of pumps or pump seal technology. The design standards shall specify known critical parameters such as tolerance, manufacturer, materials of construction, previous usage, or other applicable identified critical parameters; [§63.1035(d)(7)(i)(A)]
  - (2) Require that all equipment orders specify the design standard (or minimum tolerances) for the pump or the pump seal; [§63.1035(d)(7)(i)(B)]
  - (3) Provide for an audit procedure for quality control of purchased equipment to ensure conformance with purchase specifications. The audit program may be conducted by the permittee of the plant site or process unit or affected facility, or by a designated representative; and [§63.1035(d)(7)(i)(C)]
  - (4) Detail off-line pump maintenance and repair procedures. These procedures shall include provisions to ensure that rebuilt or refurbished pumps and pump seals will meet the design specifications for the pump category and will operate so that emissions are minimized. [§63.1035(d)(7)(i)(D)]
- ii) The quality assurance program shall be established no later than the start of the third year of the quality improvement program for plant sites with 400 or more valves or 100 or more employees; and no later than the start of the fourth year of the quality improvement program for plant sites with less than 400 valves and less than 100 employees. [§63.1035(d)(7)(ii)]
- l) *Pump or pump seal replacement.* Three years after the start of the quality improvement program for plant sites with 400 or more valves or 100 or more employees and at the start of the fourth year of the quality improvement program for plant sites with less than 400 valves and less than 100 employees, the permittee shall replace, as described in §63.1035(d)(8)(i) and (d)(8)(ii), the pumps or pump seals that are not superior emission performance technology with pumps or pump seals that have been identified as superior emission performance technology and that comply with the quality assurance standards for the pump category. Superior emission performance technology is that category or design of pumps or pump seals with emission performance that when combined with appropriate process, operating, and maintenance practices, will result in less than 10% leaking pumps for specific applications in the process unit, affected facility, or plant site. Superior emission performance technology includes material or design changes to the existing pump, pump seal, seal support system, installation of multiple mechanical seals or equivalent, or pump replacement. [§63.1035(d)(8)]
  - i) Pumps or pump seals shall be replaced at the rate of 20% per year based on the total number of pumps in light liquid service. The calculated value shall be rounded to the nearest nonzero integer value. The minimum number of pumps or pump seals shall be one. Pump replacement shall continue until all pumps subject to the

requirements of §63.1026 are pumps determined to be superior performance technology. [§63.1035(d)(8)(i)]

- ii) The permittee may delay replacement of pump seals or pumps with superior technology until the next planned process unit or affected facility shutdown, provided the number of pump seals and pumps replaced is equivalent to the 20% or greater annual replacement rate. [§63.1035(d)(8)(ii)]
- iii) The pumps shall be maintained as specified in the quality assurance program. [§63.1035(d)(8)(iii)]

5. *QIP recordkeeping.* In addition to the records required by §63.1035(d)(2), the permittee shall maintain records for the period of the quality improvement program for the process unit or affected facility as specified in §63.1035(e)(1) through (e)(6). [§63.1035(e)]
- a) When using a pump quality improvement program as specified in §63.1035, record the information specified in §63.1035(e)(1)(i) through (e)(1)(iii). [§63.1035(e)(1)]
    - i) The rolling average percent leaking pumps. [§63.1035(e)(1)(i)]
    - ii) Documentation of all inspections conducted under the requirements of §63.1035(d)(4), and any recommendations for design or specification changes to reduce leak frequency. [§63.1035(e)(1)(ii)]
    - iii) The beginning and ending dates while meeting the requirements of §63.1035(d). [§63.1035(e)(1)(iii)]
  - b) If a leak is not repaired within 15 calendar days after discovery of the leak, the reason for the delay and the expected date of successful repair. [§63.1035(e)(2)]
  - c) Records of all analyses required in §63.1035(d). The records will include the information specified in §63.1035(e)(3)(i) through (e)(3)(iv). [§63.1035(e)(3)]
    - i) A list identifying areas associated with poorer than average performance and the associated service characteristics of the stream, the operating conditions and maintenance practices. [§63.1035(e)(3)(i)]
    - ii) The reasons for rejecting specific candidate superior emission performing pump technology from performance trials. [§63.1035(e)(3)(ii)]
    - iii) The list of candidate superior emission performing valve or pump technologies, and documentation of the performance trial program items required under §63.1035(d)(6)(iii). [§63.1035(e)(3)(iii)]
    - iv) The beginning date and duration of performance trials of each candidate superior emission performing technology. [§63.1035(e)(3)(iv)]
  - d) All records documenting the quality assurance program for pumps as specified in §63.1035(d)(7), including records indicating that all pumps replaced or modified during the period of the quality improvement program comply with the quality assurance. [§63.1035(e)(4)]
  - e) Records documenting compliance with the 20% or greater annual replacement rate for pumps as specified in §63.1035(d)(8). [§63.1035(e)(5)]
  - f) Information and data to show the corporation has fewer than 100 employees, including employees providing professional and technical contracted services. [§63.1035(e)(6)]

**MACT SS Testing:**

- 1. *Performance tests.* Where §63.990 require, or the permittee elects to conduct, a performance test of a control device, the requirements of §63.997(b) through (d) apply. [§63.997(a)]
- 2. *Prior test results and waivers.* Initial performance tests are required only as specified in MACT SS or MACT FFFF. [§63.997(b)]

- a) Unless requested by the Administrator, the permittee is not required to conduct a performance test under MACT SS if a prior performance test was conducted using the same methods specified in §63.1257(b)(8), and either no process changes have been made since the test, or the permittee can demonstrate that the results of the performance test, with or without adjustments, reliably demonstrate compliance despite process changes. The permittee may request permission to substitute a prior performance test by written application to the Administrator as specified in §63.999(a)(1)(iv). [§63.997(b)(1) and §63.2460(c)(2)(ii)]
  - b) Individual performance tests may be waived upon written application to the Administrator, per §63.999(a)(1)(iii), if, in the Administrator's judgment, the source is meeting the relevant standard(s) on a continuous basis, the source is being operated under an extension or waiver of compliance, or the permittee has requested an extension or waiver of compliance and the Administrator is still considering that request. [§63.997(b)(2)]
  - c) Approval of any waiver granted under §63.997 shall not abrogate the Administrator's authority under the Act or in any way prohibit the Administrator from later canceling the waiver. The cancellation will be made only after notification is given to the permittee. [§63.997(b)(3)]
3. *Performance tests schedule.* [§63.997(c)]
- a) The Administrator may require the permittee to conduct performance tests at the regulated source at any time when the action is authorized by §114 of the Act. [§63.997(c)(2)]
  - b) If the permittee elects to use a recovery device to replace an existing control device at a later date, or elects to use a different flare, nonflare control device, or recovery device to replace an existing flare, nonflare control device, or final recovery device at a later date, the permittee shall notify the Administrator, either by amendment of the regulated source's title V permit or, if title V is not applicable, by submission of the notice specified in §63.999(c)(7) before implementing the change. Upon implementing the change, a compliance demonstration or performance test shall be performed according to the provisions of §63.997(c)(3)(i) through (v), as applicable, within 180 days. The compliance assessment report shall be submitted to the Administrator within 60 days of completing the determination, as provided in §63.999(a)(1)(ii). [§63.997(c)(3)]
4. *Performance testing facilities.* If required to do performance testing, the permittee shall provide performance testing facilities as specified in §63.997(d)(1) through (5). [§63.997(d)]
5. *Performance test procedures.* Where §63.990 require the permittee to conduct a performance test of a control device, the permittee shall follow the requirements of §63.997(e)(1)(v), as applicable. [§63.997(e), §63.2450(e)(4)(vii), and §63.2460(c)(2)(ii)]
- a) *Specific procedures.* Where §63.990 require the permittee to conduct a performance test of a control device, the permittee shall conduct that performance test using the procedures in §63.997(e)(2)(i) through (iv), as applicable. The regulated material concentration and percent reduction may be measured as either total organic regulated material or as TOC minus methane and ethane according to the procedures specified. [§63.997(e)(2)]

**MACT FFFF Recordkeeping:**

1. The permittee must keep the records specified in §63.2525(a) through (t). [§63.2525]
  - a) Except as specified in §63.2450(e)(4), §63.2480(f), §63.2485(p) and (q) and §63.2525(t), each applicable record required by 40 CFR Part 63, Subpart A and in

- referenced MACT F, MACT G, MACT SS, MACT UU, MACT WW, and MACT GGG and in referenced 40 CFR Part 65, Subpart F. [§63.2525(a)]
- b) Records of each operating scenario as specified in §63.2525(b)(1) through (8). [§63.2525(b)]
- 1) A description of the process and the type of process equipment used. [§63.2525(b)(1)]
  - 2) An identification of related process vents, including their associated emissions episodes if not complying with the alternative standard in §63.2505; wastewater point of determination (POD); storage tanks; and transfer racks. [§63.2525(b)(2)]
  - 3) The applicable control requirements of MACT FFFF, including the level of required control, and for vents, the level of control for each vent. [§63.2525(b)(3)]
  - 4) The control device or treatment process used, as applicable, including a description of operating and/or testing conditions for any associated control device. [§63.2525(b)(4)]
  - 5) The process vents, wastewater POD, transfer racks, and storage tanks (including those from other processes) that are simultaneously routed to the control device or treatment process(s). [§63.2525(b)(5)]
  - 6) The applicable monitoring requirements of MACT FFFF and any parametric level that assures compliance for all emissions routed to the control device or treatment process. [§63.2525(b)(6)]
  - 7) Calculations and engineering analyses required to demonstrate compliance. [§63.2525(b)(7)]
  - 8) For reporting purposes, a change to any of these elements not previously reported, except for §63.2525(b)(5), constitutes a new operating scenario. [§63.2525(b)(8)]
- c) A schedule or log of operating scenarios for processes with batch vents from batch operations updated each time a different operating scenario is put into effect [§63.2525(c)]
- d) The information specified in §63.2525(d)(1) and (2) for Group 1 batch process vents in compliance with a percent reduction emission limit in Table 2 to MACT FFFF if some of the vents are controlled to less the percent reduction requirements. [§63.2525(d)]
- 1) Records of whether each batch operated was considered a standard batch. [§63.2525(d)(1)]
  - 2) The estimated uncontrolled and controlled emissions for each batch that is considered to be a nonstandard batch. [§63.2525(d)(2)]
- e) A record of each time a safety device is opened to avoid unsafe conditions in accordance with §63.2450(p). [§63.2525(f)]
- f) Records of the results of each CPMS calibration check and the maintenance performed, as specified in §63.2450(k)(1). [§63.2525(g)]
- g) For each PUG, the permittee must keep records specified in §63.2525(i)(1) through (5). [§63.2525(i)]
- 1) Descriptions of the MCPU and other process units in the initial PUG required by §63.2535(l)(1)(v). [§63.2525(i)(1)]
  - 2) Rationale for including each MCPU and other process units in the initial PUG (*i.e.*, identify the overlapping equipment between process units) required by §63.2535(l)(1)(v). [§63.2525(i)(2)]
  - 3) Calculations used to determine the primary product for the initial PUG required by §63.2535(l)(2)(iv). [§63.2525(i)(3)]
  - 4) Descriptions of process units added to the PUG after the creation date and rationale

- for including the additional process units in the PUG as required by §63.2535(l)(1)(v). [§63.2525(i)(4)]
- 5) The calculation of each primary product redetermination required by §63.2535(l)(2)(iv). [§63.2525(i)(5)]
- h) §63.2525(h) no longer applies. Instead, for each deviation from an emission limit, operating limit, or work practice standard, the permittee must keep a record of the information specified in §63.2525(l)(1) through (3). The records shall be maintained as specified in §63.10(b)(1). [§63.2525(l)]
- 1) If an affected unit does not meet an applicable standard, record the number of deviations. For each deviation record the date, time, and duration of each deviation. [§63.2525(l)(1)]
  - 2) For each deviation from an applicable standard, record and retain a list of the affected sources or equipment, an estimate of the quantity of each regulated pollutant emitted over an emission limit and a description of the method used to estimate the emissions. [§63.2525(l)(2)]
  - 3) Record any actions taken to minimize emissions in accordance with §63.2450(u) and any corrective actions taken to return the affected unit to its normal or usual manner of operation. [§63.2525(l)(3)]
- i) For each flow event from a bypass line subject to the requirements in §63.2450(e)(6), the permittee must maintain records sufficient to determine whether the detected flow included flow requiring control. For each flow event from a bypass line requiring control that is released either directly to the atmosphere or to a control device not meeting the requirements specified in Tables 1 through 7 to MACT FFFF, the permittee must include an estimate of the volume of gas, the concentration of organic HAP in the gas and the resulting emissions of organic HAP that bypassed the control device using process knowledge and engineering estimates. [§63.2525(n)]
- j) For each maintenance vent opening subject to the requirements in §63.2450(v), the permittee must keep the applicable records specified in §63.2525(p)(1) through (5). [§63.2525(p)]
- 1) The permittee must maintain standard site procedures used to deinventory equipment for safety purposes (e.g., hot work or vessel entry procedures) to document the procedures used to meet the requirements in §63.2450(v). The current copy of the procedures must be retained and available on-site at all times. Previous versions of the standard site procedures, as applicable, must be retained for five years. [§63.2525(p)(1)]
  - 2) If complying with the requirements of §63.2450(v)(1)(i) and the lower explosive limit at the time of the vessel opening exceeds 10%, identification of the maintenance vent, the process units or equipment associated with the maintenance vent, the date of maintenance vent opening, and the lower explosive limit at the time of the vessel opening. [§63.2525(p)(2)]
  - 3) If complying with the requirements of §63.2450(v)(1)(ii) and either the vessel pressure at the time of the vessel opening exceeds 5 psig or the lower explosive limit at the time of the active purging was initiated exceeds 10%, identification of the maintenance vent, the process units or equipment associated with the maintenance vent, the date of maintenance vent opening, the pressure of the vessel or equipment at the time of discharge to the atmosphere and, if applicable, the lower explosive limit of the vapors in the equipment when active purging was initiated. [§63.2525(p)(3)]

- 4) If complying with the requirements of §63.2450(v)(1)(iii), records of the estimating procedures used to determine the total quantity of VOC in the equipment and the type and size limits of equipment that contain less than 50 pounds of VOC at the time of maintenance vent opening. For each maintenance vent opening that contains greater than 50 pounds of VOC for which the deinventory procedures specified in §63.2525(p)(1) are not followed or for which the equipment opened exceeds the type and size limits established in the records specified in this paragraph, records that identify the maintenance vent, the process units or equipment associated with the maintenance vent, the date of maintenance vent opening, and records used to estimate the total quantity of VOC in the equipment at the time the maintenance vent was opened to the atmosphere. [§63.2525(p)(4)]
- 5) If complying with the requirements of §63.2450(v)(1)(iv), identification of the maintenance vent, the process units or equipment associated with the maintenance vent, records documenting actions taken to comply with other applicable alternatives and why utilization of this alternative was required, the date of maintenance vent opening, the equipment pressure and lower explosive limit of the vapors in the equipment at the time of discharge, an indication of whether active purging was performed and the pressure of the equipment during the installation or removal of the blind if active purging was used, the duration the maintenance vent was open during the blind installation or removal process, and records used to estimate the total quantity of VOC in the equipment at the time the maintenance vent was opened to the atmosphere for each applicable maintenance vent opening. [§63.2525(p)(5)]
- k) For each pressure relief device subject to the pressure release managements work practice standards in §63.2480(e), the permittee must keep the records specified in §63.2525(q)(1) through (3). [§63.2525(q)]
  - 1) Records of the prevention measures implemented as required in §63.2480(e)(3)(ii). [§63.2525(q)(1)]
  - 2) Records of the number of releases during each calendar year and the number of those releases for which the root cause was determined to be a *force majeure event*. Keep these records for the current calendar year and the past five calendar years. [§63.2525(q)(2)]
  - 3) For each release to the atmosphere, the permittee must keep the records specified in §63.2525(q)(3)(i) through (iv). [§63.2525(q)(3)]
    - (i) The start and end time and date of each pressure release to the atmosphere. [§63.2525(q)(3)(i)]
    - (ii) Records of any data, assumptions, and calculation used to estimate of the mass quantity of each organic HAP released during the event. [§63.2525(q)(3)(ii)]
    - (iii) Records of the root cause analysis and corrective action analysis conducted as required in §63.2480(e)(3)(iii), including an identification of the affected facility, a statement noting whether the event resulted from the same root cause(s) identified in a previous analysis and either a description of the recommended corrective action(s) or an explanation of why corrective action is not necessary under §63.2480(e)(7)(i). [§63.2525(q)(3)(iii)]
    - (iv) For any corrective action analysis for which implementation of corrective actions are required in §63.2480(e)(7), a description of the corrective action(s) completed within the first 45 days following the discharge and, for action(s) not already completed, a schedule for implementation, including proposed commencement

- and completion dates. [§63.2525(q)(3)(iv)]
- l) Any records required to be maintained by 40 CFR Part 63 that are submitted electronically via EPA's CEDRI may be maintained in electronic format. This ability to maintain electronic copies does not affect the requirement for facilities to make records, data, and reports available upon request to a delegated air agency or EPA as part of an on-site compliance evaluation. [§63.2525(t)]
  - m) The referenced provisions specified in §63.2525(u)(1) through (8) do not apply when demonstrating compliance with §63.2525(a). [§63.2525(u)]
    - 1) §63.103(c)(2)(i). [§63.2525(u)(1)]
    - 2) §63.103(c)(2)(ii). [§63.2525(u)(2)]
    - 3) The phrase "start-up, shutdown and malfunction and" from §63.103(c)(3). [§63.2525(u)(3)]
    - 4) The phrase "other than startups, shutdowns, or malfunctions (e.g., a temperature reading of -200°C on a boiler)," from §63.152(g)(1)(i). [§63.2525(u)(4)]
    - 5) The phrase "other than a startup, shutdown, or malfunction" from §63.152(g)(1)(ii)(C). [§63.2525(u)(5)]
    - 6) The phrase "other than startups, shutdowns, or malfunctions" from §63.152(g)(1)(iii). [§63.2525(u)(6)]
    - 7) The phrase "other than a startup, shutdown, or malfunction" from §63.152(g)(2)(iii). [§63.2525(u)(7)]
    - 8) §63.152(g)(2)(iv)(A). [§63.2525(u)(8)]

**General Recordkeeping Requirements:**

- 1. The permittee shall maintain all records required by this permit for not less than five years and shall make them available to Missouri Department of Natural Resources' personnel upon request. [10 CSR 10-6.065(5)(C)1.C(II)(b)I and 10 CSR 10-6.065(5)(C)1.G(V)]

**MACT SS Recordkeeping Requirements:**

- 1. *Compliance assessment, monitoring, and compliance records* — [§63.998(a)]
  - a) *Nonflare control device performance test records*. [§63.998(a)(2)]
    - 1) *Availability of performance test records*. Upon request, the permittee shall make available to the Administrator such records as may be necessary to determine the conditions of performance tests performed pursuant to §63.990(b). [§63.998(a)(2)(i)]
    - 2) *Nonflare control device performance test records*. [§63.998(a)(2)(ii)]
      - (i) *General requirements*. The permittee shall keep up-to-date, readily accessible continuous records of the data specified in §63.998(a)(2)(ii)(B) through (C), as applicable, measured during each performance test performed pursuant to §63.990(b), and include that data in the Notification of Compliance Status required under §63.999(b). The same data specified in §63.998 shall be submitted in the reports of all subsequently required performance tests where either the emission control efficiency of a combustion device, or the outlet concentration of TOC or regulated material is determined. [§63.998(a)(2)(ii)(A)]
      - (ii) *Other nonflare control devices*. Where the permittee seeks to use an absorber, condenser, or carbon adsorber as a control device, the information specified in §63.998(a)(2)(ii)(C)(1) through (5) shall be recorded, as applicable. [§63.998(a)(2)(ii)(C)]
        - (a) Where an absorber is used as the control device, the exit specific gravity and average exit temperature of the absorbing liquid averaged over the same time

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- period as the performance test (both measured while the vent stream is normally routed and constituted); or [§63.998(a)(2)(ii)(C)(1)]
- (b) Where a condenser is used as the control device, the average exit (product side) temperature averaged over the same time period as the performance test while the vent stream is routed and constituted normally; or [§63.998(a)(2)(ii)(C)(2)]
- (c) As an alternative to §63.998(a)(2)(ii)(C)(1), (2), or (3), the concentration level or reading indicated by an organic monitoring device at the outlet of the absorber, condenser, or carbon adsorber averaged over the same time period as the performance test while the vent stream is normally routed and constituted. [§63.998(a)(2)(ii)(C)(4)]
- (d) For an absorber, condenser, or carbon adsorber used as a control device, the percent reduction of regulated material achieved by the control device or concentration of regulated material (ppmv, by compound) at the outlet of the control device. [§63.998(a)(2)(ii)(C)(5)]
- b) *Halogen concentration records.* Record the halogen concentration in the vent stream determined according to the procedures specified in MACT FFFF. Submit this record in the Notification of Compliance Status, as specified in §63.999(b)(4). If the permittee designates the vent stream as halogenated, then this shall be recorded and reported in the Notification of Compliance Status report. [§63.998(a)(4)]
2. *Continuous records and monitoring system data handling* — [§63.998(b)]
- a) *Continuous records.* Where MACT SS requires a continuous record, the permittee shall maintain a record as specified in §63.998(b)(1)(i) through (iv), as applicable: [§63.998(b)(1)]
- 1) A record of values measured at least once every 15 minutes or each measured value for systems which measure more frequently than once every 15 minutes; or [§63.998(b)(1)(i)]
  - 2) A record of block average values for 15-minute or shorter periods calculated from all measured data values during each period or from at least one measured data value per minute if measured more frequently than once per minute. [§63.998(b)(1)(ii)]
  - 3) Where data is collected from an automated continuous parameter monitoring system, the permittee may calculate and retain block hourly average values from each 15-minute block average period or from at least one measured value per minute if measured more frequently than once per minute, and discard all but the most recent three valid hours of continuous (15-minute or shorter) records, if the hourly averages do not exclude periods of CPMS breakdown or malfunction. An automated CPMS records the measured data and calculates the hourly averages using a computerized data acquisition system. [§63.998(b)(1)(iii)]
  - 4) A record as required by an alternative approved under MACT FFFF. [§63.998(b)(1)(iv)]
- b) *Excluded data.* Monitoring data recorded during periods identified in §63.998(b)(2)(i) and (ii) shall not be included in any average computed to determine compliance with an emission limit in MACT FFFF. [§63.998(b)(2) and §63.2450(e)(4)(viii)]
- 1) Monitoring system breakdowns, repairs, preventive maintenance, calibration checks, and zero (low-level) and high-level adjustments; and [§63.998(b)(2)(i)]
  - 2) Periods of non-operation of the process unit (or portion thereof), resulting in cessation of the emissions to which the monitoring applies. [§63.998(b)(2)(ii)]
- c) *Records of daily averages.* In addition to the records specified in §63.998(a), the

permittee shall keep records as specified in §63.998(b)(3)(i) and (ii) and submit reports as specified in §63.999(c), unless an alternative recordkeeping system has been requested and approved under MACT FFFF. [§63.998(b)(3)]

- 1) Except as specified in §63.998(b)(3)(ii), daily average values of each continuously monitored parameter shall be calculated from data meeting the specifications of §63.998(b)(2) for each operating day and retained for five years. [§63.998(b)(3)(i)]
    - (i) The daily average shall be calculated as the average of all values for a monitored parameter recorded during the operating day. The average shall cover a 24-hour period if operation is continuous, or the period of operation per operating day if operation is not continuous. If values are measured more frequently than once per minute, a single value for each minute may be used to calculate the daily average instead of all measured values. [§63.998(b)(3)(i)(A)]
    - (ii) The operating day shall be the period defined in the operating permit or in the Notification of Compliance Status. It may be from midnight to midnight or another daily period. [§63.998(b)(3)(i)(B)]
  - 2) If all recorded values for a monitored parameter during an operating day are within the range established in the Notification of Compliance Status or in the operating permit, the permittee may record that all values were within the range and retain this record for five years rather than calculating and recording a daily average for that operating day. In such cases, the permittee may not discard the recorded values as allowed in §63.998(b)(1)(iii). [§63.998(b)(3)(ii)]
- d) *Alternative recordkeeping.* For any parameter with respect to any item of equipment associated with a process vent, the permittee may implement the recordkeeping requirements in §63.998(b)(5)(i) or (ii) as alternatives to the recordkeeping provisions listed in §63.998(b)(1) through (3). The permittee shall retain each record required by §63.998(b)(5)(i) or (ii) as provided in MACT FFFF. [§63.998(b)(5)]
- 1) The permittee may retain only the daily average value and is not required to retain more frequently monitored operating parameter values, for a monitored parameter with respect to an item of equipment if the requirements of §63.998(b)(5)(i)(A) through (F) are met. The permittee shall notify the Administrator in the Notification of Compliance Status as specified in §63.999(b)(5) or, if the Notification of Compliance Status has already been submitted, in the Periodic Report immediately preceding implementation of the requirements of this paragraph, as specified in §63.999(c)(6)(iv). [§63.998(b)(5)(i)]
    - (i) The monitoring system can detect unrealistic or impossible data during periods of operation (e.g., a temperature reading of  $-200^{\circ}\text{C}$  on a boiler) and will alert the operator by alarm or other means. The permittee shall record the occurrence. All instances of the alarm or other alert in an operating day constitute a single occurrence. [§63.998(b)(5)(i)(A) and §63.2450(e)(4)(ix)]
    - (ii) The monitoring system generates a running average of the monitoring values, updated at least hourly throughout each operating day, that have been obtained during that operating day, and the capability to observe this average is readily available to the Administrator on-site during the operating day. The permittee shall record the occurrence of any period meeting the criteria in §63.998(b)(5)(i)(B)(1) through (3). All instances in an operating day constitute a single occurrence. [§63.998(b)(5)(i)(B)]
      - (a) The running average is above the maximum or below the minimum established limits; [§63.998(b)(5)(i)(B)(1)]

- (b) The running average is based on at least six one-hour average values; and [§63.998(b)(5)(i)(B)(2)]
- (c) The running average reflects a period of operation. [§63.998(b)(5)(i)(B)(3) and §63.2450(e)(4)(x)]
- (iii) The monitoring system can detect unchanging data during periods of operation, except in circumstances where the presence of unchanging data is the expected operating condition based on past experience (e.g., pH in some scrubbers), and will alert the operator by alarm or other means. The permittee shall record the occurrence. All instances of the alarm or other alert in an operating day constitute a single occurrence. [§63.998(b)(5)(i)(C) and §63.2450(e)(4)(xi)]
- (iv) The monitoring system will alert the permittee by an alarm, if the running average parameter value calculated under §63.998(b)(5)(i)(B) reaches a set point that is appropriately related to the established limit for the parameter that is being monitored. [§63.998(b)(5)(i)(D)]
- (v) The permittee shall verify the proper functioning of the monitoring system, including its ability to comply with the requirements of §63.998(b)(5)(i), at the times specified in §63.998(b)(5)(i)(E)(1) through (3). The permittee shall document that the required verifications occurred. [§63.998(b)(5)(i)(E)]
  - (a) Upon initial installation. [§63.998(b)(5)(i)(E)(1)]
  - (b) Annually after initial installation. [§63.998(b)(5)(i)(E)(2)]
  - (c) After any change to the programming or equipment constituting the monitoring system that might reasonably be expected to alter the monitoring system's ability to comply with the requirements of §63.998. [§63.998(b)(5)(i)(E)(3)]
- (vi) The permittee shall retain the records identified in §63.998(b)(5)(i)(F)(1) through (4). [§63.998(b)(5)(i)(F)]
  - (a) Identification of each parameter, for each item of equipment, for which the permittee has elected to comply with the requirements of §63.998(b)(5)(i). [§63.998(b)(5)(i)(F)(1)]
  - (b) A description of the applicable monitoring system(s), and of how compliance will be achieved with each requirement of §63.998(b)(5)(i)(A) through (E). The description shall identify the location and format (e.g., on-line storage; log entries) for each required record. If the description changes, the permittee shall retain both the current and the most recent superseded description. The description, and the most recent superseded description, shall be retained as provided in MACT FFFF, except as provided in §63.998(b)(5)(i)(F)(1). [§63.998(b)(5)(i)(F)(2)]
  - (c) A description, and the date, of any change to the monitoring system that would reasonably be expected to affect its ability to comply with the requirements of §63.998(b)(5)(i). [§63.998(b)(5)(i)(F)(3)]
  - (d) If subject to §63.998(b)(5)(i)(F)(2), the permittee shall retain the current description of the monitoring system as long as the description is current, but not less than five years from the date of its creation. The current description shall be retained on-site at all times or be accessible from a central location by computer or other means that provides access within two hours after a request. The permittee shall retain the most recent superseded description at least until five years from the date of its creation. The superseded description shall be retained on-site (or accessible from a central location by computer

that provides access within two hours after a request) at least six months after being superseded. Thereafter, the superseded description may be stored off-site. [§63.998(b)(5)(i)(F)(4)]

- 2) If the permittee has elected to implement the requirements of §63.998(b)(5)(i), and a period of six consecutive months has passed without an excursion as defined in §63.998(b)(6)(i), the permittee is no longer required to record the daily average value for that parameter for that unit of equipment, for any operating day when the daily average value is less than the maximum, or greater than the minimum established limit. With approval by the Administrator, monitoring data generated prior to the compliance date of this subpart shall be credited toward the period of six consecutive months if the parameter limit and the monitoring were required and/or approved by the Administrator. [§63.998(b)(5)(ii)]
    - (i) If the permittee elects not to retain the daily average values, the permittee shall notify the Administrator in the next Periodic Report, as specified in §63.999(c)(6)(i). The notification shall identify the parameter and unit of equipment. [§63.998(b)(5)(ii)(A)]
    - (ii) If there is an excursion as defined in §63.998(b)(6)(i) on any operating day after the permittee has ceased recording daily averages as provided in §63.998(b)(5)(ii), the permittee shall immediately resume retaining the daily average value for each operating day and shall notify the Administrator in the next Periodic Report, as specified in §63.999(c). The permittee shall continue to retain each daily average value until another period of six consecutive months has passed without an excursion as defined in §63.998(b)(6)(i). [§63.998(b)(5)(ii)(B)]
    - (iii) The permittee shall retain the records specified in §63.998(b)(5)(i)(A) through (F) for the duration specified in MACT FFFF. For any week, if compliance with §63.998(b)(5)(i)(A) through (D) does not result in retention of a record of at least one occurrence or measured parameter value, the permittee shall record and retain at least one parameter value during a period of operation. [§63.998(b)(5)(ii)(C) and §63.2450(e)(4)(xii)]
  - 3) For the purposes of §63.998, an excursion means that the daily average value of monitoring data for a parameter is greater than the maximum, or less than the minimum established value. [§63.998(b)(6)(i) and §63.2450(e)(4)(xiii)]
  - 4) One excused excursion for each control device or recovery device for each semiannual period is allowed. [§63.998(b)(6)(ii) and §63.2450(e)(4)(xiv)]
3. *Nonflare control and recovery device regulated source monitoring records* — [§63.998(c)]
- a) *Monitoring system records.* For process vents, the permittee shall keep the records specified in this paragraph, as well as records specified elsewhere in MACT SS. [§63.998(c)(1)]
    - 1) For a CPMS used to comply with 40 CFR Part 63, a record of the procedure used for calibrating the CPMS. [§63.998(c)(1)(i)]
    - 2) For a CPMS used to comply with MACT SS, records of the information specified in §63.998(c)(ii)(A) through (C) and (H), as indicated in MACT FFFF. [§63.998(c)(1)(ii) and §63.2450(e)(4)(xv)]
      - (i) The date and time of completion of calibration and preventive maintenance of the CPMS. [§63.998(c)(1)(ii)(A)]
      - (ii) The “as found” and “as left” CPMS readings, whenever an adjustment is made that affects the CPMS reading and a “no adjustment” statement otherwise.

- [§63.998(c)(1)(ii)(B)]
- (iii) The start time and duration or start and stop times of any periods when the CPMS is inoperative. [§63.998(c)(1)(ii)(C)]
- (iv) Records of the total duration of operating time. [§63.998(c)(1)(ii)(H)]
- b) *Monitoring records for recovery devices, absorbers, condensers, carbon adsorbers or other noncombustion systems used as control devices.* [§63.998(c)(3)]
- 1) If using a recovery device to achieve and maintain a TRE index value greater than the control applicability level specified in MACT FFFF but less than 4.0 or using an absorber, condenser, carbon adsorber or other non-combustion system as a control device, the permittee shall keep readily accessible, continuous records of the equipment operating parameters specified to be monitored under §63.990(c) (absorber, condenser, and carbon adsorber monitoring), or as approved by the Administrator in accordance with MACT FFFF. [§63.998(c)(3)(i)]
- 2) The permittee shall keep records of the daily average value of each continuously monitored parameter for each operating day determined according to the procedures specified in §63.998(b)(3)(i). [§63.998(c)(3)(ii)]
- 3) The permittee shall keep up-to-date, readily accessible records of periods of operation during which the parameter boundaries are exceeded. The parameter boundaries are established pursuant to §63.996(c)(6). [§63.998(c)(3)(iii)]
4. *Other records* — [§63.998(d)]
- a) *Closed vent system records.* For closed vent systems the permittee shall record the information specified in §63.998(d)(1)(i) through (iv), as applicable. [§63.998(d)(1)]
- 1) For closed vent systems collecting regulated material from a regulated source, the permittee shall record the identification of all parts of the closed vent system, that are designated as unsafe or difficult to inspect, an explanation of why the equipment is unsafe or difficult to inspect, and the plan for inspecting the equipment required by §63.983(b)(2)(ii) or (iii). [§63.998(d)(1)(i)]
- 2) For each closed vent system that contains bypass lines that could divert a vent stream away from the control device and to the atmosphere, the permittee shall keep a record of the information specified in either §63.998(d)(1)(ii)(A) or (B), as applicable. [§63.998(d)(1)(ii)]
- (i) Hourly records of whether the flow indicator specified under §63.983(a)(3)(i) was operating and whether a diversion was detected at any time during the hour, as well as records of the times of all periods when the vent stream is diverted from the control device, or the flow indicator is not operating. [§63.998(d)(1)(ii)(A)]
- (ii) Where a seal mechanism is used to comply with §63.983(a)(3)(ii), hourly records of flow are not required. In such cases, the permittee shall record that the monthly visual inspection of the seals or closure mechanisms has been done and shall record the occurrence of all periods when the seal mechanism is broken, the bypass line valve position has changed, or the key for a lock-and-key type lock has been checked out, and records of any car-seal that has been broken. [§63.998(d)(1)(ii)(B)]
- 3) For a closed vent system collecting regulated material from a regulated source, when a leak is detected as specified in §63.983(d)(2), the information specified in §63.998(d)(1)(iii)(A) through (F) shall be recorded and kept for five years. [§63.998(d)(1)(iii)]
- (i) The instrument and the equipment identification number and the operator's name, initials, or identification number. [§63.998(d)(1)(iii)(A)]

- (ii) The date the leak was detected and the date of the first attempt to repair the leak. [§63.998(d)(1)(iii)(B)]
- (iii) The date of successful repair of the leak. [§63.998(d)(1)(iii)(C)]
- (iv) The maximum instrument reading measured by the procedures in §63.983(c) after the leak is successfully repaired or determined to be nonreparable. [§63.998(d)(1)(iii)(D)]
- (v) "Repair delayed" and the reason for the delay if a leak is not repaired within 15 days after discovery of the leak. The permittee may develop a written procedure that identifies the conditions that justify a delay of repair. In such cases, reasons for delay of repair may be documented by citing the relevant sections of the written procedure. [§63.998(d)(1)(iii)(E)]
- (vi) Copies of the Periodic Reports as specified in §63.999(c), if records are not maintained on a computerized database capable of generating summary reports from the records. [§63.998(d)(1)(iii)(F)]
- 4) For each instrumental or visual inspection conducted in accordance with §63.983(b)(1) for closed vent systems collecting regulated material from a regulated source during which no leaks are detected, the permittee shall record that the inspection was performed, the date of the inspection, and a statement that no leaks were detected. [§63.998(d)(1)(iv)]
- b) *Equipment leaks records.* The permittee shall maintain records of the information specified in §63.998(d)(4)(i) and (ii) for closed vent systems and control devices if specified by the equipment leak provisions in MACT FFFF. The records specified in §63.998(d)(4)(i) shall be retained for the life of the equipment. The records specified in §63.998(d)(4)(ii) shall be retained for five years. [§63.998(d)(4)]
  - 1) The design specifications and performance demonstrations specified in §63.998(d)(4)(i)(A) through (C). [§63.998(d)(4)(i)]
    - (i) Detailed schematics, design specifications of the control device, and piping and instrumentation diagrams. [§63.998(d)(4)(i)(A)]
    - (ii) The dates and descriptions of any changes in the design specifications. [§63.998(d)(4)(i)(B)]
    - (iii) A description of the parameter or parameters monitored, as required in MACT FFFF, to ensure that control devices are operated and maintained in conformance with their design and an explanation of why that parameter (or parameters) was selected for the monitoring. [§63.998(d)(4)(i)(C)]
  - 2) Records of operation of closed vent systems and control devices, as specified in §63.998(d)(4)(ii)(A) through (C). [§63.998(d)(4)(ii)]
    - (i) Dates and durations when the closed vent systems and control devices required are not operated as designed as indicated by the monitored parameters. [§63.998(d)(4)(ii)(A)]
    - (ii) Dates and durations during which the monitoring system or monitoring device is inoperative. [§63.998(d)(4)(ii)(B)]
    - (iii) Dates and durations of start-ups and shutdowns of control devices required in MACT FFFF. [§63.998(d)(4)(ii)(C)]
- c) *Records of monitored parameters outside of range.* The permittee shall record the occurrences and the cause of periods when the monitored parameters are outside of the parameter ranges documented in the Notification of Compliance Status report. This information shall also be reported in the Periodic Report. [§63.998(d)(5)]

**MACT UU Recordkeeping Requirements:**

1. *Recordkeeping system.* The permittee may comply with the recordkeeping requirements for these regulated sources in one recordkeeping system. The recordkeeping system shall identify each record by regulated source and the type of program being implemented (e.g., quarterly monitoring, quality improvement) for each type of equipment. The records required by MACT UU are summarized in §63.1038(b) and (c). [§63.1038(a)]
2. *General equipment leak records.* [§63.1038(b)]
  - a) As specified in §63.1022(a) and (b), the permittee shall keep general and specific equipment identification if the equipment is not physically tagged, and the permittee is electing to identify the equipment subject to this subpart through written documentation such as a log or other designation. [§63.1038(b)(1)]
  - b) The permittee shall keep a written plan as specified in §63.1022(c)(4) for any equipment that is designated as unsafe- or difficult-to-monitor. [§63.1038(b)(2)]
  - c) The permittee shall maintain a record of the identity and an explanation as specified in §63.1022(d)(2) for any equipment that is designated as unsafe-to-repair. [§63.1038(b)(3)]
  - d) As specified in §63.1022(e), the permittee shall maintain the identity of compressors operating with an instrument reading of less than 500 ppm. [§63.1038(b)(4)]
  - e) The permittee shall keep records associated with the determination that equipment is in heavy liquid service as specified in §63.1022(f). [§63.1038(b)(5)]
  - f) The permittee shall keep records for leaking equipment as specified in §63.1023(e)(2). [§63.1038(b)(6)]
  - g) The permittee shall keep records for leak repair as specified in §63.1024(f) and records for delay of repair as specified in §63.1024(d). [§63.1038(b)(7)]
3. *Specific equipment leak records.* [§63.1038(c)]
  - a) For valves, the permittee shall maintain the records specified in §63.1038(c)(1)(i) and (c)(1)(ii). [§63.1038(c)(1)]
    - 1) The monitoring schedule for each process unit as specified in §63.1025(b)(3)(vi). [§63.1038(c)(1)(i)]
    - 2) The valve subgrouping records specified in §63.1025(b)(4)(iv), if applicable. [§63.1038(c)(1)(ii)]
  - b) For pumps, the permittee shall maintain the records specified in §63.1038(c)(2)(i) through (c)(2)(iii). [§63.1038(c)(2)]
    - 1) Documentation of pump visual inspections as specified in §63.1026(b)(4). [§63.1038(c)(2)(i)]
    - 2) Documentation of dual mechanical seal pump visual inspections as specified in §63.1026(e)(1)(v). [§63.1038(c)(2)(ii)]
    - 3) For the criteria as to the presence and frequency of drips for dual mechanical seal pumps, records of the design criteria and explanations and any changes and the reason for the changes, as specified in §63.1026(e)(1)(i). [§63.1038(c)(2)(iii)]
  - c) For connectors, the permittee shall maintain the monitoring schedule for each process unit as specified in §63.1027(b)(3)(v). [§63.1038(c)(3)]
  - d) For agitators, the permittee shall maintain the following records: [§63.1038(c)(4)]
    - 1) Documentation of agitator seal visual inspections as specified in §63.1028; and [§63.1038(c)(4)(i)]
    - 2) For the criteria as to the presence and frequency of drips for agitators, the permittee shall keep records of the design criteria and explanations and any changes and the reason for the changes, as specified in §63.1028(e)(1)(vi). [§63.1038(c)(4)(ii)]

- e) For pressure relief devices in gas and vapor or light liquid service, the permittee shall keep records of the dates and results of monitoring following a pressure release, as specified in §63.1030(c)(3). [§63.1038(c)(5)]
- f) For compressors, the permittee shall maintain the records specified in §63.1038(c)(6)(i) and (c)(6)(ii). [§63.1038(c)(6)]
  - 1) For criteria as to failure of the seal system and/or the barrier fluid system, record the design criteria and explanations and any changes and the reason for the changes, as specified in §63.1031(d)(2). [§63.1038(c)(6)(i)]
  - 2) For compressors operating under the alternative compressor standard, record the dates and results of each compliance test as specified in §63.1031(f)(2). [§63.1038(c)(6)(ii)]
- g) For a pump QIP program, the permittee shall maintain the records specified in §63.1038(c)(7)(i) through (c)(7)(v). [§63.1038(c)(7)]
  - 1) Individual pump records as specified in §63.1035(d)(2). [§63.1038(c)(7)(i)]
  - 2) Trial evaluation program documentation as specified in §63.1035(d)(6)(iii). [§63.1038(c)(7)(ii)]
  - 3) Engineering evaluation documenting the basis for judgement that superior emission performance technology is not applicable as specified in §63.1035(d)(6)(vi). [§63.1038(c)(7)(iii)]
  - 4) Quality assurance program documentation as specified in §63.1035(d)(7). [§63.1038(c)(7)(iv)]
  - 5) QIP records as specified in §63.1035(e). [§63.1038(c)(7)(v)]
- h) For process units complying with the batch process unit alternative, the permittee shall maintain the records specified in §63.1038(c)(8)(i) and (c)(8)(ii). [§63.1038(c)(8)]
  - 1) Pressure test records as specified in §63.1036(b)(7). [§63.1038(c)(8)(i)]
  - 2) Records for equipment added to the process unit as specified in §63.1036(d). [§63.1038(c)(8)(ii)]
- i) For process units complying with the enclosed-vented process unit alternative, the permittee shall maintain the records for enclosed-vented process units as specified in §63.1037(b). [§63.1038(c)(9)]

**MACT G Process Wastewater Provisions - Recordkeeping:**

- 1. The permittee shall keep in a readily accessible location the records specified in §63.147(b)(1) through (8). [§63.147(b)]
  - a) *Requirements for Group 2 wastewater streams.* This paragraph does not apply to Group 2 wastewater streams that are used to comply with §63.138(g). For all other Group 2 wastewater streams, the permittee shall keep in a readily accessible location the records specified in §63.147(b)(8)(i) through (iv). [§63.147(b)(8)]
    - 1) Process unit identification and description of the process unit. [§63.147(b)(8)(i)]
    - 2) Stream identification code. [§63.147(b)(8)(ii)]
    - 3) For existing sources, concentration of Tables 8 and 9 to MACT FFFF compound(s) in ppm, by weight. For new sources, concentration of Tables 8 and 9 to MACT FFFF compound(s) in ppm, by weight. Include documentation of the methodology used to determine concentration. [§63.147(b)(8)(iii) and §63.2485(b)]
    - 4) Flow rate in L/min. [§63.147(b)(8)(iv)]

**MACT FFFF Notifications:**

1. *General.* Except as specified in §63.2515(d), the permittee must submit all the notifications in §63.6(h)(4) and (5), §63.7(b) and (c), §63.8(e) and (f)(4) and (6), and §63.9(b) through (h) that apply by the dates specified. [§63.2515(a)]
2. *Notification of performance test.* If the permittee is required to conduct a performance test, the permittee must submit a notification of intent to conduct a performance test at least 60 calendar days before the performance test is scheduled to begin as required in §63.7(b)(1). [§63.2515(c)]
3. *Supplement to Notification of Compliance Status.* The permittee must also submit supplements to the Notification of Compliance Status as specified in §63.2520(d)(3) through (5). [§63.2515(d)]

**MACT FFFF Reporting:**

1. The permittee must submit each report in Table 11 to MACT FFFF that applies. [§63.2520(a)]
2. Unless the Director has approved a different schedule for submission of reports under §63.10(a), the permittee must submit each report by the date in Table 11 to MACT FFFF and according to §63.2520(b)(1) through (5). [§63.2520(b)]
  - a) For each affected source that is subject to permitting regulations pursuant to 40 CFR Part 70, and if the permitting authority has established dates for submitting semiannual reports pursuant to §70.6(a)(3)(iii)(A), the permittee may submit the first and subsequent compliance reports according to the dates the permitting authority has established instead of according to the dates in §63.2520(b)(1) through (4). [§63.2520(b)(5)]
3. *Compliance report.* The compliance report must contain the information specified in §63.2520(e)(1) through (17). On or after August 12, 2023 or once the reporting template for MACT FFFF has been available on the CEDRI website for one year, whichever date is later<sup>1</sup>, the permittee must submit all subsequent reports to EPA via CEDRI, which can be accessed through EPA's CDX (<https://cdx.epa.gov/>). EPA will make all the information submitted through CEDRI available to the public without further notice to the permittee. Do not use CEDRI to submit information the permittee claims as confidential business information (CBI). Anything submitting using CEDRI cannot later be claimed to be CBI. The permittee must use the appropriate electronic report template on the CEDRI website (<https://www.epa.gov/electronic-reporting-air-emissions/compliance-and-emissions-data-reporting-interface-cedri>) for MACT FFFF. The date report templates become available will be listed on the CEDRI website. Unless the Administrator or Director or other authority has approved a different schedule for submission of reports under §63.9(i) and §63.10(a), the report must be submitted by the deadline specified in MACT FFFF, regardless of the method in which the report is submitted. Although EPA does not expect a person to assert a claim of CBI, if the permittee wishes to assert a CBI claim, submit a complete report, including information claimed to be CBI, to EPA. The report must be generated using the appropriate form on the CEDRI website or an alternative electronic file consistent with the extensible markup language (XML) schema listed on the CEDRI website. Submit the file on a compact disk, flash drive, or other commonly used electronic storage medium and clearly

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<sup>1</sup> As of the issuance of this operating permit, the Compliance Reporting Template for MACT FFFF is not yet available on the CEDRI website. Once the compliance reporting template is made available, the permittee will have one year to submit all subsequent reports to the EPA via CEDRI, as per §63.2520(e).

mark the medium as CBI. Mail the electronic medium to U.S. Environmental Protection Agency, Office of Air Quality Planning and Standards, Sector Policies and Programs Division, CORE CBI Office, U.S. EPA Mailroom (C404-02), Attention: Miscellaneous Organic Chemical Manufacturing Sector Lead, 4930 Old Page Rd., Durham, NC 27703. The same file with the CBI omitted must be submitted to EPA via the EPA's CDX as described in this paragraph. All CBI claims must be asserted at the time of submission. Furthermore, under CAA §114(c) emissions data is not entitled to confidential treatment, and EPA is required to make emissions data available to the public. Thus, emissions data will not be protected as CBI and will be made publicly available. The permittee may assert a claim of EPA system outage or *force majeure* for failure to timely comply with the reporting requirement in this paragraph provided the permittee meets the requirements outlined in §63.2520(i) or (j), as applicable. [§63.2520(e)]

- a) Company name and address. [§63.2520(e)(1)]
- b) Statement by a responsible official with that official's name, title, and signature, certifying the accuracy of the content of the report. If the report is submitted via CEDRI, the certifier's electronic signature during the submission process replaces the requirement in this paragraph. [§63.2520(e)(2)]
- c) Date of report and beginning and end dates of the reporting period. The permittee is no longer required to provide the date of the report when the report is submitted via CEDRI. [§63.2520(e)(3)]
- d) This paragraph no longer applies; however, for historical compliance purposes, a copy of the SSM plan must be retained and available on-site for five years after August 12, 2023. [§63.2520(e)(4)]
- e) The compliance report must contain the information on deviations, as defined in §63.2550, according to §63.2520(e)(5)(i), (ii), (iii), and (iv). [§63.2520(e)(5)]
  - i) If there are no deviations from any emission limit, operating limit or work practice standard specified in MACT FFFF, include a statement that there were no deviations from the emission limits, operating limits, or work practice standards during the reporting period. [§63.2520(e)(5)(i)]
  - ii) For each deviation from an emission limit, operating limit, and work practice standard that occurs at an affected source where the permittee is not using a continuous monitoring system (CMS) to comply with the emission limit or work practice standard in MACT FFFF, the permittee must include the information in §63.2520(e)(5)(ii)(A) through (D). This includes periods of SSM. [§63.2520(e)(5)(ii)]
    - A. The total operating time in hours of the affected source during the reporting period. [§63.2520(e)(5)(ii)(A)]
    - B. Operating logs of processes with batch vents from batch operations for the day(s) during which the deviation occurred, except operating logs are not required for deviations of the work practice standards for equipment leaks. [§63.2520(e)(5)(ii)(C)]
    - C. §63.2520(e)(5)(ii)(B) no longer applies. Instead, report information for each deviation to meet an applicable standard. For each instance, report the start date, start time, and duration in hours of each deviation. For each deviation, the report must include a list of the affected sources or equipment, an estimate of the quantity in pounds of each regulated pollutant emitted over any emission limit, a description of the method used to estimate emissions, the cause of the deviation (including unknown cause, if applicable), as applicable, and the corrective action taken. [§63.2520(e)(5)(ii)(D)]

- iii) For each deviation from an emission limit or operating limit occurring at an affected source where the permittee is using a CMS to comply with an emission limit in MACT FFFF, the permittee must include the information in §63.2520(e)(5)(iii)(A) through (N). This includes periods of SSM. [§63.2520(e)(5)(iii)]
  - A. The start date, start time, and duration in hours that each CMS was inoperative, except for zero (low-level) and high-level checks. [§63.2520(e)(5)(iii)(A)]
  - B. The start date, start time, and duration in hours that each CEMS was out-of-control and a description of the corrective actions taken. [§63.2520(e)(5)(iii)(B)]
  - C. The total duration in hours of all deviations for each CMS during the reporting period, the total operating time in hours of the affected source during the reporting period, and the total duration as a percent of the total operating time of the affected source during that reporting period. [§63.2520(e)(5)(iii)(D)]
  - D. The total duration in hours of CMS downtime for each CMS during the reporting period, and the total duration of CMS downtime as a percent of the total operating time of the affected source during that reporting period. [§63.2520(e)(5)(iii)(F)]
  - E. An identification of each HAP that is known to be in the emission stream. [§63.2520(e)(5)(iii)(G)]
  - F. A brief description of the process units. [§63.2520(e)(5)(iii)(H)]
  - G. The monitoring equipment manufacturer(s) and model number(s) and the pollutant or parameter monitored. [§63.2520(e)(5)(iii)(I)]
  - H. The date of the latest CMS certification or audit. [§63.2520(e)(5)(iii)(J)]
  - I. Operating logs of processes with batch vents from batch operations for each day(s) during which the deviation occurred. [[§63.2520(e)(5)(iii)(K)]
  - J. The operating day or operating block average values of monitored parameters for each day(s) during which the deviation occurred. [§63.2520(e)(5)(iii)(L)]
  - K. §63.2520(e)(5)(iii)(C) no longer applies. Instead, report the number of deviations to meet an applicable standard. For each instance, report the start date, start time and duration in hours of each deviation. For each deviation, the report must include a list of the affected sources or equipment, an estimate of the quantity in pounds of each regulated pollutant emitted over any emission limit, a description of the method used to estimate the emissions, and the cause of the deviation (including unknown cause, if applicable), as applicable, and the corrective action taken. [§63.2520(e)(5)(iii)(M)]
  - L. §63.2520(e)(5)(iii)(E) no longer applies. Instead, report a breakdown of the total duration in hours of the deviations during the reporting period into those that are due to control equipment problems, process problems, other known causes, and other unknown causes. [§63.2520(e)(5)(iii)(N)]
- f) If the permittee uses a CEMS, and there were no periods during which it was out-of-control as specified in §63.8(c)(7), include a statement that there were no periods during which the CEMS was out-of-control during the reporting period. [§63.2520(e)(6)]
- g) Include each new operating scenario which has been operated since the time period covered by the last compliance report and has not been submitted in the notification of compliance status report or a previous compliance report. For each new operating scenario, the permittee must report the information specified in §63.2525(b) and provide verification that the operating conditions for any associated control or treatment device have not been exceeded and that any required calculations and engineering analyses have been performed. For the purposes of this paragraph, a revised operating scenario

- for an existing process is considered to be a new operating scenario. [§63.2520(e)(7)]
- h) For process units added to a PUG, the permittee must report the description and rationale specified in §63.2525(i)(4). The permittee must report the primary product redeterminations specified in §63.2525(i)(5). [§63.2520(e)(8)]
  - i) Except as specified in §63.2450(e)(4), §63.2480(f), §63.2485(p) and (q), and §63.2520(t), applicable records and information for periodic reports as specified in the MACT F, MACT G, MACT H, MACT SS, MACT UU, MACT WW, and MACT GGG and 40 CFR Part 65, Subpart F. [§63.2520(e)(9)]
  - j) Except as specified in §63.2520(e)(10)(ii), whenever the permittee makes a process change, or changes any of the information submitted in the notification of compliance status report or a previous compliance report, that is not within the scope of an existing operating scenario, the permittee must document the change in the compliance report. A process change does not include moving within a range of conditions identified in the standard batch, and a nonstandard batch does not constitute a process change. [§63.2520(e)(10)]
    - i) The notification must include all the information in §63.2520(e)(10)(i)(A) through (C). [§63.2520(e)(10)(i)]
      - A. A description of the process change. [§63.2520(e)(10)(i)(A)]
      - B. Revisions to any of the information reported in the original notification of compliance status report under §63.2520(d). [§63.2520(e)(10)(i)(B)]
      - C. Information required by the notification of compliance status report under §63.2520(d) for changes involving the addition of processes or equipment at the affected source. [§63.2520(e)(10)(i)(C)]
    - ii) The permittee must submit a report 60 days before the scheduled implementation date of any of the changes identified in §63.2520(e)(10)(ii)(A), (B), or (C). [§63.2520(e)(10)(ii)]
      - A. Any change to the information contained in the precompliance report. [§63.2520(e)(10)(ii)(A)]
      - B. A change in the status of a control device from small to large. [§63.2520(e)(10)(ii)(B)]
      - C. A change from Group 2 to Group 1 for any emission point except for batch process vents that meet the conditions specified in §63.2460(b)(6)(i). [§63.2520(e)(10)(ii)(C)]
  - k) For bypass lines subject to the requirements §63.2450(e)(6), the compliance report must include the start date, start time, duration in hours, estimate of the volume of gas in scf, the concentration of organic HAP in the gas in ppmv and the resulting mass emissions of organic HAP in pounds that bypass a control device. For periods when the flow indicator is not operating, report the start date, start time, and duration in hours. [§63.2520(e)(12)]
  - l) For any maintenance vent release exceeding the applicable limits in §63.2450(v)(1), the compliance report must include the information specified in §63.2520(e)(14)(i) through (iv). For the purposes of this reporting requirement, if the permittee complies with §63.2450(v)(1)(iv) then the permittee must report each venting event conducted under those provisions and include an explanation for each event as to why utilization of this alternative was required. [§63.2520(e)(14)]
    - i) Identification of the maintenance vent and the equipment served by the maintenance vent. [§63.2520(e)(14)(i)]
    - ii) The date and time the maintenance vent was opened to the atmosphere.

[§63.2520(e)(14)(ii)]

- iii) The lower explosive limit in percent, vessel pressure in psig, or mass in pounds of VOC in the equipment, as applicable, at the start of atmospheric venting. If the 5-psig vessel pressure option in §63.2450(v)(1)(ii) was used and active purging was initiated while the lower explosive limit was 10% or greater, also include the lower explosive limit of the vapors at the time active purging was initiated.

[§63.2520(e)(14)(iii)]

- iv) An estimate of the mass in pounds of organic HAP released during the entire atmospheric venting event. [§63.2520(e)(14)(iv)]

- m) Compliance reports for pressure relief devices subject to the requirements in §63.2480(e) must include the information specified in §63.2520(e)(15)(i) through (iii). [§63.2520(e)(15)]

- i) For pressure relief devices in organic HAP gas or vapor service, pursuant to §63.2480(e)(1), report the instrument readings and dates for all readings of 500 ppmv or greater. [§63.2520(e)(15)(i)]

- ii) For pressure relief devices in organic HAP gas or vapor service subject to §63.2480(e)(2), report the instrument readings and dates of instrument monitoring conducted. [§63.2520(e)(15)(ii)]

- iii) For pressure relief devices in organic HAP service subject to §63.2480(e)(3), report each pressure release to the atmosphere, including the start date, start time, and duration in minutes of the pressure release and an estimate of the mass quantity in pounds of each organic HAP released; the results of any root cause analysis and corrective action analysis completed during the reporting period, including the corrective actions implemented during the reporting period; and, if applicable, the implementation schedule for planned corrective actions to be implemented subsequent to the reporting period. [§63.2520(e)(15)(iii)]

- 4. **Performance test report.** The permittee must submit performance test reports in accordance with this paragraph. Unless otherwise specified in MACT FFFF, within 60 days after the date of completing each performance test required by MACT FFFF, the permittee must submit the results of the performance test following the procedures specified in §63.2520(f)(1) through (3). [§63.2520(f)]

- a) *Data collected using test methods supported by EPA's Electronic Reporting Tool (ERT) as listed on EPA's ERT website (<https://epa.gov/electronic-reporting-air-emissions/electronic-reporting-tool-ert>) at the time of the test.* Submit the results of the performance test to EPA via CEDRI, which can be accessed through the EPA's CDX (<https://cdx.epa.gov/>). The data must be submitted in a file format generated using EPA's ERT. Alternatively, the permittee may submit an electronic file consistent with the extensible markup language (XML) schema listed on EPA's ERT website.

[§63.2520(f)(1)]

- b) *Data collected using test methods that are not supported by EPA's ERT as listed on EPA's ERT website at the time of the test.* The results of the performance test must be included as an attachment in the ERT or an alternate electronic file consistent with the XML schema listed on EPA's ERT website. Submit the ERT generated package or alternative file to EPA via CEDRI. [§63.2520(f)(2)]

- c) *Confidential business information (CBI).* EPA will make all the information submitted through CEDRI available to the public without further notice to the permittee. Do not use CEDRI to submit information the permittee wishes to claim as CBI. Anything submitted using CEDRI cannot later be claimed to be CBI. Although EPA does not expect a

person to assert a claim of CBI, if the permittee wishes to assert a CBI claim, the permittee must submit a complete file, including information claimed to be CBI, to the EPA. The file must be generated using EPA's ERT or an alternate electronic file consistent with the XML schema listed on EPA's ERT website. Submit the file on a compact disk, flash drive, or other commonly used electronic storage medium and clearly mark the medium as CBI. Mail the electronic medium to U.S. Environmental Protection Agency, Office of Air Quality Planning and Standards, Sector Policies and Programs Division, CORE CBI Office, U.S. EPA Mailroom (C404-02), Attention: Group Leader, Measurement Policy Group, 4930 Old Page Rd., Durham, NC 27703. The same file with the CBI omitted must be submitted to EPA via EPA's CDX as described in §63.2520(f)(1) and (2). All CBI claims must be asserted at the time of submission. Furthermore, under CAA §114(c) emissions data is not entitled to confidential treatment, and EPA is required to make emissions data available to the public. Thus, emissions data will not be protected as CBI and will be made publicly available. [§63.2520(f)(3)]

5. *Claims of EPA system outage.* If the permittee is required to electronically submit a report through CEDRI in EPA's CDX, the permittee may assert a claim of EPA system outage for failure to timely comply with the reporting requirements. To assert a claim of EPA system outage, the permittee must meet the requirements outlined in §63.2520(h)(1) through (7). [§63.2520(h)]
6. *Claims of force majeure.* If the permittee is required to electronically submit a report through CEDRI in EPA's CDX, the permittee may assert a claim of *force majeure* for failure to timely comply with that reporting requirement. To assert a claim of *force majeure*, the permittee must meet the requirements outline in §63.2520(i)(1) through (5). [§63.2520(i)]
7. *Reporting.* [§63.2450(m)]
  - a) When §63.2445 through §63.2490 reference other subparts in 40 CFR Part 63 that use the term "periodic report," it means "compliance report" for the purposes of MACT FFFF. The compliance report must include the information specified in §63.2520(e), as well as the information specified in the referenced subparts. [§63.2450(m)(1)]
  - b) When there are conflicts between MACT FFFF and references subparts for the due dates of reports required by MACT FFFF, reports must be submitted according to the due dates presented in MACT FFFF. [§63.2450(m)(2)]
  - c) Excused excursions, as defined in MACT G and MACT SS, are not allowed. [§63.2450(m)(3)]

**Table 11 to MACT FFFF — Requirements for Reports**

| The permittee must submit a(n) | The report must contain...               | The permittee must submit the report...                   |
|--------------------------------|------------------------------------------|-----------------------------------------------------------|
| Compliance report              | The information specified in §63.2520(e) | Semiannually according to the requirements in §63.2520(b) |

**General Reporting Requirements:**

1. The permittee shall report no later than 10 days after records indicate an exceedance of any of the emission limitations to the Air Pollution Control Program's Compliance/Enforcement Section at: [10 CSR 10-6.065(5)(C)1.C(III)(c)III]
  - a) PO Box 176, Jefferson City, MO 65102 **and** [AirComplianceReporting@dnr.mo.gov](mailto:AirComplianceReporting@dnr.mo.gov) **or**
  - b) Any means that comply with 40 CFR Part 3.
2. The permittee shall report any deviations from the requirements of this permit condition in the semi-annual monitoring report and annual compliance certification required by Section V of this permit. [10 CSR 10-6.065(5)(C)1.C(III)(b)]

**MACT SS Reporting Requirements:**

1. *Performance test notifications and reports* — [§63.999(a)]
  - a) *General requirements.* General requirements for performance test notifications and reports are specified in §63.999(a)(1)(i) through (iii). [§63.999(a)(1)]
    - i) The permittee shall notify the Administrator of the intention to conduct a performance test or flare compliance assessment at least 30 days before such a compliance demonstration is scheduled to allow the Administrator the opportunity to have an observer present. If after 30 days' notice for such an initially scheduled compliance demonstration, there is a delay (due to operational problems, etc.) in conducting the scheduled compliance demonstration, the permittee of an affected facility shall notify the Administrator as soon as possible of any delay in the original demonstration date. The permittee shall provide at least seven days prior notice of the rescheduled date of the compliance demonstration or arrange a rescheduled date with the Administrator by mutual agreement. [§63.999(a)(1)(i)]
    - ii) Unless specified differently MACT SS or MACT FFFF, performance test reports, not submitted as part of a Notification of Compliance Status report, shall be submitted to the Administrator within 60 days of completing the test or determination. [§63.999(a)(1)(ii)]
  - b) *Performance test report submittal and content requirements.* Performance test reports shall be submitted as specified in §63.999(a)(2)(i) through (iii). [§63.999(a)(2)]
    - i) For performance tests, the Notification of Compliance Status or performance test report shall include one complete test report as specified in §63.999(a)(2)(ii) for each test method used for a particular kind of emission point and other applicable information specified in §63.999(a)(2)(iii). For additional tests performed for the same kind of emission point using the same method, the results and any other information required in applicable sections of MACT SS shall be submitted, but a complete test report is not required. [§63.999(a)(2)(i)]
    - ii) A complete test report shall include a brief process description, sampling site description, description of sampling and analysis procedures and any modifications to standard procedures, quality assurance procedures, record of operating conditions during the test, record of preparation of standards, record of calibrations, raw data sheets for field sampling, raw data sheets for field and laboratory analyses, documentation of calculations, and any other information required by the test method. [§63.999(a)(2)(ii)]
    - iii) The performance test report shall also include the information specified in §63.999(a)(2)(iii)(A) through (C), as applicable. [§63.999(a)(2)(iii)]
      - A. For nonflare control device and halogen reduction device performance tests as required under §63.990(b), also submit the records specified in §63.998(a)(2)(ii), as applicable. [§63.999(a)(2)(iii)(B)]
2. *Notification of Compliance Status* — [§63.999(b)]
  - a) *Operating range for monitored parameters.* The permittee shall submit as part of the Notification of Compliance Status, the operating range for each monitoring parameter identified for each control or recovery device as determined pursuant to §63.996(c)(6). The specified operating range shall represent the conditions for which the control or recovery device is being properly operated and maintained. This report shall include the information in §63.999(b)(3)(i) through (iii), as applicable, unless the range and the operating day have been established in the operating permit. [§63.999(b)(3)]
    - i) The specific range of the monitored parameter(s) for each emission point;

[§63.999(b)(3)(i)]

- ii) The rationale for the specific range for each parameter for each emission point, including any data and calculations used to develop the range and a description of why the range indicates proper operation of the control or recovery device, as specified in §63.999(b)(3)(ii)(A), (B), or (C), as applicable. [§63.999(b)(3)(ii)]
  - A. If a performance test or TRE index value determination is required by MACT FFFF for a control or recovery device, the range shall be based on the parameter values measured during the TRE index value determination or performance test and may be supplemented by engineering assessments and/or manufacturer's recommendations. TRE index value determinations and performance testing are not required to be conducted over the entire range of permitted parameter values. [§63.999(b)(3)(ii)(A)]
  - B. If a performance test or TRE index value determination is not required by MACT FFFF for a control or recovery device, the range may be based solely on engineering assessments and/or manufacturer's recommendations. [§63.999(b)(3)(ii)(B)]
  - C. The range may be based on ranges or limits previously established under MACT FFFF. [§63.999(b)(3)(ii)(C)]
- iii) A definition of the source's operating day for purposes of determining daily average values of monitored parameters. The definition shall specify the times at which an operating day begins and ends. [§63.999(b)(3)(iii)]
- b) *Alternative recordkeeping.* The permittee shall notify the Administrator in the Notification of Compliance Status if the alternative recordkeeping requirements of §63.998(b)(5) are being implemented. If the Notification of Compliance Status has already been submitted, the notification must be in the periodic report submitted immediately preceding implementation of the alternative, as specified in §63.999(c)(6)(iv). [§63.999(b)(5)]

3. *MACT SS Periodic reports.* [§63.999(c)]

- a) Periodic reports shall include the reporting period dates, the total source operating time for the reporting period, and, as applicable, all information specified in §63.999 and in MACT FFFF, including reports of periods when monitored parameters are outside their established ranges. [§63.999(c)(1)]
- b) For closed vent systems subject to the requirements of §63.983, the permittee shall submit as part of the periodic report the information specified in §63.999(c)(2)(i) through (iii), as applicable. [§63.999(c)(2)]
  - i) The information recorded in §63.998(d)(1)(iii)(B) through (E); [§63.999(c)(2)(i)]
  - ii) Reports of the times of all periods recorded under §63.998(d)(1)(ii)(A) when the vent stream is diverted from the control device through a bypass line; and [§63.999(c)(2)(ii)]
  - iii) Reports of all times recorded under §63.998(d)(1)(ii)(B) when maintenance is performed in car-sealed valves, when the seal is broken, when the bypass line valve position is changed, or the key for a lock-and-key type configuration has been checked out. [§63.999(c)(2)(iii)]
- c) For process vents, periodic reports shall include the information specified in §63.999(c)(6)(i) through (iv). [§63.999(c)(6)]
  - i) Periodic reports shall include the daily average values of monitored parameters, calculated as specified in §63.998(b)(3)(i) for any days when the daily average value is outside the bounds as defined in §63.998(c)(2)(iii) or (c)(3)(iii), or the data

availability requirements defined in §63.999(c)(6)(i)(A) through (D) are not met, whether these excursions are excused or unexcused excursions. For excursions caused by lack of monitoring data, the duration of periods when monitoring data were not collected shall be specified. An excursion means any of the cases listed in §63.999(c)(6)(i)(A) through (C). If the permittee elects not to retain the daily average values pursuant to §63.998(b)(5)(ii)(A), the permittee shall report this in the Periodic Report. [§63.999(c)(6)(i)]

- A. When the daily average value of one or more monitored parameters is outside the permitted range. [§63.999(c)(6)(i)(A)]
- B. When the period of control or recovery device operation is four hours or greater in an operating day and monitoring data are insufficient to constitute a valid hour of data for at least 75% of the operating hours. [§63.999(c)(6)(i)(B)]
- C. When the period of control or recovery device operation is less than four hours in an operating day and more than one of the hours during the period of operation does not constitute a valid hour of data due to insufficient monitoring data. [§63.999(c)(6)(i)(C)]
- D. Monitoring data are insufficient to constitute a valid hour of data as used in §63.999(c)(6)(i)(B) and (C), if measured values are unavailable for any of the 15-minute periods within the hour. [§63.999(c)(6)(i)(D)]

- ii) Report all carbon-bed regeneration cycles during which the parameters recorded under §63.998(a)(2)(ii)(C) were outside the ranges established in the Notification of Compliance Status or in the operating permit. [§63.999(c)(6)(ii)]
- iii) If the permittee has chosen to use the alternative recordkeeping requirements of §63.998(b)(5) and has not notified the Administrator in the Notification of Compliance Status that the alternative recordkeeping provisions are being implemented as specified in §63.999(b)(5), the permittee shall notify the Administrator in the Periodic Report submitted immediately preceding implementation of the alternative. The notifications specified in §63.998(b)(5)(ii) shall be included in the next Periodic Report following the identified event. [§63.999(c)(6)(iv)]

4. *Requests for approval of monitoring alternatives* — [§63.999(d)]

- a) *Alternatives to the continuous operating parameter monitoring and recordkeeping provisions.* Requests for approval to use alternatives to continuous operating parameter monitoring and recordkeeping provisions, as provided for in §63.996(d)(1), shall be submitted as specified in MACT FFFF, and MACT FFFF will govern the review and approval of such requests. The information specified in §63.999(d)(1)(i) and (ii) shall be included. [§63.999(d)(1)]
  - i) A description of the proposed alternative system; and [§63.999(d)(1)(i)]
  - ii) Information justifying the permittee's request for an alternative method, such as the technical or economic infeasibility, or the impracticality, of the regulated source using the required method. [§63.999(d)(1)(ii)]
- b) *Monitoring a different parameter than those listed.* Requests for approval to monitor a different parameter than those established in §63.996(c)(6) or to set unique monitoring parameters, as provided for in §63.996(d)(2), shall be submitted as specified as specified in MACT FFFF, and MACT FFFF will govern the review and approval of such requests. The information specified in §63.999(d)(2)(i) through (iii) shall be included in the request. [§63.999(d)(2)]
  - i) A description of the parameter(s) to be monitored to ensure the control technology or

- pollution prevention measure is operated in conformance with its design and achieves the specified emission limit, percent reduction, or nominal efficiency, and an explanation of the criteria used to select the parameter(s); [§63.999(d)(2)(i)]
- ii) A description of the methods and procedures that will be used to demonstrate that the parameter indicates proper operation of the control device, the schedule for this demonstration, and a statement that the permittee will establish a range for the monitored parameter(s) as part of the Notification of Compliance Status if required under MACT FFFF, unless this information has already been submitted; and [§63.999(d)(2)(ii)]
  - iii) The frequency and content of monitoring, recording, and reporting, if monitoring and recording is not continuous, or if reports of daily average values when the monitored parameter value is outside the established range will not be included in periodic reports under §63.999(c). The rationale for the proposed monitoring, recording, and reporting system shall be included. [§63.999(d)(2)(iii)]

**MACT UU Reporting Requirements:**

1. *Periodic Reports.* The permittee shall report the information specified in §63.1039(b)(1) through (b)(8), as applicable, in the Periodic Report specified in MACT FFFF. [§63.1039(b)]
  - a) For the equipment specified in §63.1039(b)(1)(i) through (b)(1)(v), report in a summary format by equipment type, the number of components for which leaks were detected and for valves, pumps and connectors show the percent leakers, and the total number of components monitored. Also include the number of leaking components that were not repaired as required by §63.1024, and for valves and connectors, identify the number of components that are determined by §63.1025(c)(3) to be nonrepairable. [§63.1039(b)(1)]
    - i) Valves in gas and vapor service and in light liquid service pursuant to §63.1025(b) and (c). [§63.1039(b)(1)(i)]
    - ii) Pumps in light liquid service pursuant to §63.1026(b) and (c). [§63.1039(b)(1)(ii)]
    - iii) Connectors in gas and vapor service and in light liquid service pursuant to §63.1027(b) and (c). [§63.1039(b)(1)(iii)]
    - iv) Agitators in gas and vapor service and in light liquid service pursuant to §63.1028(c). [§63.1039(b)(1)(iv)]
    - v) Compressors pursuant to §63.1031(d). [§63.1039(b)(1)(v)]
  - b) Where any delay of repair is utilized pursuant to §63.1024(d), report that delay of repair has occurred and report the number of instances of delay of repair. [§63.1039(b)(2)]
  - c) If applicable, report the valve subgrouping information specified in §63.1025(b)(4)(iv). [§63.1039(b)(3)]
  - d) For pressure relief devices in gas and vapor service pursuant to §63.1030(b) and for compressors pursuant to §63.1031(f) that are to be operated at a leak detection instrument reading of less than 500 ppm, report the results of all monitoring to show compliance conducted within the semiannual reporting period. [§63.1039(b)(4)]
  - e) Report, if applicable, the initiation of a monthly monitoring program for valves pursuant to §63.1025(b)(3)(i). [§63.1039(b)(5)]
  - f) Report, if applicable, the initiation of a quality improvement program for pumps pursuant to §63.1035. [§63.1039(b)(6)]
  - g) Where the alternative means of emissions limitation for batch processes is utilized, report the information listed in §63.1036(f). [§63.1039(b)(7)]
  - h) Report any revisions to items reported in an earlier Initial Compliance Status Report if the method of compliance has changed since the last report. [§63.1039(b)(8)]

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**MACT G Process Wastewater Provisions – Reporting:**

1. The permittee shall submit the information specified in §63.146(b)(1) through (b)(9) as part of the Notification of Compliance Status required by §63.152(b). [§63.146(b)]
  - a) *Requirements for Group 2 wastewater streams.* This paragraph does not apply to Group 2 wastewater streams that are used to comply with §63.138(g). For Group 2 wastewater streams, the permittee shall include the information specified in §63.146(b)(1)(i) through (iv) in the Notification of Compliance Status Report. This information may be submitted in any form. Table 15 of MACT G is an example. [§63.146(b)(1)]
    - i) Process unit identification and description of the process unit. [§63.146(b)(1)(i)]
    - ii) Stream identification code. [§63.146(b)(1)(ii)]
    - iii) For existing sources, concentration of Tables 8 and 9 to MACT FFFF compound(s) in ppm, by weight. For new sources, concentration of Tables 8 and 9 to MACT FFFF compound(s) in ppm, by weight. Include documentation of the methodology used to determine concentration. [§63.146(b)(1)(iii) and §63.2485(b)]
    - iv) Flow rate in L/min. [§63.146(b)(1)(iv)]

| <b>PERMIT CONDITION 002</b>                                     |                                                  |                                                                      |
|-----------------------------------------------------------------|--------------------------------------------------|----------------------------------------------------------------------|
| <b>10 CSR 10-6.060 Construction Permits Required</b>            |                                                  |                                                                      |
| <b>Construction Permit 122014-009, Issued December 24, 2014</b> |                                                  |                                                                      |
| <b>Emission Unit</b>                                            | <b>Description</b>                               | <b>Control Equipment</b>                                             |
| EP-100                                                          | Biodiesel Process Vent (91 ft <sup>3</sup> /min) | Closed Vent System to Precondenser, condenser, and scrubber (CD-100) |
| EP-101                                                          | Light Liquid Valves                              | None                                                                 |
| EP-102                                                          | Light Liquid Pumps                               | None                                                                 |
| EP-103                                                          | Gas Pressure Relief Valves                       | Closed Vent System to Precondenser, condenser, and scrubber (CD-100) |
|                                                                 | ME Wastewater                                    | None                                                                 |
| EP-104                                                          | Connectors                                       | None                                                                 |
|                                                                 | Agitators                                        | None                                                                 |
| EP-106                                                          | Haul Roads                                       | None                                                                 |
| EP-107                                                          | Methanol Storage Tank (76,310 gallons)           | Closed Vent System to Precondenser, condenser, and scrubber (CD-100) |
|                                                                 | Catalyst Tank (35,720 gallons)                   |                                                                      |

**Control Equipment – Condenser/Scrubber System Requirements:**

1. Special Condition 2.A: The permittee shall ensure that the condenser/scrubber system is in use at all times when the equipment listed below is in operation and shall be operated and maintained in accordance with the manufacturer's specifications.

**Equipment Controlled by Condenser/Scrubber (EP-100)**

| <b>Description</b>               |
|----------------------------------|
| Fatty Acid Stripping             |
| Glycerin Stripping               |
| Methyl Ester Drying              |
| Methanol and Water Rectification |

2. Special Condition 2.B: The permittee shall monitor and record the operating pressure of the scrubber using a continuous internal pressure monitor.
3. Special Condition 2.C: The permittee shall monitor and record the operating pressure drop across each scrubber at least once every 24 hours using Attachment D or an equivalent form, such as an electronic form, approved by the Air Pollution Control Program. The scrubber shall be equipped with a gauge or meter that indicates the pressure drop across the scrubber. The operating pressure drop shall be maintained within the design conditions specified by the manufacturer's performance warranty.
4. Special Condition 2.D: The permittee shall monitor and record the flow rate through the scrubber at least once every 24 hours using Attachment D or an equivalent form, such as an electronic form, approved by the Air Pollution Control Program. The scrubber shall be equipped with a flow meter that indicates the flow through the scrubber. The flow rate shall be maintained within the design conditions specified by the manufacturer's performance warranty.
5. Special Condition 2.E: The permittee shall maintain an operating and maintenance log for the scrubber which shall include the following: (use Attachment C or an equivalent form, such as an electronic form, approved by the Air Pollution Control Program)

- a) Incidents of malfunction, with impact on emissions, duration of event, probable cause, and corrective actions.
- b) Maintenance activities, with inspection schedule, repair action, and replacements, etc.; and
- c) A written record of regular inspection schedule, the date and results of all inspections including any actions or maintenance activities that result from the inspections.

**Requirements for Haul Roads**

1. Special Condition 3.A: The permittee shall pave all haul roads with material such as asphalt, concrete, and/or other material(s) after receiving approval from the Air Pollution Control Program. The pavement will be applied in accordance with industry standards for such pavement to achieve "Control of Fugitive Emissions" while the plant is operating. (Note: This condition has been satisfied)
2. Special Condition 3.B: The permittee shall conduct maintenance and/or repair of the surfaces as necessary to ensure that the physical integrity of the pavement is adequate to achieve control of fugitive emissions from these areas while the plant is operating.
3. Special Condition 3.C: The permittee shall periodically water, wash, and/or otherwise clean all the paved portions of the haul roads as necessary to achieve control of fugitive emissions from these areas while the plant is operating.

**Recordkeeping:**

1. The permittee shall maintain all records required by this permit for not less than five years and shall make them available to Department of Natural Resources' personnel upon request. [10 CSR 10-6.065(5)(C)1.C(II)(b)I and 10 CSR 10-6.065(5)(C)1.G(V)]

**Reporting:**

1. The permittee shall report no later than 10 days after records indicate an exceedance of any of the emission limitations to the Air Pollution Control Program's Compliance/Enforcement Section at: [10 CSR 10-6.065(5)(C)1.C(III)(c)III]
  - a) PO Box 176, Jefferson City, MO 65102 **and** [AirComplianceReporting@dnr.mo.gov](mailto:AirComplianceReporting@dnr.mo.gov) **or**
  - b) Any means that comply with 40 CFR Part 3.
2. The permittee shall report any deviations from the requirements of this permit condition in the semi-annual monitoring report and annual compliance certification required by Section V of this permit. [10 CSR 10-6.065(5)(C)1.C(III)(b)]

## **IV. Core Permit Requirements**

The installation shall comply with each of the following requirements. Consult the appropriate sections in the CFR, CSR, Missouri's SIP, and local ordinances for the full text of the applicable requirements. All citations, unless otherwise noted, are to the regulations in effect on the date of permit issuance. The following are only excerpts from the regulation or code and are provided for summary purposes only.

### **10 CSR 10-6.045 Open Burning Requirements**

1. This regulation prohibits the open burning of the following materials:
  - a) Petroleum-based materials, including but not limited to, tires, asphalt roofing material, carpet, and used oils;
  - b) Asbestos containing materials;
  - c) Trade waste, except untreated wood;
  - d) Construction or demolition waste, except untreated wood;
  - e) Salvage operation waste;
  - f) Household waste on or from properties with five or more residential units, such as mobile home parks or multi-family dwellings;
  - g) Household waste originated from another's property; or
  - h) Durable goods.
2. Open burning of certain materials, in certain areas of the state, and/or during specific times of the year are subject to additional requirements. The permittee should review 10 CSR 10-6.045 and local ordinances to see if additional restrictions apply prior to conducting open burning.

### **10 CSR 10-6.050 Start-up, Shutdown, and Malfunction Conditions**

1. In the event of a malfunction, which results in excess emissions that exceeds one hour, the permittee shall notify the Air Pollution Control Program in the form of a written report submitted within two business days. The written report shall include, at a minimum, the following: [10 CSR 10-6.050(3)(A)]
  - a) Name and location of installation; [10 CSR 10-6.050(3)(A)1]
  - b) Name and telephone number of person responsible for the installation; [10 CSR 10-6.050(3)(A)2]
  - c) Name of the person who first discovered the malfunction and precise time and date that the malfunction was discovered; [10 CSR 10-6.050(3)(A)3]
  - d) Identity of the equipment causing the excess emissions; [10 CSR 10-6.050(3)(A)4]
  - e) Time and duration of the period of excess emissions; [10 CSR 10-6.050(3)(A)5]
  - f) Cause of the excess emissions; [10 CSR 10-6.050(3)(A)6]
  - g) Air pollutants involved; [10 CSR 10-6.050(3)(A)7]
  - h) Estimate of the magnitude of the excess emissions expressed in the units of the applicable requirement and the operating data and calculations used in estimating the magnitude; [10 CSR 10-6.050(3)(A)8]
  - i) Measures taken to mitigate the extent and duration of the excess emissions; and [10 CSR 10-6.050(3)(A)9]
  - j) Measures taken to remedy the situation which caused the excess emissions and the measures taken or planned to prevent the recurrence of these situations. [10 CSR 10-6.050(3)(A)10]

2. The permittee shall notify the Air Pollution Control Program at least 10 days prior to any maintenance, start-up, or shutdown activity, which is expected to cause an excess release of emissions that exceeds one hour. If notification cannot be given 10 days prior to any maintenance, start-up, or shutdown activity, which is expected to cause an excess release of emissions that exceeds one hour, notification shall be given as soon as practicable prior to the maintenance, start-up, or shutdown activity. If prior notification is not given for any maintenance, start-up, or shutdown activity which resulted in an excess release of emissions that exceeded one hour, notification shall be given within two business days of the release. In all cases, the notification shall be a written report and include, at a minimum, the following: [10 CSR 10-6.050(3)(B)]
  - a) Name and location of installation; [10 CSR 10-6.050(3)(B)1]
  - b) Name and telephone number of person responsible for the installation; [10 CSR 10-6.050(3)(B)2]
  - c) Identity of the equipment involved in the maintenance, start-up, or shutdown activity; [10 CSR 10-6.050(3)(B)3]
  - d) Time and duration of the period of excess emissions; [10 CSR 10-6.050(3)(B)4]
  - e) Type of activity and the reason for the maintenance, start-up, or shutdown; [10 CSR 10-6.050(3)(B)5]
  - f) Type of air contaminant involved; [10 CSR 10-6.050(3)(B)6]
  - g) Estimate of the magnitude of the excess emissions expressed in the units of the applicable emission control regulation and the operating data and calculations used in estimating the magnitude; [10 CSR 10-6.050(3)(B)7]
  - h) Measures taken to mitigate the extent and duration of the excess emissions; and [10 CSR 10-6.050(3)(B)8]
  - i) Measures taken to remedy the situation which caused the excess emissions and the measures taken or planned to prevent the recurrence of these situations. [10 CSR 10-6.050(3)(B)9]
3. Information regarding the type and amount of emissions and time of the episode shall be recorded and kept on file. This data shall be included in emissions reported on any required EIQ. [10 CSR 10-6.050(4)(A)]
4. The information submitted according to 10 CSR 10-6.050(3)(A), (3)(B), and (3)(C)2 shall be kept on file at the installation for a period of five years. This data shall be included in emissions reported on any required EIQ. The information shall be available to the Director upon request. [10 CSR 10-6.050(4)(B)]

#### **10 CSR 10-6.060 Construction Permits Required**

1. The permittee shall obtain a construction permit prior to any of the following activities: [10 CSR 10-6.060(1)(D)]
  - a) The beginning of actual construction, modification, or major modification of any installation subject to this rule; [10 CSR 10-6.060(1)(D)1]
  - b) Operation after construction, modification, or major modification; or [10 CSR 10-6.060(1)(D)2]
  - c) Operation of any emission unit that has been permanently shut down. [10 CSR 10-6.060(1)(D)3]

**10 CSR 10-6.080 NESHAP and  
40 CFR Part 61, Subpart M - National Emission Standard for Asbestos**

The permittee shall follow the procedures and requirements of 40 CFR Part 61, Subpart M for any activities occurring at this installation which would be subject to provisions for 40 CFR Part 61, Subpart M - *National Emission Standard for Asbestos*.

**10 CSR 10-6.110 Reporting of Emission Data, Emission Fees, and Process Information**

1. The permittee shall pay an annual emission fee per ton of primary PM<sub>10</sub>, SO<sub>2</sub>, NO<sub>x</sub>, VOC, HAP (excluding HAP reported as PM<sub>10</sub> or VOC), and Lead based on previous calendar year emissions and in accordance with 10 CSR 10-6.110(3)(A)2 through (3)(A)7. [10 CSR 10-6.110(3)(A)1 **State Enforceable Only**]
2. Public Availability of Emission Data and Process Information. Any information obtained pursuant to the rule(s) of the Missouri Air Conservation Commission that would not be entitled to confidential treatment under 10 CSR 10-6.210 shall be made available to any member of the public upon request. [10 CSR 10-6.110(3)(D)]
3. All data collected and recorded in accordance with the provisions of this rule shall be retained by the permittee for not less than five years after the end of the calendar year in which the data was collected, and all these records shall be made available upon the Director's request. [10 CSR 10-6.110(4)]
4. The permittee shall collect information as required in 10 CSR 10-6.110(4). The information required in the emissions report is listed in Table 3 of this rule. All data elements must be reported initially, and only changed data elements must be reported subsequently. To ensure permit consistency, the Air Pollution Control Program Emissions Inventory Unit will help identify and quantify the data elements in Table 3 of this rule. [10 CSR 10-6.110(4)(A)]
5. The permittee shall submit a Full Emissions Report annually. [10 CSR 10-6.110(4)(B)2.A]
6. The Full Emissions Report shall be submitted either electronically via MoEIS, which requires Form 1.0 signed by an authorized company representative, or on EIQ paper forms. Alternate methods of reporting the emissions, such as spreadsheet file, can be submitted for approval by the Director. [10 CSR 10-6.110(4)(C)1]
7. The Full Emissions Report is due April 1 after each reporting year. If the Full Emissions Report is filed electronically via MoEIS, this due date is extended to May 1. [10 CSR 10-6.110(4)(C)3]

**10 CSR 10-6.130 Controlling Emissions during Episodes of High Air Pollution Potential**

The permittee shall submit a purple or maroon plan within 60 days of a request from the Director. [10 CSR 10-6.130(4)]

**10 CSR 10-6.150 Circumvention**

The permittee shall not cause or permit the installation or use of any device or any other means which, without resulting in reduction in the total amount of air contaminant emitted, conceal, or dilute an emission or air contaminant which violates a rule of the Missouri Air Conservation Commission. [10 CSR 10-6.150(1)]

**10 CSR 10-6.165 Restriction of Emission of Odors**

**This requirement is a State Only permit requirement.**

The permittee may not cause, permit, or allow the emission of odorous matter in concentrations and frequencies or for durations that odor can be perceived when one volume of odorous air is diluted with seven volumes of odor-free air for two separate trials not less than

15 minutes apart within the period of one hour. This odor evaluation shall be taken at a location outside of the installation's property boundary. [10 CSR 10-6.165(3)]

### **10 CSR 10-6.170**

#### **Restriction of Particulate Matter to the Ambient Air Beyond the Premises of Origin**

##### **Emission Limitation:**

1. The permittee shall not cause or allow fugitive particulate matter emissions to- [10 CSR 10-6.170(3)(A)1]
  - a) Remain visible in the ambient air beyond the property line of origin. [10 CSR 10-6.170(3)(A)1.B]
  - b) Go beyond the premises of origin in such quantities that the particulate matter may be found on surfaces beyond the property line of origin due to the following activities: [10 CSR 10-6.170(3)(A)1.A]
    - i) Handling, transporting, or storing of any material; [10 CSR 10-6.170(3)(A)1.A(I)]
    - ii) Construction, repair, cleaning, or demolition of a building or its appurtenances; [10 CSR 10-6.170(3)(A)1.A(II)]
    - iii) Construction or use of a road, driveway, or open area; or [10 CSR 10-6.170(3)(A)1.A(III)]
    - iv) Operation of a commercial or industrial facility. [10 CSR 10-6.170(3)(A)1.A(IV)]
2. The nature or origin of the particulate matter shall be determined to a reasonable degree of certainty by a technique proven to be accurate and approved by the Director. [10 CSR 10-6.170(3)(A)2]
3. Should it be determined that noncompliance has occurred, the Director may require reasonable control measures as may be necessary. These measures may include, but are not limited to, the following: [10 CSR 10-6.170(3)(B)]
  - a) Revision of procedures involving construction, repair, cleaning, and demolition of buildings and their appurtenances that produce particulate matter emissions; [10 CSR 10-6.170(3)(B)1]
  - b) Paving or frequent cleaning of roads, driveways, and parking lots; [10 CSR 10-6.170(3)(B)2]
  - c) Application of dust-free surfaces; [10 CSR 10-6.170(3)(B)3]
  - d) Application of water; and [10 CSR 10-6.170(3)(B)4]
  - e) Planting and maintenance of vegetative ground cover. [10 CSR 10-6.170(3)(B)5]

##### **Monitoring: [10 CSR 10-6.065(5)(C)1.C]**

1. The permittee shall conduct inspections of its facilities sufficient to determine compliance with this regulation. If the permittee discovers a violation, the permittee shall undertake corrective action to eliminate the violation.
2. The permittee shall maintain the following monitoring schedule:
  - a) The permittee shall conduct weekly observations for a minimum of eight consecutive weeks after permit issuance.
  - b) Should no violation of this regulation be observed during this period then-
    - i) The permittee may observe once every two weeks for a period of eight weeks.
    - ii) If a violation is noted, monitoring reverts to weekly.
    - iii) Should no violation of this regulation be observed during this period then-
      - (1) The permittee may observe once per month.
      - (2) If a violation is noted, monitoring reverts to weekly.

- c) If the permittee reverts to weekly monitoring at any time, monitoring frequency will progress in an identical manner to the initial monitoring frequency.

**Recordkeeping: [10 CSR 10-6.065(5)(C)1.C]**

1. The permittee shall document all inspections using Attachment B or an equivalent form such as an electronic form, approved by the Air Pollution Control Program, noting the following:
  - a) Whether air emissions (except water vapor) remain visible in the ambient air beyond the property line of origin.
  - b) Whether equipment malfunctions contributed to an exceedance.
  - c) Any violations and any corrective actions undertaken to correct the violation.

**10 CSR 10-6.180 Measurement of Emissions of Air Contaminants**

1. The Director may require the permittee to conduct tests to determine the quantity or nature, or both, of their air contaminant emissions. [10 CSR 10-6.180(3)(A)]
  - a) The Director may specify test methods to be used and observe testing as it is performed. [10 CSR 10-6.180(3)(A)1]
  - b) All tests must be performed by qualified personnel. [10 CSR 10-6.180(3)(A)2]
  - c) The Director shall be provided a copy of the test results in writing and signed by the person responsible for the tests. [10 CSR 10-6.180(3)(A)3]
2. The Director may conduct tests of emissions of air contaminants from any source. Upon the Director's request, the permittee shall provide necessary ports in stacks or ducts and other safe and proper sampling and testing facilities, exclusive of instruments and sensing devices as may be necessary for proper determination of the emission of air contaminants. [10 CSR 10-6.180(3)(B)]

**10 CSR 10-6.250 Asbestos Abatement Projects**

**Certification, Accreditation, and Business Exemption Requirements**

**This is a State Only permit requirement.**

The permittee shall conduct all asbestos abatement projects within the procedures established for certification and accreditation by 10 CSR 10-6.250. This rule requires individuals who work in asbestos abatement projects to be certified by the Air Pollution Control Program. This rule requires training providers who offer training for asbestos abatement occupations to be accredited by the Air Pollution Control Program. This rule requires persons who hold exemption status from certain requirements of this rule to allow the Department to monitor training provided to employees.

**10 CSR 10-6.280 Compliance Monitoring Usage**

1. For submission of compliance certificates, the permittee may use the following in addition to any specified compliance methods: [10 CSR 10-6.280(3)(A)]
  - a) Monitoring methods outlined in 40 CFR Part 64; [10 CSR 10-6.280(3)(A)1]
  - b) Monitoring method(s) approved within this operating permit; and [10 CSR 10-6.280(3)(A)2]
  - c) Any other monitoring methods approved by the Director. [10 CSR 10-6.280(3)(A)3]
2. Any credible evidence may be used for the purpose of establishing whether a permittee has violated or is in violation of an applicable requirement. Information from the use of the following methods is presumptively credible evidence of whether a violation has occurred at an installation: [10 CSR 10-6.280(3)(B)]

- a) Monitoring methods outlined in 40 CFR Part 64; [10 CSR 10-6.280(3)(B)1]
  - b) A monitoring method approved within this operating permit; and [10 CSR 10-6.280(3)(B)2]
  - c) Compliance test methods specified in 10 CSR 10-6.280 cited as the authority for the emission limitations. [10 CSR 10-6.280(3)(B)3]
3. The following testing, monitoring, or information gathering methods are presumptively credible testing, monitoring, or information gathering methods: [10 CSR 10-6.280(5)]
- a) Applicable monitoring or testing methods, cited in:
    - i) 10 CSR 10-6.030 *Sampling Methods for Air Pollution Sources*;
    - ii) 10 CSR 10-6.040 *Reference Methods*;
    - iii) 10 CSR 10-6.070 *NSPS*;
    - iv) 10 CSR 10-6.075 *MACT Regulations*; and
    - v) 10 CSR 10-6.080 *NESHAP*; or [10 CSR 10-6.280(5)(A)]
  - b) Other testing, monitoring, or information gathering methods, if approved by the Director, that produce information comparable to that produced by any method listed in 10 CSR 10-6.280(3)(B) or (5)(A). [10 CSR 10-6.280(5)(B)]

#### **40 CFR Part 82 Protection of Stratospheric Ozone (Title VI)**

##### **This is a Federal Only permit requirement.**

1. The permittee shall comply with the ozone-depleting substances (ODS) Phaseout Program, Exemption Programs, Imports & Exports Program, ODS Destruction Program, Stationary Refrigeration and Air Condition Program, Motor Vehicle Air Conditioning Program, Halons Program, Nonessential Products Ban Program, Labeling Program, and Significant New Alternatives Policy (SNAP) Program requirements at 40 CFR Part 82.
- a) The ODS Phaseout Program bans the production and import of Class I ODS and phases out the production and import of Class II ODS.
  - b) The Exemption Programs exclude specific uses of ODS from the phaseout, such as CFC-MDIs and methyl bromide.
  - c) The Imports & Exports Program establishes recordkeeping, reporting, and other requirements for importing virgin and used Class I and Class II substances.
  - d) The ODS Destruction Program establishes acceptable practices for destroying ODS in the United States.
  - e) The Stationary Refrigeration and Air Conditioning Program establishes service practices, technician certification, sales restrictions, and other requirements for these appliances and systems.
  - f) The Motor Vehicle Air Conditioning Program establishes servicing, technician certification, small can sales restrictions, and other requirements for these air-conditioning systems.
  - g) The Halons Program establishes installation, handling, and emissions requirements for halon fire suppression systems.
  - h) The Nonessential Products Ban Program concerns the ban on sale and introduction into interstate commerce of certain products manufactured with or containing ODS.
  - i) The Labeling Program establishes labeling requirements for ODS containers or products manufactured with ODS.
  - j) The SNAP Program identifies and evaluates substitutes for ozone-depleting substances.

## V. General Permit Requirements

The installation shall comply with each of the following requirements. Consult the appropriate sections in the CFR, CSR, and Missouri's SIP for the full text of the applicable requirements. All citations, unless otherwise noted, are to the regulations in effect as of the date that this permit is issued.

### Permit Duration and Extension of Expired Permits

1. This permit is issued for a term of five years, commencing on the date of issuance. [10 CSR 10-6.065(5)(C)1.B]
2. The permittee shall file a complete application for renewal of this operating permit at least six months before the date of permit expiration. In no event shall this time be greater than 18 months. [10 CSR 10-6.065(5)(B)1.A(III)]
3. This permit will expire at the end of this period unless renewed. If a timely and complete application for a permit renewal is submitted, but the Air Pollution Control Program fails to take final action to issue or deny the renewal permit before the end of the term of the previous permit, the previous permit shall not expire until the renewal permit is issued or denied. Any permit shield granted under the previous permit shall continue in effect during this period of time. However, the Administrator may invoke its authority under §505(e) of the Act to terminate or revoke and reissue the permit. [10 CSR 10-6.065(5)(E)3.C]

### General Record Keeping and Reporting Requirements

1. Record Keeping
  - a) All required monitoring data and support information shall be retained for a period of at least five years from the date of the monitoring sample, measurement, report, or application. Support information includes all calibration and maintenance records and all original strip-chart recordings when used for continuous monitoring instrumentation, and copies of all reports required by the permit. Where appropriate, records may be maintained in computerized form. [10 CSR 10-6.065(5)(C)1.C(II)(b)I]
  - b) Affected sources under Title IV of the Act have a three-year monitoring data record retention period as required in 40 CFR 75. [10 CSR 10-6.065(5)(C)1.C(II)(b)II]
2. Reporting
  - a) All reports shall be submitted to the Air Pollution Control Program's Compliance and Enforcement Section at:
    - i) P. O. Box 176, Jefferson City, MO 65102 and [AirComplianceReporting@dnr.mo.gov](mailto:AirComplianceReporting@dnr.mo.gov)
    - or
    - ii) Any means that comply with 40 CFR Part 3.
  - b) The permittee shall submit a report of all required monitoring by: [10 CSR 10-6.065(5)(C)1.C(III)(a)]
    - i) October 1st for monitoring which covers the January through June time period, and
    - ii) April 1st for monitoring which covers the July through December time period.
  - c) Each report shall identify any deviations from emission limitations, monitoring, record keeping, reporting, or any other requirements of this permit; this includes deviations or 40 CFR Part 64 exceedances. [10 CSR 10-6.065(5)(C)1.C(III)(b)]
  - d) Submit supplemental reports as required or as needed. All reports of deviations shall identify the cause or probable cause of the deviations and any corrective actions or preventative measures taken. [10 CSR 10-6.065(5)(C)1.C(III)(c)]

- i) Notice of any deviation resulting from an emergency (or upset) condition as defined at 10 CSR 10-6.065(5)(C)7.A shall be submitted to the permitting authority either verbally or in writing within two working days after the date on which the emission limitation is exceeded due to the emergency. The notice must contain a description of the emergency, the steps taken to mitigate emissions, and the corrective actions taken. [10 CSR 10-6.065(5)(C)1.C(III)(c)I]
- ii) Any deviation that poses an imminent and substantial danger to public health, safety, or the environment shall be reported as soon as practicable. [10 CSR 10-6.065(5)(C)1.C(III)(c)II]
- iii) Any other deviations identified in the permit as requiring more frequent reporting than the permittee's semiannual report shall be reported on the schedule specified in this permit. [10 CSR 10-6.065(5)(C)1.C(III)(c)III]
- e) Every report submitted shall be certified by the responsible official, except that, if a report of a deviation must be submitted within 10 days after the deviation, the report may be submitted without a certification if the report is resubmitted with an appropriate certification within 10 days after that, together with any corrected or supplemental information required concerning the deviation. [10 CSR 10-6.065(5)(C)1.C(III)(d)]
- f) The permittee may request confidential treatment of information submitted in any report of deviation. [10 CSR 10-6.065(5)(C)1.C(III)(e)]

### **Risk Management Plan Under §112(r)**

If the installation is required to develop and register a risk management plan pursuant to §112(r) of the Act, the permittee shall verify that it has complied with the requirement to register the plan. The contents of the risk management plan itself need not be incorporated as a permit term. [10 CSR 10-6.065(5)(C)1.D]

### **Severability Clause**

In the event of a successful challenge to any part of this permit, all uncontested permit conditions shall continue to be in force. All terms and conditions of this permit remain in effect pending any administrative or judicial challenge to any portion of the permit. If any provision of this permit is invalidated, the permittee shall comply with all other provisions of the permit. [10 CSR 10-6.065(5)(C)1.F]

### **General Requirements**

1. The permittee must comply with all the terms and conditions of this permit. Any noncompliance with a permit condition constitutes a violation and is grounds for enforcement action, permit termination, permit revocation and reissuance, permit modification, or denial of a permit renewal application. Note: The grounds for termination of a permit under 10 CSR 10-6.065(5)(C)1.G(I) are the same as the grounds for revocation as stated in 10 CSR 10-6.065(5)(E)8.A(I). [10 CSR 10-6.065(5)(C)1.G(I)]
2. The permittee may not use as a defense in an enforcement action that it would have been necessary for the permittee to halt or reduce the permitted activity in order to maintain compliance with the conditions of the permit [10 CSR 10-6.065(5)(C)1.G(II)]
3. The permit may be modified, revoked, reopened, reissued, or terminated for cause. Except as provided for minor permit modifications, the filing of an application or request for a permit modification, revocation and reissuance, or termination, or the filing of a notification of planned changes or anticipated noncompliance, does not stay any permit condition. [10 CSR 10-6.065(5)(C)1.G(III)]

4. This permit does not convey any property rights of any sort or grant any exclusive privilege. [10 CSR 10-6.065(5)(C)1.G(IV)]
5. The permittee shall furnish to the Air Pollution Control Program, upon receipt of a written request and within a reasonable time, any information that the Air Pollution Control Program reasonably may require to determine whether cause exists for modifying, reopening, reissuing, or revoking the permit or to determine compliance with the permit. Upon request, the permittee also shall furnish to the Air Pollution Control Program copies of records required to be kept by the permittee. The permittee may make a claim of confidentiality for any information, or records submitted pursuant to 10 CSR 10-6.065(5)(C)1. [10 CSR 10-6.065(5)(C)1.G(V)]

### **Incentive Programs Not Requiring Permit Revisions**

No permit revision will be required for any installation changes made under any approved economic incentive, marketable permit, emissions trading, or other similar programs or processes provided for in this permit. [10 CSR 10-6.065(5)(C)1.H]

### **Reasonably Anticipated Operating Scenarios [10 CSR 10-6.065(5)(C)1.I]**

There are no reasonably anticipated operating scenarios.

### **Compliance Requirements**

1. Any document (including reports) required to be submitted under this permit shall contain a certification signed by the responsible official. [10 CSR 10-6.065(5)(C)3.A]
2. Upon presentation of credentials and other documents as may be required by law, the permittee shall allow authorized officials of the Air Pollution Control Program or their authorized agents, to perform the following (subject to the installation's right to seek confidential treatment of information submitted to, or obtained by, the Air Pollution Control Program): [10 CSR 10-6.065(5)(C)3.B]
  - a) Enter upon the permittee's premises where a permitted installation is located or an emissions-related activity is conducted, or where records must be kept under the conditions of this permit; [10 CSR 10-6.065(5)(C)3.B(I)]
  - b) Have access to and copy, at reasonable times, any records that must be kept under the conditions of this permit; [10 CSR 10-6.065(5)(C)3.B(II)]
  - c) Inspect, at reasonable times and using reasonable safety practices, any facilities, equipment (including monitoring and air pollution control equipment), practices, or operations regulated or required under this permit; and [10 CSR 10-6.065(5)(C)3.B(III)]
  - d) As authorized by the Missouri Air Conservation Law, Chapter 643, RSMo or the Act, sample or monitor, at reasonable times, substances or parameters for the purpose of assuring compliance with the terms of this permit, and all applicable requirements as outlined in this permit. [10 CSR 10-6.065(5)(C)3.B(IV)]
3. All progress reports required under an applicable schedule of compliance shall be submitted semiannually (or more frequently if specified in the applicable requirement). These progress reports shall contain the following: [10 CSR 10-6.065(5)(C)3.D]
  - a) Dates for achieving the activities, milestones, or compliance required in the schedule of compliance, and dates when these activities, milestones, or compliance were achieved, and [10 CSR 10-6.065(5)(C)3.D(I)]
  - b) An explanation of why any dates in the schedule of compliance were not or will not be met, and any preventative or corrective measures adopted. [10 CSR 10-6.065(5)(C)3.D(II)]

4. The permittee shall submit an annual certification that it complies with all the federally enforceable terms and conditions contained in this permit, including emissions limitations, standards, or work practices. [10 CSR 10-6.065(5)(C)3.E]
  - a) These certifications shall be submitted annually by April 1st, unless the applicable requirement specifies more frequent submission. All deviations and 40 CFR Part 64 exceedances and excursions must be included in the compliance certifications. These certifications shall be submitted to: [10 CSR 10-6.065(5)(C)3.E(I)]
    - i) The Missouri Compliance Coordinator, Air Branch, Enforcement and Compliance Assurance Division, EPA Region 7 at 11201 Renner Blvd., Lenexa, KS 66219 **and**
    - ii) The Air Pollution Control Program's Compliance and Enforcement Section at:
      - (1) PO Box 176, Jefferson City, MO 65102 **and**  
[AirComplianceReporting@dnr.mo.gov](mailto:AirComplianceReporting@dnr.mo.gov) **or**
      - (2) Any means that comply with 40 CFR Part 3.
  - b) The compliance certification shall include the following: [10 CSR 10-6.065(5)(C)3.E(III)]
    - i) The identification of each term or condition of the permit that is the basis of the certification; [10 CSR 10-6.065(5)(C)3.E(III)(a)]
    - ii) The current compliance status, as shown by monitoring data and other information reasonably available to the permittee; [10 CSR 10-6.065(5)(C)3.E(III)(b)]
    - iii) Whether compliance was continuous or intermittent; [10 CSR 10-6.065(5)(C)3.E(III)(c)]
    - iv) The method(s) used for determining the compliance status of the installation, both currently and over the reporting period; and [10 CSR 10-6.065(5)(C)3.E(III)(d)]
    - v) Such other facts as the Air Pollution Control Program will require in order to determine the compliance status of this installation. [10 CSR 10-6.065(5)(C)3.E(III)(e)]

### **Permit Shield**

1. Compliance with the conditions of this permit shall be deemed compliance with all applicable requirements as of the date that this permit is issued, provided that: [10 CSR 10-6.065(5)(C)6.A]
  - a) The applicable requirements are included and specifically identified in this permit, or [10 CSR 10-6.065(5)(C)6.A(I)]
  - b) The permitting authority, in acting on the permit revision or permit application, determines in writing that other requirements, as specifically identified in this permit, are not applicable to the installation, and this permit expressly includes that determination or a concise summary of it. [10 CSR 10-6.065(5)(C)6.A(II)]
2. Be aware that there are exceptions to this permit protection. The permit shield does not affect the following: [10 CSR 10-6.065(5)(C)6.B]
  - a) The provisions of §303 of the Act or §643.090, RSMo concerning emergency orders; [10 CSR 10-6.065(5)(C)6.B(I)]
  - b) Liability for any violation of an applicable requirement which occurred prior to, or was existing at, the time of permit issuance; [10 CSR 10-6.065(5)(C)6.B(II)]
  - c) The applicable requirements of the acid rain program; [10 CSR 10-6.065(5)(C)6.B(III)]
  - d) The authority of EPA and the Air Pollution Control Program to obtain information; or [10 CSR 10-6.065(5)(C)6.B(IV)]
  - e) Any other permit or extra-permit provisions, terms, or conditions expressly excluded from the permit shield provisions. [10 CSR 10-6.065(5)(C)6.B(V)]

## Emergency Provisions

1. An emergency or upset means any condition arising from sudden and not reasonably foreseeable events beyond the control of the permittee, including acts of God, which require immediate corrective action to restore normal operation and that causes the installation to exceed a technology-based emission limitation under the permit due to unavoidable increases in emissions attributable to the emergency or upset. An emergency or upset does not include noncompliance caused by improperly designed equipment, lack of preventive maintenance, careless or improper operation, or operator error. [10 CSR 10-6.065(5)(C)7.A]
2. To establish an emergency- or upset-based defense, the permittee must demonstrate, through properly signed, contemporaneous operating logs or other relevant evidence, the following: [10 CSR 10-6.065(5)(C)7.B]
  - a) That an emergency or upset occurred and that the permittee can identify the source of the emergency or upset, [10 CSR 10-6.065(5)(C)7.B(I)]
  - b) That the installation was being operated properly, [10 CSR 10-6.065(5)(C)7.B(II)]
  - c) That the permittee took all reasonable steps to minimize emissions that exceeded technology-based emissions limitations or requirements in this permit, and [10 CSR 10-6.065(5)(C)7.B(III)]
  - d) That the permittee submitted notice of the emergency to the Air Pollution Control Program within two working days of the time when emission limitations were exceeded due to the emergency. This notice must contain a description of the emergency, any steps taken to mitigate emissions, and any corrective actions taken. [10 CSR 10-6.065(5)(C)7.B(IV)]

## Operational Flexibility

1. An installation that has been issued a Part 70 operating permit is not required to apply for or obtain a permit revision in order to make any of the changes to the permitted installation described in 10 CSR 10-6.065(5)(C)8.A if the changes are not Title I modifications, the changes do not cause emissions to exceed emissions allowable under the permit, and the changes do not result in the emission of any air contaminant not previously emitted. Emissions allowable under the permit means a federally enforceable permit term or condition determined at issuance to be required by an applicable requirement that establishes an emissions limit (including a work practice standard) or a federally enforceable emissions cap that the source has assumed to avoid an applicable requirement to which the source would otherwise be subject. At least seven days in advance of these changes, except as allowed for emergency or upset conditions, the permittee shall notify: [10 CSR 10-6.065(5)(C)8]
  - a) The Missouri Compliance Coordinator, Air Branch, Enforcement and Compliance Assurance Division, EPA Region 7 at 11201 Renner Blvd., Lenexa, KS 66219 **and**
  - b) The Air Pollution Control Program's Compliance and Enforcement Section at:
    - i) PO Box 176, Jefferson City, MO 65102 **and** [AirComplianceReporting@dnr.mo.gov](mailto:AirComplianceReporting@dnr.mo.gov)
    - or**
    - ii) Any means that comply with 40 CFR Part 3.
  - c) §502(b)(10) changes. Changes that, under §502(b)(10) of the Act, contravene an express permit term may be made without a permit revision, except for changes that would violate applicable requirements of the Act or contravene federally enforceable monitoring (including test methods), record keeping, reporting, or compliance requirements of this permit. [10 CSR 10-6.065(5)(C)8.A]

- i) Before making a change under this provision, the permittee shall provide advance written notice, describing the change to be made, the date on which the change will occur, any changes in emissions, and any permit terms and conditions that are affected. The permittee shall maintain a copy of the notice with this permit and the Air Pollution Control Program shall place a copy with the permit in the public file. Written notice shall be provided to EPA and the Air Pollution Control Program at least seven days before the change is to be made. If less than seven days' notice is provided because of a need to respond more quickly to these unanticipated conditions, the permittee shall provide notice to EPA and the Air Pollution Control Program as soon as possible after learning of the need to make the change. This notice shall be sent to: [10 CSR 10-6.065(5)(C)8.A(I)]
  - (1) The Missouri Compliance Coordinator, Air Branch, Enforcement and Compliance Assurance Division, EPA Region 7 at 11201 Renner Blvd., Lenexa, KS 66219 **and**
  - (2) The Air Pollution Control Program's Compliance and Enforcement Section at:
    - (a) PO Box 176, Jefferson City, MO 65102 **and** [AirComplianceReporting@dnr.mo.gov](mailto:AirComplianceReporting@dnr.mo.gov) **or**
    - (b) Any means that comply with 40 CFR Part 3.
- ii) The permit shield shall not apply to these changes. [10 CSR 10-6.065(5)(C)8.A(II)]

#### **Off-Permit Changes**

- 1. Except as provided in 10 CSR 10-6.065(5)(C)9.A, the permittee may make any change in its permitted operations, activities, or emissions that is not addressed in, constrained by, or prohibited by this permit without obtaining a permit revision. Insignificant activities listed in this permit, but not otherwise addressed in or prohibited by this permit, shall not be considered to be constrained by this permit for purposes of these off-permit provisions. Off-permit changes shall be subject to the following requirements and restrictions: [10 CSR 10-6.065(5)(C)9]
  - a) The change must meet all applicable requirements of the Act and may not violate any existing permit term or condition; the permittee may not change a permitted installation without a permit revision if this change is subject to any requirements under Title IV of the Act or is a Title I modification; [10 CSR 10-6.065(5)(C)9.A]
  - b) The permittee must provide contemporaneous written notice of the change. This written notice shall describe each change, including the date, any change in emissions, pollutants emitted, and any applicable requirement that would apply because of the change. This notice shall not be required for changes that are insignificant activities under 10 CSR 10-6.065(5)(B)3. This notice shall be sent to: [10 CSR 10-6.065(5)(C)9.B]
    - i) The Missouri Compliance Coordinator, Air Branch, Enforcement and Compliance Assurance Division, EPA Region 7 at 11201 Renner Blvd., Lenexa, KS 66219 **and**
    - ii) The Air Pollution Control Program's Compliance and Enforcement Section at:
      - (1) PO Box 176, Jefferson City, MO 65102 **and** [AirComplianceReporting@dnr.mo.gov](mailto:AirComplianceReporting@dnr.mo.gov) **or**
      - (2) Any means that comply with 40 CFR Part 3.
  - c) The permittee shall keep a record describing all changes made at the installation that result in emissions of a regulated air pollutant subject to an applicable requirement and the emissions resulting from these changes; and [10 CSR 10-6.065(5)(C)9.C]
  - d) The permit shield shall not apply to these changes. [10 CSR 10-6.065(5)(C)9.D]

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**Responsible Official [10 CSR 10-6.065(2)(W)]**

The application utilized in the preparation of this permit was signed by Lou Rickers, Chief Operations Officer. If this person terminates employment or is reassigned different duties such that a different person becomes the responsible person to represent and bind the installation in environmental permitting affairs, the permittee shall notify the Director of the Air Pollution Control Program of the change. Said notification shall be in writing and shall be submitted within 30 days of the change. The notification shall include the name and title of the new person assigned by the permittee to represent and bind the installation in environmental permitting affairs. All representations, agreement to terms and conditions and covenants made by the former responsible person that were used in the establishment of limiting permit conditions on this permit will continue to be binding on the installation until such time that a revision to this permit is obtained that would change said representations, agreements, and covenants.

**Reopening-Permit for Cause**

1. This permit shall be reopened for cause if: [10 CSR 10-6.065(5)(E)6.A]
  - a) The Air Pollution Control Program receives notice from EPA that a petition for disapproval of this permit pursuant to 40 CFR 70.8(d) has been granted, provided that the reopening may be stayed pending judicial review of that determination; [10 CSR 10-6.065(5)(E)6.A(I)]
  - b) The Air Pollution Control Program or EPA determines that this permit contains a material mistake or that inaccurate statements were made in establishing the emissions limitations, standards, or other terms of this permit; [10 CSR 10-6.065(5)(E)6.A(II)]
  - c) Additional applicable requirements under the Act become applicable to the installation; however, reopening on this ground is not required if—: [10 CSR 10-6.065(5)(E)6.A(III)]
    - i) This permit has a remaining term of less than three years; [10 CSR 10-6.065(5)(E)6.A(III)(a)]
    - ii) The effective date of the requirement is later than the date on which this permit is due to expire; or [10 CSR 10-6.065(5)(E)6.A(III)(b)]
    - iii) The additional applicable requirements are implemented in a general permit that is applicable to the installation and the installation receives authorization for coverage under that general permit; [10 CSR 10-6.065(5)(E)6.A(III)(c)]
  - d) The installation is an affected source under the acid rain program and additional requirements (including excess emissions requirements) become applicable to that source, provided that, upon approval by EPA, excess emissions offset plans shall be deemed to be incorporated into this permit; or [10 CSR 10-6.065(5)(E)6.A(IV)]
  - e) The Air Pollution Control Program or EPA determines that the permit must be reopened and revised to assure compliance with applicable requirements. [10 CSR 10-6.065(5)(E)6.A(V)]

**Statement of Basis**

This permit is accompanied by a statement setting forth the legal and factual basis for the permit conditions (including references to applicable statutory or regulatory provisions). This Statement of Basis, while referenced by the permit, is not an actual part of the permit. [10 CSR 10-6.065(5)(E)1.C]

## **VI. Attachments**

Attachments follow. Attachment A defines a list of abbreviations and acronyms commonly used throughout this permit.

## Attachment A

### Abbreviations and Acronyms

|                                                               |                                                                                                  |
|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>°C</b> .....degrees Celsius                                | <b>m/s</b> .....meters per second                                                                |
| <b>°F</b> ..... degrees Fahrenheit                            | <b>mg</b> .....milligrams                                                                        |
| <b>AAQIA</b> ..... ambient air quality impact analysis        | <b>kgal</b> ..... 1,000 gallons                                                                  |
| <b>acfm</b> ..... actual cubic feet per minute                | <b>MW</b> .....megawatt                                                                          |
| <b>BACT</b> ..... Best Available Control Technology           | <b>MHDR</b> .....maximum hourly design rate                                                      |
| <b>BLDS</b> ..... Bag leak detection system                   | <b>MMBtu</b> .....million British thermal units                                                  |
| <b>BMPs</b> ..... Best Management Practices                   | <b>MMgal</b> .....million gallons                                                                |
| <b>Btu</b> ..... British thermal unit                         | <b>mmHg</b> .....millimeters mercury                                                             |
| <b>CAM</b> ..... Compliance Assurance Monitoring              | <b>MMscf</b> .....Million standard cubic feet                                                    |
| <b>CAS</b> ..... Chemical Abstracts Service                   | <b>NAAQS</b> ....National Ambient Air Quality Standards                                          |
| <b>CEMS</b> ..... Continuous Emission Monitor System          | <b>NESHAP</b> ..National Emissions Standards for HAP                                             |
| <b>CFR</b> ..... Code of Federal Regulations                  | <b>NO<sub>x</sub></b> .....nitrogen oxides                                                       |
| <b>CO</b> ..... carbon monoxide                               | <b>NSPS</b> .....New Source Performance Standards                                                |
| <b>CO<sub>2</sub></b> ..... carbon dioxide                    | <b>NSR</b> .....New Source Review                                                                |
| <b>CO<sub>2e</sub></b> ..... carbon dioxide equivalent        | <b>PM</b> .....particulate matter                                                                |
| <b>COMS</b> ..... Continuous Opacity Monitoring System        | <b>PM<sub>2.5</sub></b> .....particulate matter less than 2.5 microns<br>in aerodynamic diameter |
| <b>CSR</b> ..... Code of State Regulations                    | <b>PM<sub>10</sub></b> .....particulate matter less than 10 microns<br>in aerodynamic diameter   |
| <b>dscf</b> ..... dry standard cubic feet                     | <b>ppm</b> .....parts per million                                                                |
| <b>dscm</b> .....dry standard cubic meter                     | <b>PSD</b> .....Prevention of Significant Deterioration                                          |
| <b>EIQ</b> ..... Emission Inventory Questionnaire             | <b>psi</b> .....pounds per square inch                                                           |
| <b>EP</b> ..... Emission Point                                | <b>PTE</b> .....potential to emit                                                                |
| <b>EPA</b> ..... Environmental Protection Agency              | <b>RACT</b> .....Reasonable Available Control<br>Technology                                      |
| <b>EU</b> ..... Emission Unit                                 | <b>RAL</b> .....Risk Assessment Level                                                            |
| <b>FGD</b> ..... flue gas desulfurization                     | <b>SCC</b> .....Source Classification Code                                                       |
| <b>FIRE</b> .....EPA's Factor Information Retrieval<br>System | <b>scfm</b> .....standard cubic feet per minute                                                  |
| <b>fps</b> ..... feet per second                              | <b>SCR</b> .....selective catalytic reduction                                                    |
| <b>ft</b> ..... feet                                          | <b>SDS</b> .....Safety Data Sheet                                                                |
| <b>GACT</b> ..... Generally Available Control Technology      | <b>SIC</b> .....Standard Industrial Classification                                               |
| <b>GHG</b> ..... Greenhouse Gas                               | <b>SIP</b> .....State Implementation Plan                                                        |
| <b>gpm</b> ..... gallons per minute                           | <b>SMAL</b> .....Screening Model Action Levels                                                   |
| <b>gr</b> ..... grains                                        | <b>SO<sub>x</sub></b> .....sulfur oxides                                                         |
| <b>GWP</b> ..... Global Warming Potential                     | <b>SO<sub>2</sub></b> .....sulfur dioxide                                                        |
| <b>HAP</b> ..... Hazardous Air Pollutant                      | <b>tph</b> .....tons per hour                                                                    |
| <b>hr</b> ..... hour                                          | <b>tpy</b> .....tons per year                                                                    |
| <b>HP</b> ..... horsepower                                    | <b>VMT</b> .....vehicle miles traveled                                                           |
| <b>lb</b> ..... pound                                         | <b>VOC</b> .....Volatile Organic Compounds                                                       |
| <b>lb/hr</b> ..... pounds per hour                            |                                                                                                  |
| <b>MACT</b> ..... Maximum Achievable Control<br>Technology    |                                                                                                  |
| <b>MERV</b> ..... minimum efficiency reporting value          |                                                                                                  |
| <b>µg/m<sup>3</sup></b> ..... micrograms per cubic meter      |                                                                                                  |

**Attachment B**

**Fugitive Emission Observations**

| Date | Time | Visible Emissions<br>Beyond the Property<br>Boundary |                  | Excess Emissions |                   | Initial |
|------|------|------------------------------------------------------|------------------|------------------|-------------------|---------|
|      |      |                                                      |                  | Cause            | Corrective Action |         |
|      |      | No                                                   | Yes <sup>2</sup> |                  |                   |         |
|      |      |                                                      |                  |                  |                   |         |
|      |      |                                                      |                  |                  |                   |         |
|      |      |                                                      |                  |                  |                   |         |
|      |      |                                                      |                  |                  |                   |         |
|      |      |                                                      |                  |                  |                   |         |
|      |      |                                                      |                  |                  |                   |         |
|      |      |                                                      |                  |                  |                   |         |
|      |      |                                                      |                  |                  |                   |         |
|      |      |                                                      |                  |                  |                   |         |
|      |      |                                                      |                  |                  |                   |         |
|      |      |                                                      |                  |                  |                   |         |
|      |      |                                                      |                  |                  |                   |         |
|      |      |                                                      |                  |                  |                   |         |
|      |      |                                                      |                  |                  |                   |         |
|      |      |                                                      |                  |                  |                   |         |

<sup>2</sup> If there are visible emissions, the permittee shall complete the excess emissions columns.

[illegible]

[illegible]

## STATEMENT OF BASIS

### INSTALLATION DESCRIPTION

Ag Processing, Inc. is a methyl ester (biodiesel) plant that began production on September 10, 2007. The plant produces 292 million pounds of methyl ester per year.

Biodiesel is produced from the base-catalyzed transesterification of vegetable oil with methanol. Glycerin is produced as a by-product of this process. The primary vegetable oil feedstock is refined soybean oil from the Ag Processing extraction/refining facility (adjacent facility 021-0060). The plant is designed to accept vegetable oil from alternative sources when the extraction/refining facility is unable to provide the oil. The Ag Processing extraction/refining facility is considered a separate installation for permitting purposes.

As the methyl esters are separated from the glycerin by-product, the excess methanol is recovered and recycled back into the process. Emissions from all the process equipment are vented to a single emission point and controlled by a condenser and scrubber as required by MACT FFFF. Biodiesel and crude glycerin are loaded onto tank trucks and rail cars for transport off-site. All haul roads are paved as required by Special Condition 3 of Construction Permit 122014-009. All steam for the plant is provided by an off-site source, therefore no boiler is required and there is no cooling tower associated with wastewater treatment of this plant.

The plant is adjacent to the Ag Processing, Inc. extraction/refining plant (021-0060). It is a major source of HAP emissions.

The installation is on the List of Named Installations found at 10 CSR 10-6.020(3)(B), Table 2. The installation is classified as item number 20: Chemical Process Plant. The installation's major source level is 100 tons per year and fugitive emissions are counted towards major source applicability.

### Installation Emissions Profile (tpy)

| Pollutants        | MoEIS Reported Actual Emissions |      |      |      |      | PTE <sup>3</sup> |
|-------------------|---------------------------------|------|------|------|------|------------------|
|                   | 2018                            | 2019 | 2020 | 2021 | 2022 |                  |
| PM                | N/A                             | N/A  | N/A  | N/A  | N/A  | 1.76             |
| PM <sub>10</sub>  | 0.55                            | 0.54 | 0.67 | 0.54 | 0.69 | 0.92             |
| PM <sub>2.5</sub> | 0.07                            | 0.08 | 0.10 | 0.08 | 0.10 | 0.09             |
| VOC               | 2.46                            | 2.49 | 3.19 | 3.25 | 2.25 | 37.72            |
| Combined HAP      | 1.73                            | 1.58 | 1.82 | 1.64 | 1.88 | 30.49            |
| Methanol          | 1.73                            | 1.58 | 1.82 | 1.64 | 1.88 | 30.49            |

N/A = not applicable

<sup>3</sup> Each emission unit was evaluated at 8,760 hours of uncontrolled annual operation unless otherwise noted.

- Methanol is the only HAP emitted from the equipment at this installation.
- HAPs were reported as VOC on EIQs
- Emissions from EP-100 and EP-107 are controlled by a scrubber (CD-100) which is made practically enforceable in Permit Conditions 001 and d002.

### Permit Reference Documents

These documents were relied upon in the preparation of the operating permit. Because they are not incorporated by reference, they are not an official part of the operating permit.

1. Part 70 Operating Permit Application, received March 21, 2023
2. 2018 through 2022 EIQs
3. U.S. EPA document AP-42, *Compilation of Air Pollutant Emission Factors*; Volume I, Stationary Point and Area Sources, Fifth Edition
4. WebFIRE
5. Construction Permit 122014-009

### Applicable Requirements Included in the Operating Permit but Not in the Application or Previous Operating Permits

In the operating permit application, the installation indicated they were not subject to the following regulation(s). However, in the review of the application, the agency has determined that the installation is subject to the following regulation(s) for the reasons stated.

None.

### Construction Permit History

Construction Permit 122014-009, Issued December 24, 2014:

- ♦ This minor NSR permit is for the increase in production of the plant from 250 million pounds per year to 292 million pounds per year.
- ♦ Special Condition 1 states that the special conditions of this permit supersede:
  - o All the special conditions in previously issued Construction Permit 102006-002.
- ♦ Special Condition 2 contains requirements for the use of the condenser/scrubber system (EP-100) at all times when Fatty Acid Stripping (EU-01), Glycerin Stripping (EU-02), Methyl Ester Dryer (EU-03), or Methanol and Water Rectification (EU-04) are in operation.
- ♦ Special Conditions 3 requires the pavement of all haul roads (EP-106) and maintenance and/or repair of the surfaces.
- ♦ Special Condition 5 required one time methanol stack testing that has already been conducted on April 25, 2015. The following table summarizes the test results:

**Summary of Test Results – Biodiesel Process**

|                | Units          | Run 1      | Run 2      | Run 3      |
|----------------|----------------|------------|------------|------------|
| Methanol       | µg             | 206,768.95 | 189,871.80 | 196,460.80 |
| Methanol       | lb/hr          | 0.2730     | 0.2521     | 0.2605     |
| Methanol       | ppm            | 1,449.30   | 1,338.23   | 1,383.13   |
| Sample Volume  | dry std liters | 106.938    | 106.349    | 106.467    |
| Flow Rate      | dscfm          | 37.68      | 37.68      | 37.68      |
| Run Start Time | HH:MM          | 7:24       | 10:33      | 13:40      |
| Run Stop Time  | HH:MM          | 10:24      | 13:33      | 16:40      |
| Date of Test   | mm/dd/yr       | 04/28/15   | 04/28/15   | 04/28/15   |

Construction Permit 102006-002, Issued October 3, 2006:

- ♦ This minor NSR permit is for the construction of a 250 million pounds per year biodiesel production plant.
- ♦ The special conditions in this permit were superseded by the special conditions in Construction Permit 122014-009.

## **NSPS Applicability**

40 CFR Part 60, Subpart Kb – *Standards of Performance for Volatile Organic Liquid Storage Vessels (Including Petroleum Liquid Storage Vessels) for Which Construction, Reconstruction, or Modification Commenced After July 23, 1984.*

NSPS Kb applies to volatile organic liquid storage vessels with a capacity greater than or equal to 19,800 gallons (75 m<sup>3</sup>) that were constructed after July 23, 1984. Per §60.110b(b), storage vessels with a capacity greater than or equal to 39,890 gallons (151 m<sup>3</sup>) with a maximum true vapor pressure of less than 3.5 kPa are not subject to NSPS Kb. Additionally, storage vessels with a capacity between 19,800 gallons and 39,890 gallons with a maximum true vapor pressure of less than 15.0 kPa are not subject to NSPS Kb.

The four 55,200 gallon and two 525,000-gallon biodiesel and two 55,330-gallon glycerin storage tanks are not subject to NSPS Kb because their vapor pressures are less than 3.5 kPa. The 35,720-gallon sodium methylate catalyst tank, the 26,172-gallon glycerin day tank, and the 36,320-gallon fatty acid tank are not subject to NSPS Kb because their vapor pressures are less than 15 kPa.

The 76,310-gallon methanol storage tank is subject to this rule because of the size of the tank and the vapor pressure of methanol, which is 12.9 kPa. The tank is subject to control in accordance with §60.112b(a)(3).

MACT FFFF will be used for compliance with NSPS Kb in accordance with §63.2535(c):

“The permittee complies with the provisions of MACT FFFF if the permittee has a storage tank with a fixed roof, closed-vent system, and control device in compliance with the provisions of either NSPS Kb, or NESHAP Y, except that the permittee must comply with the monitoring, recordkeeping, and reporting requirements in MACT FFFF. Alternatively, if a storage tank assigned to an MCPU is subject to control under NSPS Kb, or NESHAP Y, the permittee may elect to comply only with the requirements for Group 1 storage tanks in MACT FFFF.”

40 CFR Part 60, Subpart VV – *Standards of Performance for Equipment Leaks of VOC in the Synthetic Organic Chemicals Manufacturing Industry for Which Construction, Reconstruction, or Modification Commenced After January 5, 1981, and on or Before November 7, 2006.*

Per §60.480(e)(2): The permittee may choose to comply with the provisions of NSPS VVa to satisfy the requirements of NSPS VV for an affected facility.

40 CFR Part 60, Subpart VVa – *Standards of Performance for Equipment Leaks of VOC in the Synthetic Organic Chemicals Manufacturing Industry for Which Construction, Reconstruction, or Modification Commenced After November 7, 2006.*

NSPS VVa applies to affected facilities in the synthetic organic chemicals manufacturing industry. The facility is subject to NSPS VV, and therefore NSPS VVa, because of the production of glycerin as a by-product. MACT FFFF can be used for compliance with NSPS VV in accordance with §63.2535(k):

“After the compliance date specified in §63.2445, for an affected source with equipment that is also subject to the requirements of NSPS VV or NSPS VVa, or NESHAP V, the permittee may elect to apply MACT FFFF to all such equipment. After the compliance date specified in §63.2445, for an affected source with equipment to which MACT FFFF does not apply, but which is subject to the requirements of NSPS VV or NSPS VVa, or NESHAP V, the permittee may elect to apply MACT FFFF to all such equipment. If the permittee elects either of the methods of compliance in this paragraph, the permittee must consider all total organic compounds, minus methane and ethane, in such equipment for purposes of compliance with MACT FFFF, as if they were organic HAP. Compliance with the provisions of MACT FFFF, in the manner described in this paragraph, will constitute compliance with NSPS VV or NSPS VVa, and NESHAP V, as applicable.”

40 CFR Part 60, Subpart NNN – *Standards of Performance for Volatile Organic Compound (VOC) Emissions from Synthetic Organic Chemical Manufacturing Industry (SOCMI) Distillation Operations.*

NSPS NNN applies to affected facilities designated in §60.660(b) that are part of a process unit that produces any of the chemicals listed in §60.667 as a product, co-product, by-product, or intermediate, except as provided in §60.660(c). As defined in §60.660(b), an affected facility is any of the following for which construction, modification, or reconstruction commenced after December 30, 1983:

- 1) Each distillation unit not discharging its vent stream into a recovery system.
- 2) Each combination of a distillation unit and the recovery system into which its vent stream is discharged.
- 3) Each combination of two or more distillation units and the common recovery system into which their vent streams are discharged.

The definition of “product” as per §60.661 is any compound or chemical listed in §60.667 that is produced for sale as a final product as that chemical, or for use in the production of other chemicals or compounds. By-products, co-products, and intermediates are considered to be products.

Methanol is listed in §60.667, which is included in this process. However, the facility has a distillation unit for methanol recovery for use as a raw material. The recovered methanol is not a product, so NSPS NNN does not apply.

**40 CFR Part 60, Subpart RRR – *Standards of Performance for Volatile Organic Compound Emissions from Synthetic Organic Chemical Manufacturing Industry (SOCMI) Reactor Processes.***

NSPS RRR applies to affected facilities designated in §60.700(b) that are part of a process unit that produces any of the chemicals listed in §60.707 as a product, co-product, by-product, or intermediate, except as provided in §60.700(c). As defined in §60.700(b), an affected facility is any of the following for which construction, modification, or reconstruction commenced after June 29, 1990:

- 1) Each reactor process not discharging its vent stream into a recovery system.
- 2) Each combination of a reactor process and the recovery system into which its vent stream is discharged.
- 3) Each combination of two or more reactor processes and the common recovery system into which their vent streams are discharged.

The methyl ester facility has a reactor, which will produce an intermediate glycerin-rich phase that will be further processed to produce glycerol. Glycerol is listed in §60.707, so the facility is subject to NSPS RRR.

MACT FFFF can be used for compliance with NSPS RRR in accordance with §63.2535(h):

“After the compliance dates specified in §63.2445, for an MCPU that contains equipment subject to the provisions of MACT FFFF that are also subject to the provisions of NSPS DDD, NSPS III, NSPS NNN, or NSPS RRR, the permittee may elect to apply MACT FFFF to all equipment in the MCPU. If an MCPU subject to the provisions of MACT FFFF has equipment to which MACT FFFF does not apply but which is subject to a standard in NSPS DDD, NSPS III, NSPS NNN, or NSPS RRR, the permittee may elect to comply with the requirements for Group 1 process vents in MACT FFFF for such equipment. If the permittee elects any of these methods of compliance, the permittee must consider all total organic compounds, minus methane and ethane, in such equipment for purposes of compliance with MACT FFFF, as if they were organic HAP. Compliance with the provisions of MACT FFFF, in the manner described in this paragraph, will constitute compliance with NSPS DDD, NSPS III, NSPS NNN, or NSPS RRR, as applicable.”

**MACT Applicability**

40 CFR Part 63, Subpart EEEE - *National Emission Standards for Hazardous Air Pollutants: Organic Liquids Distribution (Non-gasoline)* does not apply. The biodiesel distributed by the Biodiesel Plant do not meet the definition of organic liquid in MACT EEEE as it does not contain 5 wt% or greater of an organic HAP listed in Table 1 of MACT EEEE.

40 CFR Part 63, Subpart FFFF – *National Emission Standards for Hazardous Air Pollutants: Miscellaneous Organic Chemical Manufacturing*

NSPS FFFF applies to this installation and is applied within the operating permit as Permit Condition 001. The Notification of Compliance Status Report for 40 CFR Part 63, Subpart FFFF - *Miscellaneous Organic Chemical Manufacturing* was signed February 7, 2008, and

submitted in accordance with §63.2520(d). A detailed review of the applicability of MACT FFFF can be found in Appendix A following the Statement of Basis.

- ♦ 40 CFR Part 63, Subpart F – *National Emission Standards for Organic HAP From the Synthetic Organic Chemical Industry* applies as referenced by MACT FFFF.
- ♦ 40 CFR Part 63, Subpart G – *National Emission Standards for Organic HAP From the Synthetic Organic Chemical Industry for Process Vents, Storage Vessels, Transfer Operations, and Wastewater* applies as referenced by MACT FFFF.
- ♦ 40 CFR Part 63, Subpart SS – *National Emission Standards for Closed Vent Systems, Control Devices, Recovery Devices, and Routing to a Fuel Gas System or a Process* applies as referenced by MACT FFFF.
- ♦ 40 CFR Part 63, Subpart UU – *National Emission Standards for Equipment Leaks – Control Level 2 Standards* applies as referenced by MACT FFFF.

### **NESHAP Applicability**

40 CFR Part 61, Subpart V – *National Emission Standard for Equipment Leaks (Fugitive Emission Sources)*

NESHAP V applies to each of the following sources that are intended to operate in volatile HAP (VHAP) service: pumps, compressors, pressure relief devices, sampling connection systems, open-ended valves or lines, valves, connectors, surge control vessels, bottoms receivers, and control devices or system required by NESHAP V.

MACT FFFF can be used for compliance with NESHAP V in accordance with §63.2535(k):

“After the compliance date specified in §63.2445, for an affected source with equipment that is also subject to the requirements of NSPS VV or NSPS VVa, or NESHAP V, the permittee may elect to apply MACT FFFF to all such equipment. After the compliance date specified in §63.2445, for an affected source with equipment to which MACT FFFF does not apply, but which is subject to the requirements of NSPS VV or NSPS VVa, or NESHAP V, the permittee may elect to apply MACT FFFF to all such equipment. If the permittee elects either of the methods of compliance in this paragraph, the permittee must consider all total organic compounds, minus methane, and ethane, in such equipment for purposes of compliance with MACT FFFF, as if they were organic HAP. Compliance with the provisions of MACT FFFF, in the manner described in this paragraph, will constitute compliance with NSPS VV or NSPS VVa, and NESHAP V, as applicable.”

### **10 CSR Division 10 Applicability**

10 CSR 10-6.220 *Restriction of Emission of Visible Air Contaminants* and 10 CSR 10-6.400 *Restriction of Emission of Particulate Matter From Industrial Processes* do not apply to EP-106 Haul Roads as fugitive emissions are exempted at 10 CSR 10-6.220(1)(K) and 10 CSR 10-6.400(1)(B)7.

## **CAM Applicability**

### **40 CFR Part 64 - CAM**

The CAM rule applies to each pollutant specific emission unit that:

- Is subject to an emission limitation or standard, and
- Uses a control device to achieve compliance, and
- Has pre-control emissions that exceed or are equivalent to the major source threshold.

A closed vent system to a precondenser, condenser, and scrubber (CD-100) are used to demonstrate compliance with the requirements of MACT FFFF applicable to EP-100 and EP-107; however, CAM does not apply per §64.2(b)(1)(i) as MACT FFFF contains emission limitations that were proposed by the Administrator after November 15, 1990 pursuant to §112 of the CAA. MACT FFFF was first proposed by the Administrator April 4, 2002.

## **Greenhouse Gas Emissions**

Note that this source may be subject to the Greenhouse Gas Reporting Rule. However, the preamble of the GHG Reporting Rule clarifies that 40 CFR Part 98 requirements do not have to be incorporated in Part 70 permits operating permits at this time. In addition, Missouri regulations do not require the installation to report CO<sub>2</sub> emissions in their EIQ; therefore, the installation's CO<sub>2</sub> emissions were not included within this permit. The applicant is required to report the data directly to EPA. The public may obtain CO<sub>2</sub> emissions data for this installation by visiting <https://www.epa.gov/ghgreporting/data-sets>.

## **Other Regulatory Determinations**

None.

## **Other Regulations Not Cited in the Operating Permit or the Above Statement of Basis**

Any regulation which is not specifically listed in either the Operating Permit or in the above Statement of Basis does not appear, based on this review, to be an applicable requirement for this installation for one or more of the following reasons:

1. The specific pollutant regulated by that rule is not emitted by the installation;
2. The installation is not in the source category regulated by that rule;
3. The installation is not in the county or specific area that is regulated under the authority of that rule;
4. The installation does not contain the type of emission unit which is regulated by that rule;
5. The rule is only for administrative purposes.

Should a later determination conclude that the installation is subject to one or more of the regulations cited in this Statement of Basis or other regulations which were not cited, the installation shall determine and demonstrate, to the Air Pollution Control Program's satisfaction, the installation's compliance with that regulation(s). If the installation is not in compliance with a regulation which was not previously cited, the installation shall submit to the Air Pollution Control Program a schedule for achieving compliance for that regulation(s).

## Appendix A

### Applicability of 40 CFR Part 63 Subpart FFFF

The Notification of Compliance Status Report for 40 CFR Part 63 Subpart FFFF, *Miscellaneous Organic Chemical Manufacturing*, was signed February 7, 2008, and submitted in accordance with §63.2520(d). The italicized items below are excerpts from the Notification of Compliance Status Report.

#### *40 CFR Part 63, Subpart FFFF - Miscellaneous Organic Chemical Manufacturing*

§63.2435 Is the installation subject to the requirements in MACT FFFF?

The Notification of Compliance Status Report signed February 7, 2008, states: *"The biodiesel production process located at the AGP facility in St. Joseph, MO is subject to 40 CFR Part 63, Subpart FFFF because it is located at a major HAP facility. The process SIC is defined as 2869, the process uses methanol (a HAP) and it is not covered by another NESHAP. There is one miscellaneous chemical process unit (MCPU) operating at this facility and it is the Biodiesel Process.*

*The biodiesel process involves reacting semi-refined or refined vegetable oils with methanol in the presence of a catalyst to produce biodiesel and crude glycerin. Because methanol is a HAP, and is used in this process, this process is subject to MACT FFFF. The entire biodiesel process line, including the methanol and catalyst storage tanks, is defined as a single MCPU. This MCPU is subject to several provisions of MACT FFFF. The applicability of each provision and the required compliance demonstrations are detailed below."*

The methyl ester (fatty acid) is subject to MACT FFFF because fatty acids have the SIC code 2869. The facility is regulated as new because it became operational after November 10, 2003.

§63.2450 What are the general requirements for complying with MACT FFFF?

The Notification of Compliance Status Report signed February 7, 2008, states; *"Per §63.2450(k)(5), the liquid flow rate will be continuously monitored on the scrubber. The gas now does not need to be monitored but needs to be either measured at the inlet of the scrubber or calculated based on the blower design (adjusted for pressure drop). There is no blower on this system and the inlet flow to the control equipment cannot be measured due to the system being under pressure. It is logical to assume that without a blower, the gas flow at the inlet of the scrubber is almost equivalent to the gas flow measured at the exit of the scrubber. Based on the engineering airflow calculation completed for the stack test, the average gas flow is 37 dscfm.*

*The liquid flow rate averages 9 gpm, with a range of 6 to 10 gpm. The gas flow at the inlet of the scrubber is assumed to be equivalent to the gas flow measured at the exit of the scrubber since there is no blower utilized in this system. Based on the stack test, the average gas flow is 37 dscfm. Using this information, the average L/G ratio during the stack test was 0.033. The range of the L/G flow will be 0.022 to 0.0336."*

## Emission Limits, Work Practice Standards, and Compliance Requirements

§63.2460 What requirements must the permittee meet for batch process vents?

The permittee must meet each emission limit in Table 2 to MACT FFFF that applies, and the permittee must meet each applicable requirement specified in §63.2460(b) and (c).

[§63.2460(a)]

The Notification of Compliance Status Report signed February 7, 2008, states: *"There are three reactors that operate separately as batch reactors with intermittent or discontinuous feed into the equipment. The procedures in §63.1257(d)(2)(i) and (iii) must be used to determine group status of the four batch vents above. These calculations can be found in Appendix B (Design Evaluation for Uncontrolled Emissions). Based on these calculations, the hourly and annual emission rates for each of these process vents are below. For calculated annual emission rate, a conservative estimate of 8000 hr/yr of operation was assumed.*

*Reactors 1,2,3-----2.92 lb/hr avg during each batch or 23,360 lb/yr each.*

*All three of the reactors' vents are Group 1 Batch vents (>10,000 lb/yr uncontrolled)."*

*"The exit vapor temperature is continuously monitored on the vent condenser, as required. For the vent condenser, the operating range for the exit temperature is 55°F to 85°F."*

The emissions testing report to determine the efficiency of the scrubber, installed to control methanol emissions, was reviewed by the Department of Natural Resources and a concurrence letter was sent dated March 20, 2008. The scrubber demonstrated a control efficiency greater than 99.99%.

§63.2455 What requirements must the permittee meet for continuous process vents?

The Notification of Compliance Status Report signed February 7, 2008, states: *"There are several process vents that operate under steady-state conditions and therefore these vents do not meet the definition of batch process vents. Along with the batch process vents and the storage tanks, all these process vents go to the common header and are vented to the pre-condenser, the final condenser, and the scrubber. Per §63.2450(c), the requirements for combined emission streams apply. Since there are Group 1 batch process vents at this facility, the requirements for Group 1 batch vents apply to all the vents in the combined emission stream and per §63.2455(b)(1), the facility is not required to determine the Group status of the continuous process vents.*

§63.2450 What are the general requirements for complying with MACT FFFF?

The permittee must comply with the emission limits and work practice standards in Tables 1 through 7 to MACT FFFF at all times, and the permittee must meet the requirements specified in §63.2445 through §63.2490 (or alternative means of compliance in §63.2495, §63.2500, or §63.2505), except as specified in §63.2450(b) through (v). The permittee must meet the notification, reporting, and recordkeeping requirements specified in §63.2515, §63.2520, and §63.2525. [§63.2450(a)(2)]

Requirements for combined emission streams. When organic HAP emissions from different emission types (e.g., continuous process vents, batch process vents, storage tanks, transfer

operations, and waste management units) are combined, the permittee must comply with the requirements of either §63.2450(c)(1) or (2). [§63.2450(c)]

Comply with the applicable requirements of MACT FFFF for each kind of organic HAP emissions in the stream (e.g., the requirements of Table 1 to MACT FFFF for continuous process vents and the requirements of Table 4 to MACT FFFF for emissions from storage tanks). [§63.2450(c)(1)]

Determine the applicable requirements based on the hierarchy presented in §63.2450(c)(2)(i) through (vi). For a combined stream, the applicable requirements are specified in the highest-listed paragraph in the hierarchy that applies to any of the individual streams that make up the combined stream. For example, if a combined stream consists of emissions from Group 1 batch process vents and any other type of emission stream, then the permittee must comply with the requirements in §63.2450(c)(2)(i) for the combined stream; compliance with the requirements of §63.2450(c)(2)(i) constitutes compliance for the other emission streams in the combined stream. Two exceptions are that the permittee must comply with the requirements in Table 3 to MACT FFFF and §63.2465 for all process vents with hydrogen halide and halogen HAP emissions, and recordkeeping requirements for Group 2 applicability or compliance are still required. (e.g., the requirement in §63.2525(e)(3) and (4) to track the number of batches produced and calculate rolling annual emissions for processes with Group 2 batch process vents). [§63.2450(c)(2)]

§63.2465 What requirements must the permittee meet for process vents that emit hydrogen halide and halogen HAP or HAP metals?

The Notification of Compliance Status Report signed February 7, 2008, states: *“Hydrochloric acid is utilized in the wash step of the process to remove any soaps that may be present. The HCl reacts with the soaps to form salts. These salts are not emitted to the atmosphere. Therefore, there are no process vents in the biodiesel process MCPU with potential HCl or chlorine emissions and there are no applicable hydrogen halide or halogen HAP requirements.”*

§63.2470 What requirements must the permittee meet for storage tanks?

The Notification of Compliance Status Report signed February 7, 2008, states: *“The methanol and catalyst storage tanks are Group 1 storage tanks, these tanks are designed to vent through the closed vent system connected to the Vent Condenser and Scrubber.*

*The biodiesel and crude glycerin storage tanks, as well as the loadouts associated with these products are not vented to any control devices. These storage tanks are not considered storage tanks regulated under MACT FFFF since they contain HAP only as impurities and are specifically excluded per definition of the storage tanks contained in §63.2550(i).*

*The HCl storage tank is less than 10,000 gallons so it does not meet the definition of a Group 1 storage tank and there are no applicable requirements. Phosphoric acid is not a HAP; therefore, the phosphoric acid storage tank is not regulated under MACT FFFF.”*

Vapor balancing alternative. As an alternative to the emission limits specified in Table 4 to MACT FFFF, the permittee may elect to implement vapor balancing in accordance with §63.1253(f), except as specified in §63.2470(e)(1) through (3). [§63.2470(e)]

*"None of the storage tanks at this facility are complying with the vapor balancing alternative."*

§63.2475 What requirements must the permittee meet for transfer racks?

The Notification of Compliance Status Report signed February 7, 2008, states: *"Per §63.2550(i), transfer racks are those used to fill tank trucks or railcars with organic liquids containing one or more organic HAP. Group 1 transfer racks are defined as having a throughput greater than 0.65 million L/yr and a rack-weighted partial pressure greater than or equal to 1.5 psia. There are no transfer racks associated with the biodiesel process MCPU that meet the definition of Group 1 transfer racks. The only Group 2 transfer racks load biodiesel and/or crude glycerin. There are no applicable requirements for Group 2 transfer racks."*

§63.2480 What requirements must the permittee meet for equipment leaks?

The Notification of Compliance Status Report signed February 7, 2008, states: *"The biodiesel process MCPU is subject to the equipment leaks provisions."*

§63.2485 What requirements must the permittee meet for wastewater streams and liquid streams in open systems with an MCPU?

The Notification of Compliance Status Report signed February 7, 2008, states: *"There is one wastewater stream from the biodiesel MCPU. The stream is identified as the "ME Wastewater" stream. The average ppmw of methanol in this stream is 6250 ppmw. Per §63.2485(c)(1), a Group 1 wastewater stream is defined as one that has more than 30,000 ppmw total for Table 8 and Table 9 compounds AND combined total annual loads of compounds in Tables 8 and 9 greater than or equal to 1 tpy. Based on sampling, this stream has less than 30,000 ppmw methanol, which is a Table 9 chemical. There are no Table 8 chemicals in this process or therefore this waste stream. This stream does not meet the definition of a Group 1 wastewater stream and is therefore categorized as a Group 2 wastewater stream."*

*As a Group 2 wastewater stream, the facility is required to comply with §63.146(b)(1) and §63.147(b)(8). §63.146(b)(1) required description of process unit, stream identification code, and concentration of Table 8 and/or Table 9 compounds (ppmw) including documentation on the methodology used, and flow rate in gpm. §63.147(b)(8) requires the facility to keep records containing this information in an accessible location.*

*Process Unit Description – The methyl ester process unit description is contained in this response to "§63.2435 Is the permittee subject to the requirements in MACT FFFF" above.*

*Stream Identification – WW1/ME Wastewater Stream*

*Concentration – Table 8 Compounds – 0 ppmw*

*Table 9 Compounds – 6250 ppmw average"*

§63.2490 What requirements must the permittee meet for the heat exchange systems?

The Notification of Compliance Status Report signed February 7, 2008, states: *"There are no heat exchanges that meet the definition of heat exchange system per §63.101."*

§63.2535 What compliance options does the permittee have if part of the plant is subject to both MACT FFFF and another subpart?

The Notification of Compliance Status Report signed February 7, 2008, states: *“The biodiesel process MCPU is subject to NSPS Kb and NSPS VV. AGP is opting to demonstrate compliance with these standards by complying with NESHAP FFFF and the provisions of §63.2535. These provisions require facilities to demonstrate compliance for total organic compounds instead of total organic HAP. Methanol is the only VOC utilized in this MCPU.”*

Applicability of process units included in a process unit group. The permittee may elect to develop and comply with the requirements for PUG in accordance with §63.2535(l)(1) through (3). [§63.2535(l)]  
*“Not applicable.”*

40 CFR Part 63, Subpart UU — National Emission Standards for Equipment Leaks—Control Level 2 Standards

§63.1039 Reporting Requirements.

The Notification of Compliance Status Report signed February 7, 2008, states: *“Below is the information required by §63.1039(a)(1)(i) through (iv). The requirements of §63.1039(a)(2) and (3) do not apply as the facility is not complying with the applicable alternative standards.”*

Process Unit or affected facility identification. [§63.1039(a)(1)(i)]

*“Ag Processing Inc.  
Biodiesel Production Facility  
900 Lower Lake Road  
St. Joseph, MO 64502*

*These responses apply to the biodiesel MCPU.”*

Number of each equipment type (e.g., valves, pumps) excluding equipment in vacuum service. [§63.1039(a)(1)(ii)]

| <i>“Equipment Type</i> | <i>Type of Service</i>    | <i>Component Count</i> |
|------------------------|---------------------------|------------------------|
| <i>Connectors</i>      | <i>Light Liquid/Vapor</i> | <i>1091</i>            |
| <i>Pumps</i>           | <i>Light Liquid</i>       | <i>21</i>              |
| <i>Agitators</i>       | <i>Vapor</i>              | <i>7</i>               |
| <i>Valves</i>          | <i>Light Liquid/Vapor</i> | <i>300</i>             |

*Vegetable oil, biodiesel, and glycerin are not VOC compounds. Therefore, the only VOC and/or HAP compound at this facility is methanol. The equipment count above reflects all the equipment that contains >5% methanol.”*

Method of compliance with the standard (e.g., “monthly leak detection and repair” or “equipped with dual mechanical seals”). [§63.1039(a)(1)(iii)]

*“An initial survey has been completed.*

*Valves shall be monitored per §63.1025. Repairs will be done in accordance with §63.1024.*

*Pumps shall be monitored per §63.1026. Weekly visual inspection shall be completed as required. If a visual leak is detected the pump shall be monitored per EPA Method 21 - Determination of VOC leaks. All repairs will be in accordance with §63.1024.*

*Connectors shall be monitored per §63.1027. Repairs will be done in accordance with §63.1024.*

*Agitators will be monitored per §63.1028. Weekly visual inspection shall be done. If a visual leak is detected the agitator shall be monitored per EPA Method 21 - Determination of VOC Leaks. All repairs will be in accordance with §63.1024.*

*Pressure Relief Valves will be monitored in accordance with §63.1030. Repairs will be made in accordance with §63.1024. LDAR requirements do not apply to the pressure relief devices which are conservation vents on storage tanks.”*

Planned schedule for requirements in §63.1025 and §63.1026. [§63.1039(a)(1)(iv)]

*“Valves in light liquid/vapor service will be monitored on an annual basis as per §63.1025. Pumps in light liquid service will be monitored monthly per §63.1026.”*

MACT FFFF lists 40 CFR Part 65, Subpart F or MACT UU as options, in Table 6 — Requirements for Equipment Leaks. Ag Processing has chosen to comply with MACT UU.

*40 CFR Part 63, Subpart UU — National Emission Standards for Equipment Leaks—Control Level 2 Standards*

*§63.1019 Applicability.*

The provisions of MACT UU apply to the control of air emissions from equipment leaks for which another subpart references the use of MACT UU for such air emission control. These air emission standards for equipment leaks are placed here for administrative convenience and only apply to facilities subject to a referencing subpart. The provisions of 40 CFR Part 63, Subpart A (General Provisions) do not apply to MACT UU except as noted in the referencing subpart. [§63.1019(a)]

MACT UU is referenced by MACT FFFF.

*40 CFR Part 63, Subpart WW — National Emission Standards for Storage Vessels (Tanks)—Control Level 2*

*§63.1060 Applicability.*

The provisions of MACT WW apply to the control of air emissions from storage vessels for which another subpart references the use of MACT WW for such air emission control. These air emissions standards for storage vessels are placed here for administrative convenience and only apply to facilities subject to a referencing subpart.

MACT WW is referenced by MACT FFFF.

*40 CFR Part 63, Subpart SS — National Emission Standards for Closed Vent Systems, Control Devices, Recovery Devices, and Routing to a Fuel Gas System or a Process*  
§63.980 Applicability.

The provisions of MACT SS include requirements for closed vent systems, control devices, and routing of air emissions to a fuel gas system or process. These provisions apply when another subpart references the use of MACT SS for such air emission control. These air emission standards are placed here for administrative convenience and only apply to those facilities subject to a referencing subpart.

MACT SS is referenced by MACT FFFF.

*40 CFR Part 63, Subpart F — National Emission Standards for Organic Hazardous Air Pollutants from the Synthetic Organic Chemical Manufacturing Industry*

§63.100 Applicability and designation of source.

MACT F provides applicability provisions, definitions, and other general provisions that are applicable to MACT G and MACT H. [§63.100(a)]

Except as provided §63.100(b)(4) and (c), the provisions of MACT F, MACT G, and MACT H apply to chemical manufacturing process units that meet all the criteria specified in §63.100(b)(1), (b)(2), and (b)(3): [§63.100(b)]

Manufacture as a primary product one or more of the chemicals listed in §63.100(b)(1)(i) or (b)(1)(ii). [§63.100(b)(1)]

One or more of the chemicals listed in Table 2 of MACT F; or [§63.100(b)(1)(i)]

One or more of the chemicals listed in §63.100(b)(1)(ii)(A) or (b)(1)(ii)(B):

[§63.100(b)(1)(ii)]

Tetrahydrobenzaldehyde (100-50-5); or [§63.100(b)(1)(ii)(A)]

Crotonaldehyde (123-73-9). [§63.100(b)(1)(ii)(B)]

Use a reactant or manufacture as a product, or co-product, one or more of the organic HAP listed in Table 2 of MACT F; [§63.100(b)(2)]

Are located at a plant site that is a major source as defined in §112(a) of the Act.

[§63.100(b)(3)]

Glycerol (glycerin) is listed in MACT F as an applicable product. It is not the primary product; it is a byproduct of methyl ester production.

MACT F is referenced by MACT FFFF.

*40 CFR Part 63, Subpart G — National Emission Standards for Organic HAP from the Synthetic Organic Chemical Manufacturing Industry for Process Vents, Storage Vessels, Transfer Operations, and Wastewater*

§63.110 Applicability.

MACT G applies to all process vents, storage vessels, transfer racks, wastewater streams, and in-process equipment subject to §63.149 within a source subject to MACT F. [§63.110(a)]

MACT G is referenced by MACT FFFF.

*40 CFR Part 63, Subpart H — National Emission Standards for Organic HAP for Equipment Leaks*

§63.160 Applicability and designation of source.

The provisions of MACT H apply to pumps, compressors, agitators, pressure relief devices, sampling connection systems, open-ended valves or lines, valves, connectors, surge control vessels, bottoms receivers, instrumentation systems, and control devices or closed vent systems required by MACT H that are intended to operate in organic HAP service 300 hours or more during the calendar year within a source subject to the provisions of a specified subpart in 40 CFR Part 63 that references MACT H. [§63.160(a)]

MACT H is referenced by MACT FFFF.

## Response to Public Comments

The draft Part 70 operating permit, Project 2023-03-045, for Ag Processing, Inc. (021-118) was placed on public notice as of July 26, 2024, for a 30-day comment period. The public notice was published on the Department of Natural Resources' Air Pollution Control Program's web page at: <https://dnr.mo.gov/air/what-were-doing/public-notice>.

The draft permit was placed on public notice with the information below:

Ag Processing, Inc. submitted a Part 70 Operating Permit application for its facility located in St. Joseph, Missouri. The department reviewed the application and proposes to issue a Part 70 Operating Permit. The department invites the public to review and offer written comments on the draft permit until **August 25, 2024**. The public can review the renewal application and draft permit below. Copies of the draft permit, all materials the applicant submitted, other than those granted confidential treatment, and a summary of other materials, if any, considered in making the preliminary permit determination are available from the Missouri Department of Natural Resources, Air Pollution Control Program, PO Box 176, Jefferson City, MO 65102-0176, or by calling **573-751-4817**. Written comments or any relevant information or materials may be submitted by mail to Stephen Hall, Missouri Department of Natural Resources, Air Pollution Control Program, PO Box 176, Jefferson City, MO 65102-0176 or by email to [apcpppermitspn@dnr.mo.gov](mailto:apcpppermitspn@dnr.mo.gov). Comments submitted by email are limited to 20 MB in size. Comments larger than 20 MB need to be submitted by mail to the address above.

An informal public hearing about the draft permit will be scheduled, if requested by **August 15, 2024**. The person(s) requesting the hearing must identify material issues regarding the preliminary permit determination. If a public hearing is scheduled, an announcement for the public hearing will be posted at <https://dnr.mo.gov/air/what-were-doing/public-notice>.

No comments were received during the public comment period.